EUGENE HECHT/ALFRED ZAJAC Adelphi University

OPTICS

ADDISON-WESLEY PUBLISHING COMPANY

READING, MASSACHUSETTS
MENLO PARK, CALIFORNIA · LONDON · AMSTERDAM · DON MILLS, ONTARIO · SYDNEY

5 Geometrical Optics—Paraxial Theory

5.1 INTRODUCTORY REMARKS

Suppose that we have an object which is either self-luminous or externally illuminated and imagine its surface as consisting of a large number of point sources. Each of these emits spherical waves, i.e. rays emanate radially in the direction of energy flow, or if you like, in the direction of the Poynting vector (Fig. 4.1). In this case, the rays *diverge* from a given point source S, whereas if the spherical wave were collapsing to a point, the rays would of course be *converging*. Generally one deals only with a small portion of a wavefront. *A point from which a portion of a spherical wave diverges, or one towards which the wave segment converges, is known as a focal point of the bundle of rays.*

Now envision the situation where we have a point source in the vicinity of some arrangement of reflecting and refracting surfaces representing an *optical system*. Of the infinity of rays emanating from S, generally speaking, only one will pass through an arbitrary point in space. Even so, it is possible to arrange for an infinite number of rays to arrive at a certain point P, as in Fig. 5.1. Thus, if for a cone of rays coming from S there is a corresponding cone of rays passing through P, the system is said to be *stigmatic* for these two points. The energy in the cone (apart from some inadvertent losses due to reflection, scattering and absorption) reaches P which is then referred to as a *perfect image* of S. The wave could conceivably arrive to form a finite patch of light or *blur spot* about P; it would still be an image of S but no longer a perfect one.

It follows from the principle of reversibility (Section 4.2.4) that a point source placed at P would equally well be imaged at S and accordingly these two are spoken of as *conjugate points*. In an *ideal optical system* every point of a three-dimensional region will be perfectly (or stigmatically) imaged in another region; the former being the *object space*, the latter the *image space*.

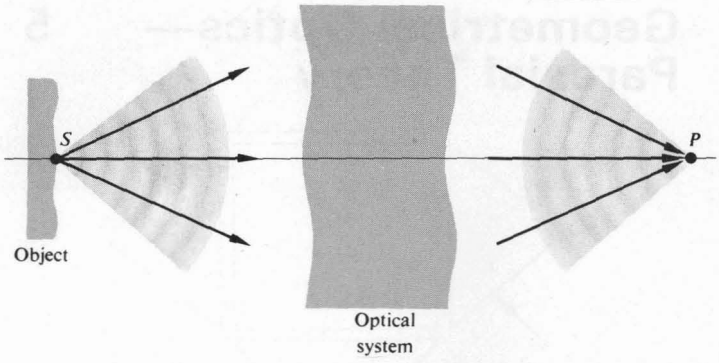


Fig. 5.1 Converging and diverging waves.

Most commonly, the function of an optical device is to collect and reshape a portion of the incident wavefront, often with the ultimate intent of forming an image of an object. Notice that inherent in realizable systems is the limitation of being unable to collect all of the emitted light; the system accepts only a segment of the wavefront. As a result, there will always be an apparent deviation from rectilinear propagation even in homogeneous media—the waves will be *diffracted*. The attainable degree of perfection in the imaging capability of a real optical system will therefore be *diffraction limited* (there will always be a blur spot). As the wavelength of the radiant energy decreases in comparison to the physical dimensions of the optical system, the effects of diffraction become less significant. In the conceptual limit as $\lambda_0 \to 0$ rectilinear propagation obtains in homogeneous media and we have the idealized domain of *geometrical optics*.* Behavior which is specifically attributable to the wave nature of light (e.g. interference and diffraction) would no longer be observable. There are many situations in which the great simplicity arising from the approximation of geometrical optics more than compensates for its inaccuracies. In short, *the subject treats the controlled manipulation of wavefronts (or rays) by means of the interpositioning of reflecting and or refracting bodies, neglecting any diffraction effects.*

5.2 LENSES

No doubt the most widely used optical device is the lens and that notwithstanding the fact that we see the world through a pair of them. Lenses date back to the burning glasses of antiquity and indeed who can say when man first peered through the liquid lens formed by a droplet of water?

As an initial step toward an understanding of what lenses do and how they manage to do it, let's examine what happens when light impinges on the curved surface of a transparent dielectric medium.

5.2.1 Refraction at Aspherical Surfaces

Imagine that we have a point source S whose spherical waves arrive at a boundary between two transparent media as shown in Fig. 5.2. We would like to determine the shape which the interface must have in order that the wave traveling within the second medium converges to a point P, there forming a perfect image of S. Practical reasons for wanting to focus a diverging wave to a point will become evident as we proceed.

The time it takes for each and every portion of a wavefront leaving S to converge at P must be identical. Or in the words of Section 4.2.3, "the distance between corresponding points on any and all rays will be traversed in that same time." Another way to say essentially the same thing from the perspective of Fermat's principle is that if a great many different rays are to go from S to P (i.e. if point A in Fig. 5.3 can be anywhere on the interface) each ray must traverse the same

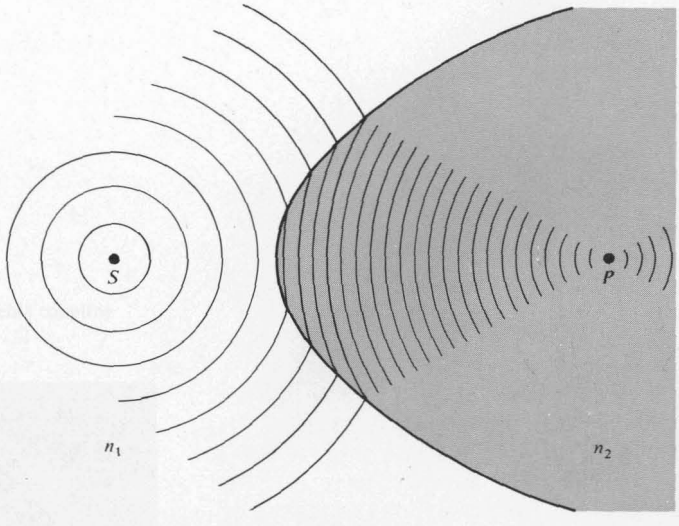


Fig. 5.2 Reshaping a spherical wave at a refracting interface ($n_1 < n_2$).

* *Physical optics* deals with situations in which the nonzero wavelength of light must be reckoned with. Analogously, when the de Broglie wavelength of a material object is negligible, we have *classical mechanics*; when it is not, we have the domain of *quantum mechanics* (see Chapter 13).

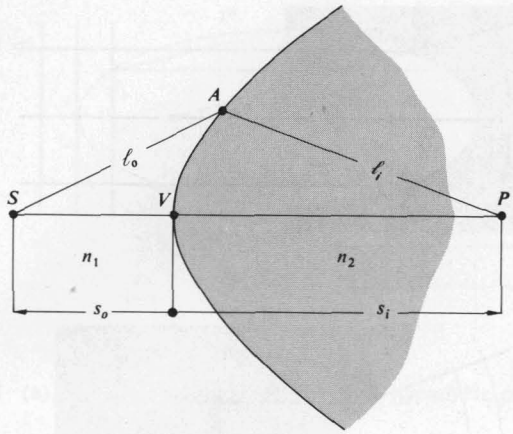


Fig. 5.3 The Cartesian oval.

optical path length. Thus, for example, if S is in a medium of index n_1 and P is in an optically more dense medium of index n_2,

$$\ell_o n_1 + \ell_i n_2 = s_o n_1 + s_i n_2 \qquad (5.1)$$

where s_o and s_i are the *object* and *image distances* measured from the *vertex* or *pole* V respectively. Once we choose s_o and s_i the right-hand side of this equation becomes fixed, and so

$$\ell_o n_1 + \ell_i n_2 = \text{constant}. \qquad (5.2)$$

This is the equation of a *Cartesian oval* whose significance in optics was studied extensively by René Descartes in the early sixteen hundreds (Problem 5.1). Hence, when the boundary between two media has the shape of a Cartesian oval of revolution about the $\overline{SP}$ or *optical axis*, S and P will be conjugate points, i.e. a point source at either location will be perfectly imaged at the other. What's actually occuring physically is rather easy to comprehend. Since $n_2 > n_1$ those regions of the wavefront traveling in the optically more dense medium move slower than those regions traversing the rarer material. Consequently, as the wave begins to pass through the vertex of the oval, the segment immediately about the optical axis is slowed down from c/n_1 to c/n_2. Regions of the same wavefront remote from the axis are still in the first medium traveling with a greater speed, c/n_1. Thus the wavefronts bend and if the boundary is properly configured (in the form of a Cartesian ovoid), the wavefronts will be inverted from diverging to converging spherical segments.

In addition to focusing a spherical wave, we would like to be able to perform a few other reshaping operations using refracting interfaces; some of these are illustrated in Fig. 5.4 We shall consider these only briefly and more for pedagogica than practical reasons. The surfaces in Fig. 5.4(a) and (b) are ellipsoidal while (c) and (d) are hyperboloidal. Notice that in all cases, the rays either diverge from or converge toward the foci. The arrowheads were omitted to indicate that the rays can go either way. In other words, an incident plane wave will converge to the farthest focus of an ellipsoic just as a spherical wave emitted from that focus will emerge as a plane wave. Furthermore, as you might expect, if we let the point S in Fig. 5.2 move out to infinity the ovoid would gradually metamorphose into an ellipsoid.

Rather than deriving expressions for these surfaces let's just justify the above remarks. To that end, examine Fig. 5.5 which relates back to Fig. 5.4(a). The optical path lengths from any point D on the planar wavefront Σ to the focus F_1 must all be equal to the same constant C, that is

$$(\overline{F_1A})n_2 + (\overline{AD})n_1 = C$$

or

$$(\overline{F_1A}) + (\overline{AD})n_{12} = C/n_2. \qquad (5.3)$$

To see that this relationship is indeed satisfied by an ellipsoid of revolution recall that if Σ corresponds to the directrix of the ellipse, $(\overline{F_2A}) = e(\overline{AD})$ where e is the eccentricity. Thus if $e = n_{12}$, the left-hand side of Eq. (5.3) becomes $(\overline{F_1A}) + (\overline{F_2A})$ which is most certainly constant for an ellipse. Here the eccentricity is less than one ($e = n_1/n_2$) but had it been greater than one (i.e. $n_1 > n_2$) the curve would have been a hyperbola instead [compare (a) with (c) and (b) with (d) in Fig. 5.4]. If all of this brings back memories of analytic geometry, you might keep in mind that that subject was originated by Descartes.

The knowledge we have at hand now may be used to construct lenses such that both the object and image points can be in the same medium, which is usually air. The first such device to be considered [Fig. 5.6(a)] is a *double convex hyperbolic* lens which utilizes the response characterized in Fig. 5.4(c). A diverging spherical wave becomes planar after traversing the first hyperbolic surface and then spherically converging on leaving the lens. Alternatively, if the second surface is made planar so that we have a *hyperbolic planar convex* lens as in Fig. 5.6(b), the plane waves within the lens will strike the back surface perpendicularly and emerge unaltered. Another arrangement which will convert diverging spherical waves into plane waves is illustrated in Fig. 5.6(c). This is a *sphero-elliptic convex* lens where F_1 is simultaneously at the center of the spherical

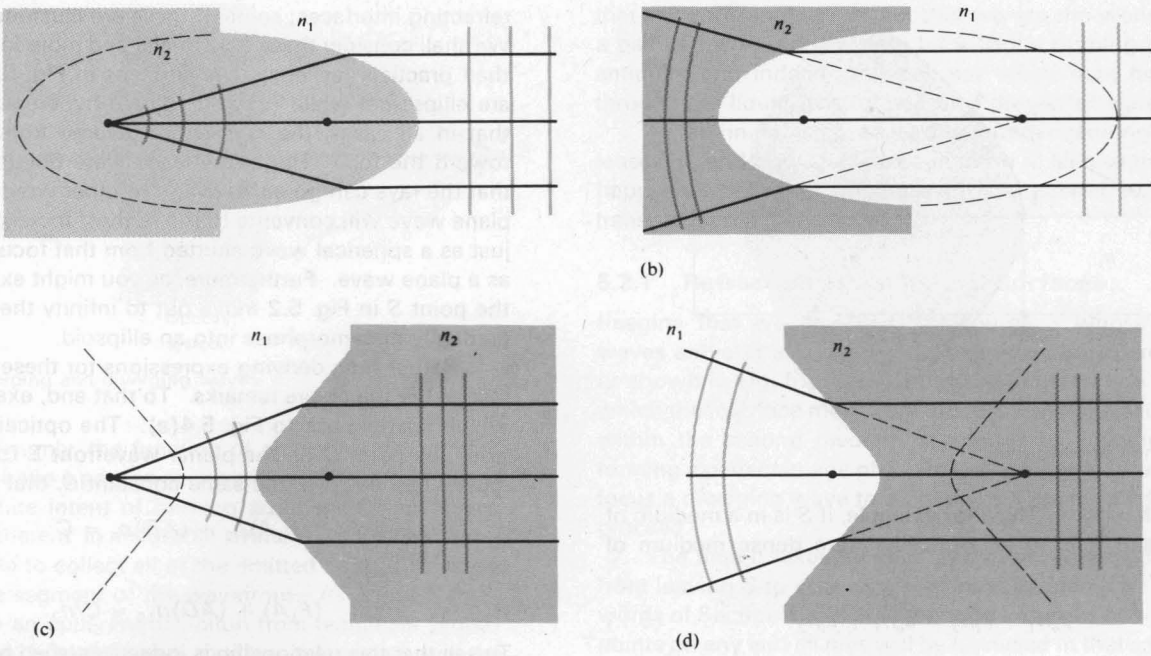


Fig. 5.4 Ellipsoidal and hyperboloidal refracting surfaces ($n_2 > n_1$).

surface and at the focus of the ellipsoid. Rays from F_1 strike the first surface perpendicularly and are therefore undeviated by it. As in Fig. 5.4(a), the exiting wavefronts are planar. Each of the elements thus far examined have been thicker at their midpoints than they were at their edges and are, for that reason, said to be *convex* (from the Latin *convexus* meaning arched). In contrast, the *planar hyperbolic concave* lens (from the Latin *concavus* meaning hollow; and most easily remembered because it contains the word cave) is thinner at the middle than at the edges, as is evident in Fig. 5.6(d). A

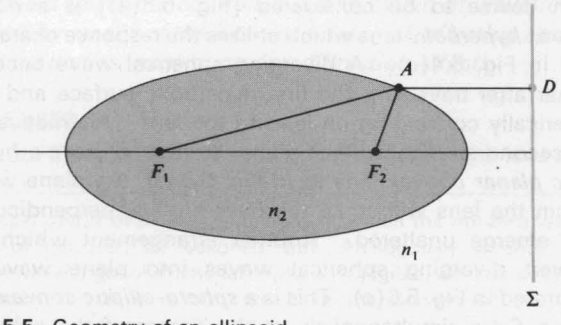


Fig. 5.5 Geometry of an ellipsoid.

number of other arrangements are possible and a few will be considered in the problems (5.2). Note that each of these lenses will work just as well in reverse where the waves shown emerging can instead be thought of as entering from the right.

If a point source is positioned on the optical axis at the point F_1 of the lens in Fig. 5.6(a), rays will *converge* to the conjugate point F_2. A luminous image of the source would appear on a screen placed at F_2, an image which is therefore said to be *real*. On the other hand, in Fig. 5.6(d) the point source is at infinity and the rays emerging from the system this time are *diverging*. They appear to come from a point F_2 but no actual luminous image will appear on a screen at that location. The image here is spoken of as *virtual* just as is the familiar image generated by a plane mirror.

5.2.2 Refraction at Spherical Surfaces

Imagine that we have two pieces of material, one with a concave and the other a convex spherical surface, both having the same radius. It is a unique property of the sphere that such pieces will fit together in intimate contact regardless of their mutual orientation. Thus if we take two roughly spherical objects of suitable curvature, one a grinding tool

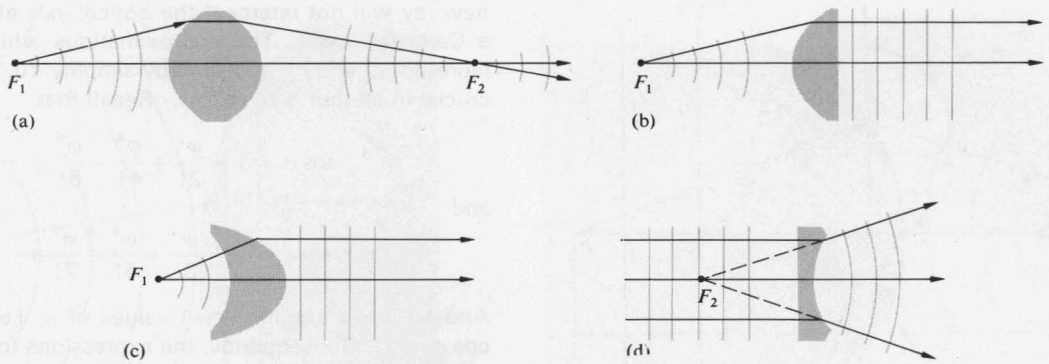


Fig. 5.6 (a) A double hyperbolic lens. (b) A hyperbolic planar convex lens. (c) A sphero-elliptic lens. (d) A planar hyperbolic lens.

and the other a disc of glass, separate them with some abrasive and then randomly move them with respect to each other, we can anticipate that any high spots on either object will wear away. As they wear, both pieces will gradually become ever more spherical (Fig. 5.7). Such surfaces are now commonly generated in batches by automatic grinding and polishing machines. In contrast, high quality aspherical shapes require considerably more effortful working and reworking and accordingly are used, as a rule, only when the concomitant high costs are justifiable. (Molded plastic aspherics are becoming more common in applications not requiring great precision.) It should therefore come as no surprise that the vast majority of quality lenses in use today have spherical surfaces. Our intent here is to establish techniques for using such surfaces whereby a great many object points can be satisfactorily imaged simultaneously in light composed of a broad frequency range. Image errors, known as *aberrations*, will occur but it is possible with the present technology to construct high-quality spherical lens systems whose aberrations are so well controlled that image fidelity is limited only by diffraction.

Fig. 5.7 Polishing a spherical lens. [Photo courtesy Optical Society of America.]

Now that we know why and where we are going, let's move on. Figure 5.8 depicts a wave from the point source S impinging on a spherical interface of radius R centered at C. The ray ($\overline{SA}$) will be refracted at the interface toward the local normal ($n_2 > n_1$) and therefore toward the optical axis. Assume that at some point P it will cross the axis as will all other rays incident at the same angle θ_i (Fig. 5.9). Fermat's principle maintains that the optical path length (O.P.L.) will be stationary, i.e. its derivative with respect to the position variable will be zero. For the ray in question

$$(\text{O.P.L.}) = n_1\ell_o + n_2\ell_i. \tag{5.4}$$

Using the law of cosines in triangles SAC and ACP along with the fact that $\cos\varphi = -\cos(180 - \varphi)$ we get

$$\ell_o = [R^2 + (s_o + R)^2 - 2R(s_o + R)\cos\varphi]^{1/2}$$

and

$$\ell_i = [R^2 + (s_i - R)^2 + 2R(s_i - R)\cos\varphi]^{1/2}.$$

The O.P.L. can be rewritten as

$$(\text{O.P.L.}) = n_1[R^2 + (s_o + R)^2 - 2R(s_o + R)\cos\varphi]^{1/2} + n_2[R^2 + (s_i - R)^2 + 2R(s_i - R)\cos\varphi]^{1/2}.$$

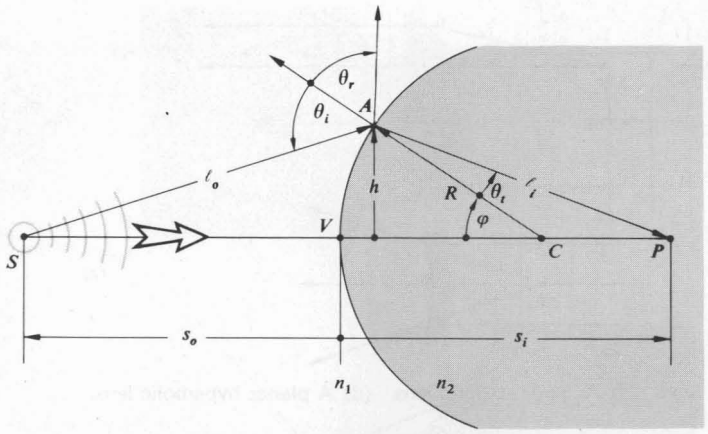


Fig. 5.8 Refraction at a spherical interface.

All of the quantities in the diagram, i.e. s_i, s_o, R etc. are positive numbers and these form the basis of a *sign convention* which is gradually unfolding and to which we shall return time and again (see Table 5.1). Inasmuch as the point A moves at the end of a fixed radius (i.e. R = constant), φ is the position variable and thus setting $d(\text{O.P.L.})/d\varphi = 0$ via Fermat's principle we have

$$\frac{n_1 R(s_o + R)\sin\varphi}{2\ell_o} - \frac{n_2 R(s_i - R)\sin\varphi}{2\ell_i} = 0,$$

from which it follows that

$$\frac{n_1}{\ell_o} + \frac{n_2}{\ell_i} = \frac{1}{R}\left(\frac{n_2 s_i}{\ell_i} - \frac{n_1 s_o}{\ell_o}\right). \tag{5.5}$$

Table 5.1 Sign convention for spherical refracting surfaces and thin lenses* (light entering from the left).

s_o, f_o	+ left of V
x_o	+ left of F_o
s_i, f_i	+ right of V
x_i	+ right of F_i
R	+ if C is right of V
y_o, y_i	+ above optical axis

* This table anticipates the imminent introduction of a few quantities not yet spoken of.

This is the relationship which must hold amongst the parameters for a ray going from S to P by way of refraction at the spherical interface. Although this expression has the attribute of being exact it is rather complicated. We already know that if A is moved to a new location by changing φ the new ray will not intercept the optical axis at P—this is not a Cartesian oval. The approximations which are used to represent ℓ_o and ℓ_i, and thereby simplify Eq. (5.5), are quite crucial in all that is to follow. Recall that

$$\cos\varphi = 1 - \frac{\varphi^2}{2!} + \frac{\varphi^4}{4!} - \frac{\varphi^6}{6!} + \cdots \tag{5.6}$$

and

$$\sin\varphi = \varphi - \frac{\varphi^3}{3!} + \frac{\varphi^5}{5!} - \frac{\varphi^7}{7!} + \cdots. \tag{5.7}$$

And so if we assume small values of φ, i.e. A close to V, $\cos\varphi \approx 1$. Consequently, the expressions for ℓ_o and ℓ_i yield $\ell_o \approx s_o$, $\ell_i \approx s_i$ and to that approximation

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}. \tag{5.8}$$

We could have begun this derivation with Snell's law rather than Fermat's principle (Problem 5.3) in which case small values of φ would have led to $\sin\varphi \approx \varphi$ and Eq. (5.8) once again. This approximation delineates the domain of what is called *first-order theory*—we'll examine *third-order theory* ($\sin\varphi \approx \varphi - \varphi^3/3!$) in the next chapter. Rays which do arrive at shallow angles with respect to the optical axis (such that φ and h are appropriately small) are known as *paraxial rays*. The *emerging wavefront segment corresponding to these paraxial rays is essentially spherical and will form a "perfect" image at its center P located at* s_i. Notice that Eq. (5.8) is certainly independent of the location of A over a small area about the symmetry axis, namely the *paraxial region*. Gauss in 1841 was the first to give a systematic exposition of the formation of images under the above approximation and the result is variously known as *first-order, paraxial,* or *Gaussian optics*. If the optical system is

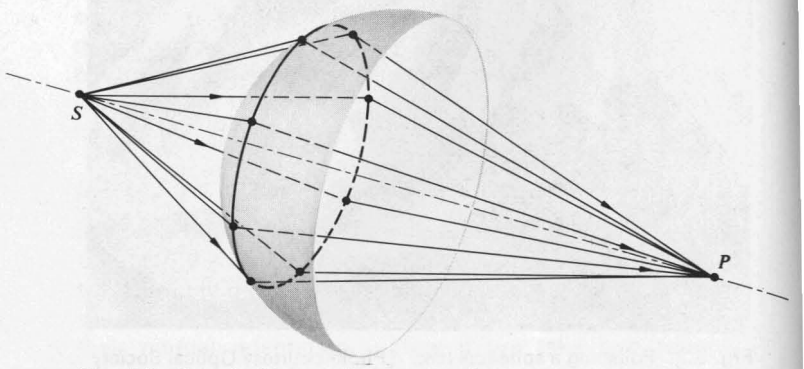


Fig. 5.9 Rays incident at the same angle.

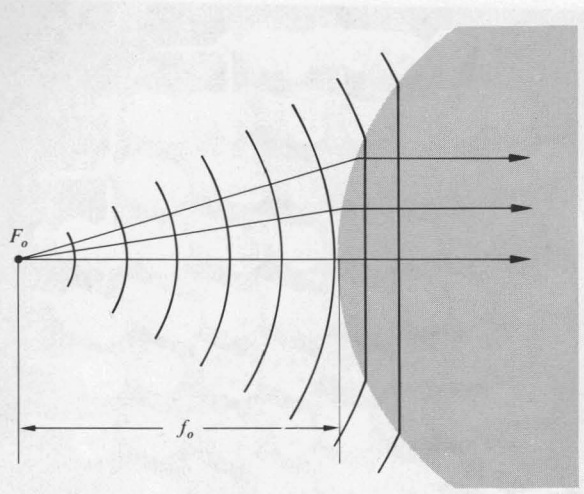


Fig. 5.10 Plane waves propagating beyond a spherical interface—the object focus.

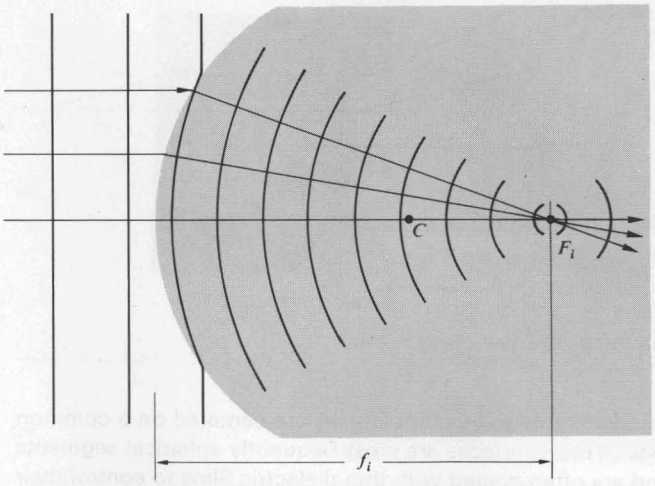


Fig. 5.11 The reshaping of plane into spherical waves at a spherical interface—the image focus.

well corrected, an incident spherical wave will emerge in a form very closely resembling a spherical wave. Consequently as the perfection of the system increases it more closely approaches first-order theory. Deviations from that of paraxial analysis will provide a convenient measure of the quality of an actual optical device.

If the point F_o in Fig. 5.10 is imaged at infinity ($s_i = \infty$) we have

$$\frac{n_1}{s_o} + \frac{n_2}{\infty} = \frac{n_2 - n_1}{R}.$$

That special object distance is defined as the *first focal length* or the *object focal length* $s_o \equiv f_o$, so that

$$f_o = \frac{n_1}{n_2 - n_1} R. \tag{5.9}$$

The point F_o itself is known as the *first* or *object focus*. Similarly the *second or image focus* is the axial point F_i where the image is formed when $s_o = \infty$, that is

$$\frac{n_1}{\infty} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}.$$

Defining the *second or image focal length* f_i as equal to s_i in this special case (Fig. 5.11) we have

$$f_i = \frac{n_2}{n_2 - n_1} R. \tag{5.10}$$

Recall that an image was virtual when the rays diverged from it (Fig. 5.12). Analogously, *an object is virtual when the rays converge toward it* (Fig. 5.13). Observe that the virtual object is now on the right-hand side of the vertex and therefore s_o will be taken as a negative quantity. Moreover the surface is concave and its radius will also be negative as required by Eq. (5.9) since f_o would be negative. In the same way the virtual image distance appearing to the left of V is negative.

5.2.3 Thin Lenses

Lenses take a wide range of forms, e.g. there are acoustic and microwave lenses; some of the latter are made of glass or wax in easily recognizable shapes while still others are far more subtle in appearance (Fig. 5.14). In the traditional sense, *a lens is an optical system consisting of two or more refracting interfaces where at least one of these is curved.*

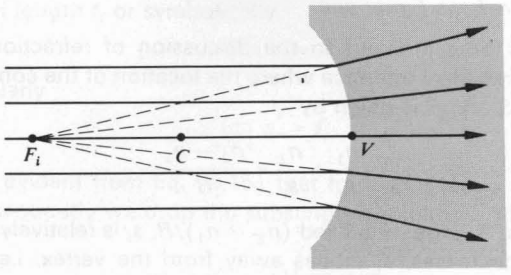


Fig. 5.12 A virtual image point.

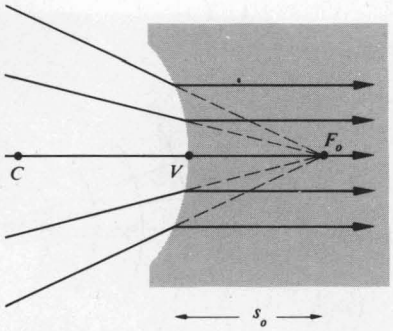


Fig. 5.13 A virtual object point.

Generally the nonplanar surfaces are centered on a common axis. These surfaces are most frequently spherical segments and are often coated with thin dielectric films to control their transmission properties (see Section 9.9). When a lens consists of one element, i.e. when it has only two refracting surfaces, it is a *simple lens*. The presence of more than one element makes it a *compound lens*. A lens is also classified as to whether it is *thin* or *thick*, i.e. whether its thickness is effectively negligible or not. We will limit ourselves, for the most part, to *centered systems* (for which all surfaces are rotationally symmetric about a common axis) of spherical surfaces. Under these restrictions, the simple lens can take the diverse forms shown in Fig. 5.15. Lenses which are variously known as *convex*, *converging* or *positive* are thicker at the center and so tend to decrease the radius of curvature of the wavefronts, i.e. the wave becomes more converging as it traverses the lens. This is, of course, assuming that the index of the lens is greater than that of the media in which it is immersed. *Concave*, *diverging* or *negative* lenses, on the other hand, are thinner at the center and tend to advance that portion of the wavefront causing it to become more diverging than it was upon entry.

i) Thin-Lens Equations

Return for a moment to the discussion of refraction at a single spherical interface where the location of the conjugate points S and P is given by

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}. \qquad [5.8]$$

When s_o is large for a fixed $(n_2 - n_1)/R$, s_i is relatively small. As s_o decreases, s_i moves away from the vertex, i.e. both θ_i and θ_t increase until finally $s_o = f_o$ and $s_i = \infty$. At that

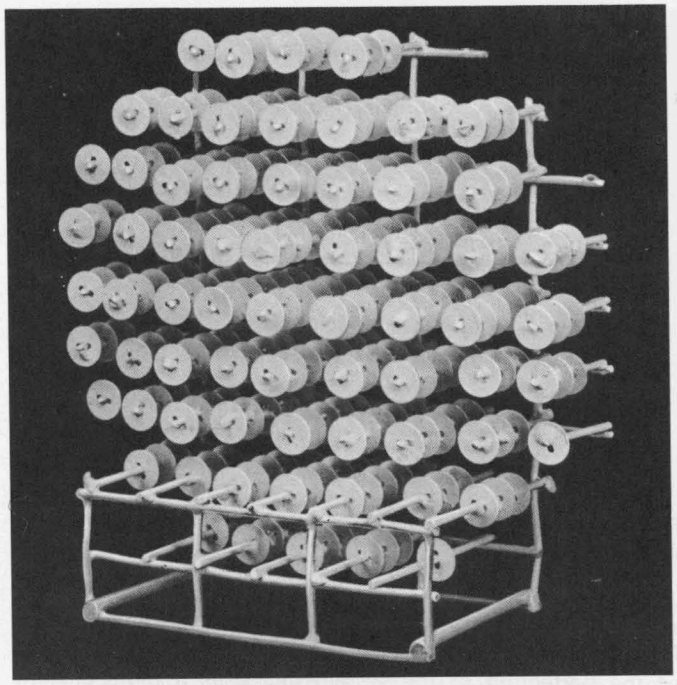


Fig. 5.14 A lens for short-wavelength radio waves. The disks serve to refract these waves much as rows of atoms refract light. [Photo courtesy Optical Society of America.]

point, $n_1/s_o = (n_2 - n_1)/R$ so that if s_o gets any smaller, s_i will have to be negative if Eq. (5.8) is to hold. In other words, the image becomes virtual (Fig. 5.16). Let's now locate the conjugate points for the lens of index n_l surrounded by a medium of index n_m as in Fig. 5.17 where we have simply ground another end on the piece in Fig. 5.16(c). This certainly isn't the most general set of circumstances but it is the most common and, even more cogently, it is the simplest.* We know from Eq. (5.8) that the paraxial rays issuing from S at s_{o1} will meet at P' a distance, we now call s_{i1}, from V_1 given by

$$\frac{n_m}{s_{o1}} + \frac{n_l}{s_{i1}} = \frac{n_l - n_m}{R_1}. \qquad (5.11)$$

Thus as far as the second surface is concerned, it "sees" rays coming toward it from P' which serves as its object point a

* See Jenkins and White, *Fundamentals of Optics*, p. 57 for a derivation containing three different indices.

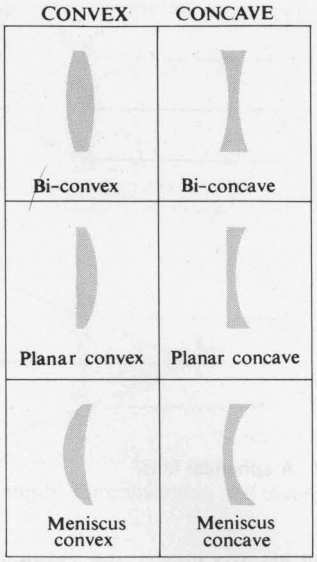


Fig. 5.15 Cross sections of various centered spherical simple lenses.

distance s_{o2} away. Furthermore, the rays arriving at that second surface are in the medium of index n_l. Thus, the object space for the second interface which contains P' has an index n_l. Note that the rays from P' to that surface are indeed straight lines. Considering the fact that

$$|s_{o2}| = |s_{i1}| + d$$

since s_{o2} is on the left and therefore positive $s_{o2} = |s_{o2}|$ while s_{i1} is also on the left and therefore negative, i.e. $-s_{i1} = |s_{i1}|$, we have

$$s_{o2} = -s_{i1} + d. \tag{5.12}$$

Thus Eq. (5.8) applied at the second surface yields

$$\frac{n_l}{(-s_{i1}+d)} + \frac{n_m}{s_{i2}} = \frac{n_m - n_l}{R_2}. \tag{5.13}$$

Here $n_l > n_m$ and $R_2 < 0$ so that the right-hand side is positive. Adding Eqs. (5.11) and (5.13) results in

$$\frac{n_m}{s_{o1}} + \frac{n_m}{s_{i2}} = (n_l - n_m)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) + \frac{n_l d}{(s_{i1}-d)s_{i1}}. \tag{5.14}$$

If the lens is thin enough ($d \to 0$), the last term on the right is effectively zero. As a further simplification, assume the surrounding medium to be air (i.e. $n_m \approx 1$). Accordingly, we have the very useful *thin-lens equation* often referred to

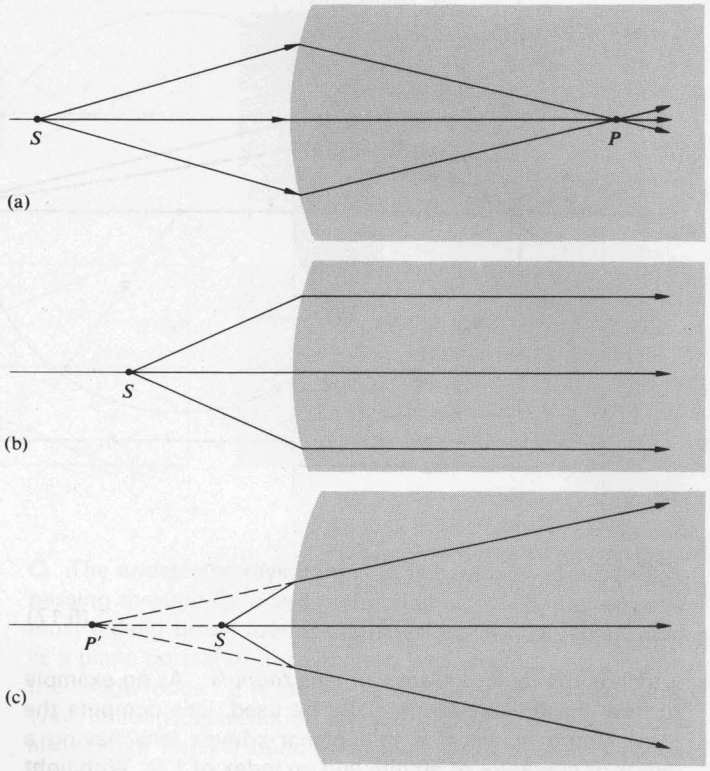


Fig. 5.16 Refraction at a spherical interface.

as the *lensmaker's formula*:

$$\frac{1}{s_o} + \frac{1}{s_i} = (n_l - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right), \tag{5.15}$$

where we let $s_{o1} = s_o$ and $s_{i2} = s_i$. The points V_1 and V_2 tend to coalesce as $d \to 0$ so that s_o and s_i can be measured from either the vertices or the lens center.

Just as in the case of the single spherical surface, if s_o is moved out to infinity, the image distance becomes the focal length f_i or symbolically

$$\lim_{s_o \to \infty} s_i = f_i.$$

Similarly

$$\lim_{s_i \to \infty} s_o = f_o.$$

It is evident from Eq. (5.15) that for a thin lens $f_i = f_o$ and consequently we drop the subscripts altogether. Thus

$$\frac{1}{f} = (n_l - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \tag{5.16}$$

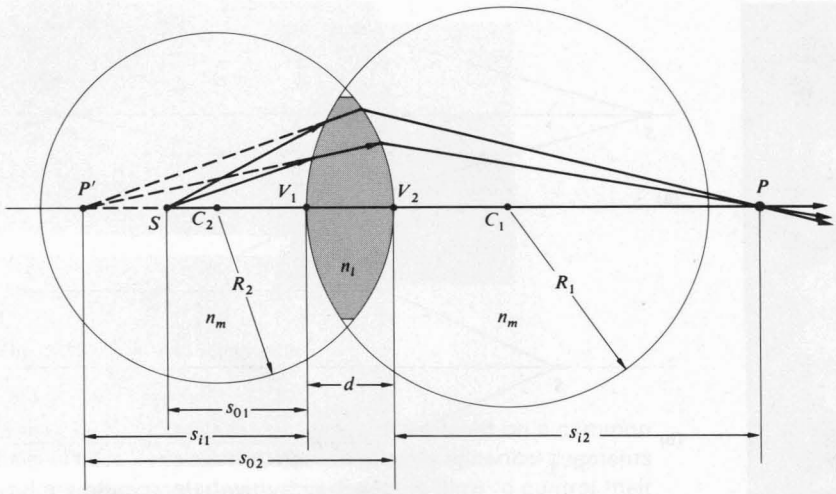


Fig. 5.17 A spherical lens.

and

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}, \tag{5.17}$$

which is the famous *Gaussian lens formula*. As an example of how these expressions might be used, let's compute the focal length in air of a thin planar-convex lens having a radius of curvature of 50 mm and an index of 1.5. With light entering on the planar surface ($R_1 = \infty$, $R_2 = -50$),

$$\frac{1}{f} = (1.5 - 1)\left(\frac{1}{\infty} - \frac{1}{-50}\right),$$

while if instead it arrives at the curved surface ($R_1 = +50$, $R_2 = \infty$),

$$\frac{1}{f} = (1.5 - 1)\left(\frac{1}{+50} - \frac{1}{\infty}\right),$$

and in either case $f = 100$ mm. If an object is alternately placed at distances 600 mm, 200 mm, 150 mm, 100 mm, and 50 mm from the lens on either side, we can find the image points from Eq. (5.17). Hence

$$\frac{1}{600} + \frac{1}{s_i} = \frac{1}{100}$$

and $s_i = 120$ mm. Similarly, the other image distances are 200 mm, 300 mm, ∞, and -100 mm respectively. Interestingly enough, when $s_o = \infty$, $s_i = f$; as s_o decreases s_i increases positively until $s_o = f$ and s_i is negative thereafter. You can qualitatively check this out with a simple convex lens and a small electric light—the "high-intensity" variety that uses auto lamps is probably the most convenient. Standing as far as you can from the source, project a clear image of it onto a white sheet of paper. You should be able to see the lamp quite clearly and not just as a blur. That image distance approximates f. Now move the lens in toward S adjusting s_i to produce a clear image. It will surely increase. As $s_o \to f$, a clear image of the filament can be projected, but only on an ever-increasingly distant screen. For $s_o < f$ there will just be a blur where the farthest wall intersects the diverging cone of rays—the image is virtual.

ii) Focal Points and Planes

Figure 5.18 summarizes pictorially some of the situations described analytically by Eq. 5.16. Observe that if a lens of index n_l is in a medium of index n_m

$$\frac{1}{f} = (n_{lm} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right). \tag{5.18}$$

The focal lengths in Fig. 5.18(a) and (b) are equal because the same medium exists on either side of the lens. Since $n_l > n_m$ it follows that $n_{lm} > 1$. In both cases $R_1 > 0$ and $R_2 < 0$ so that each focal length is positive. We have a real object in (a) and a real image in (b). In (c), $n_l < n_m$ and consequently f is negative. In (d) and (e), $n_{lm} > 1$ but $R_1 < 0$ while $R_2 > 0$ and so f is again negative and the object in one case, and the image in the other, are virtual. The last situation shows $n_{lm} < 1$ yielding an $f > 0$.

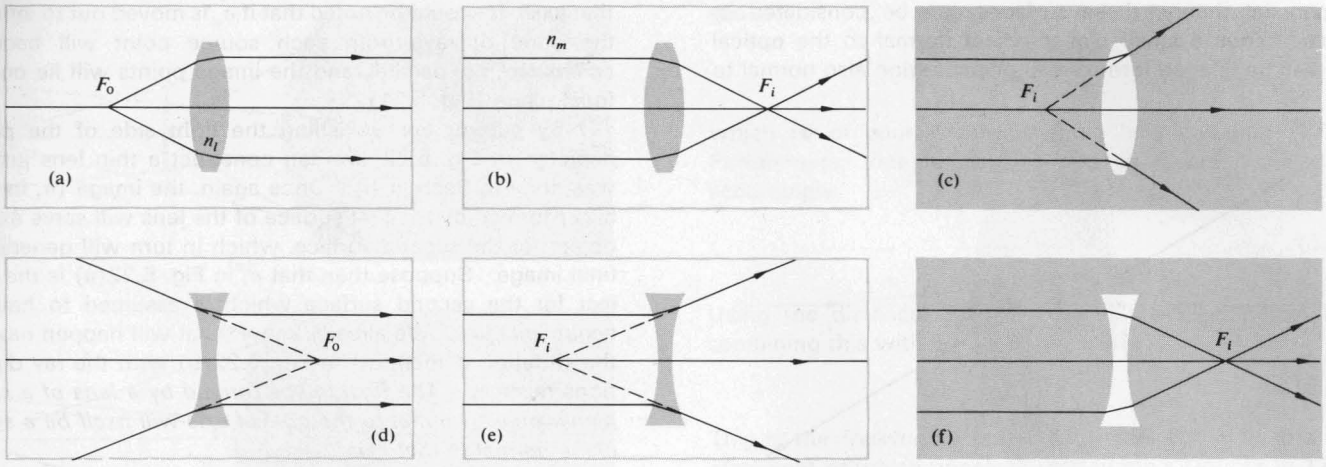


Fig. 5.18 Focal lengths for converging and diverging lenses.

Notice that in each instance it is particularly convenient to draw a ray through the center of the lens which, because it is perpendicular to both surfaces, is undeviated. Suppose however that an off-axis paraxial ray emerges from the lens parallel to its incident direction as in Fig. 5.19. We maintain that all such rays will pass through the point defined as the *optical center* of the lens O. To see this, draw two parallel planes, one on each side tangent to the lens at any pair of points A and B. This can easily be done by selecting A and B such that the radii $\overline{AC}_1$ and $\overline{BC}_2$ are themselves parallel. It is to be shown that the paraxial ray traversing $\overline{AB}$ enters and leaves the lens in the same direction. It is evident from the diagram that triangles AOC_1 and BOC_2 are similar, in the geometrical sense, and therefore their sides are proportional. Hence, $|R_1|(\overline{OC}_2) = |R_2|(\overline{OC}_1)$ and since the radii are constant, the location of O is constant, independent of A and B. As we saw earlier (Problem 4.15 and Fig. 4.44), a ray traversing a medium bounded by parallel planes will be displaced laterally, but will suffer no angular deviation. This displacement is proportional to the thickness, which for a thin lens is negligible. *Rays passing through O may, accordingly, be drawn as straight lines.* It is customary when dealing with thin lenses simply to place O midway between the vertices.

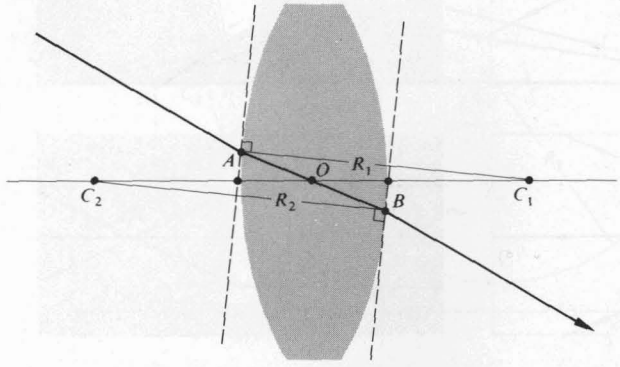
Recall that a bundle of parallel paraxial rays incident on a spherical refracting surface comes to a focus at a point on the optical axis (Fig. 5.11). As shown in Fig. 5.20, this implies that several such bundles entering in a narrow cone will be focused on a spherical segment σ, also centered on C. The undeviated rays normal to the surface, and therefore passing through C, locate the foci on σ. Since the ray cone must indeed be narrow, σ can satisfactorily be represented as a plane normal to the symmetry axis and passing through the image focus. It is known as a *focal plane*. In the same way, limiting ourselves to paraxial theory, a lens will focus all incident parallel bundles of rays* onto a surface called the *second or back focal plane* as in Fig. 5.21. Here each point on σ is located by the undeviated ray through O. Similarly, the *first or front focal plane* contains the object focus F_o.

iii) Finite Imagery

Thus far we've dealt with the mathematical abstraction of a single-point source, but now let's suppose there to be a great many such points combining to form a continuous finite object. For the moment, imagine the object to be a segment of a sphere, σ_o, centered on C as in Fig. 5.22. If σ_o is close to the spherical interface, point S will have a virtual image $P(s_i < 0$ and therefore on the left of $V)$. With S farther away, its image will be real ($s_i > 0$ and therefore on the right-hand side). In either case, each point on σ_o has a conjugate point on σ_i lying on a straight line through C. Within the restrictions

* Perhaps the earliest literary reference to the focal properties of a lens appears in Aristophanes' play, *The Clouds*, which dates back to 423 B.C. In it Strepsiades plots to use a burning glass to focus the sun's rays onto a wax tablet and thereby melt out the record of a gambling debt.

of paraxial theory, these surfaces can be considered as planar. Thus a small planar object normal to the optical axis will be imaged into a small planar region also normal to that axis. It should be noted that if σ_o is moved out to infinity, the cone of rays from each source point will become *collimated*, i.e. parallel, and the image points will lie on the focal plane (Fig. 5.21).



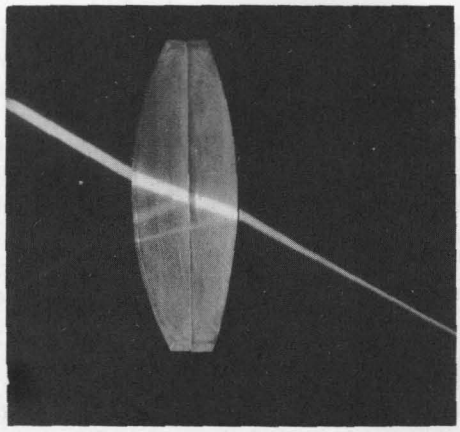


Fig. 5.19 The optical center of a lens. [Photo by E.H.].

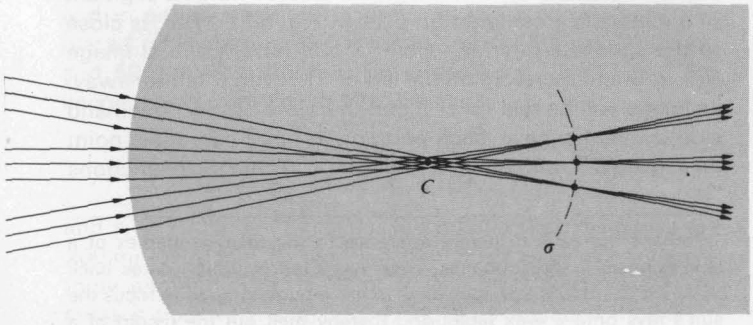


Fig. 5.20 Focusing of several ray bundles.

By cutting and polishing the right side of the piece depicted in Fig. 5.22, we can construct a thin lens just as was done in Section (i). Once again, the image (σ_i in Fig. 5.22) formed by the first surface of the lens will serve as the object for the second surface, which in turn will generate a final image. Suppose then that σ_i in Fig. 5.22(a) is the object for the second surface which is assumed to have a negative radius. We already know what will happen next—the situation is identical to Fig. 5.22(b) with the ray directions reversed. The *final image formed by a lens of a small planar object normal to the optical axis will itself be a small plane normal to that axis.*

The location, size and orientation of an image produced by a lens can be determined, particularly simply, using ray diagrams. To find the image of the object in Fig. 5.23, we must locate the image point corresponding to each object point. Since all rays issuing from a source point in a paraxial cone will arrive at the image point, any two such rays will suffice to fix that point. Knowing the positions of the focal points, there are three rays which are especially easy to apply. Two of these make use of the fact that a ray passing through the focal point will emerge from the lens parallel to the optical axis and vice versa; the other is the undeviated ray through *O*. Incidentally, this technique dates back to the work of Robert Smith as long ago as 1738.

This graphical procedure can be made even simpler by replacing the thin lens with a plane passing through its

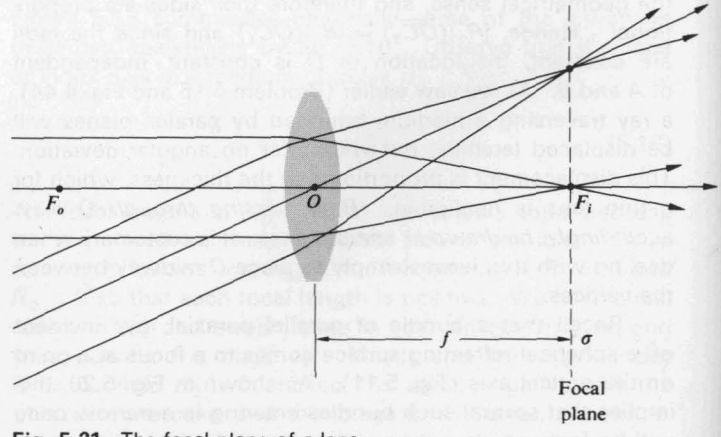


Fig. 5.21 The focal plane of a lens.

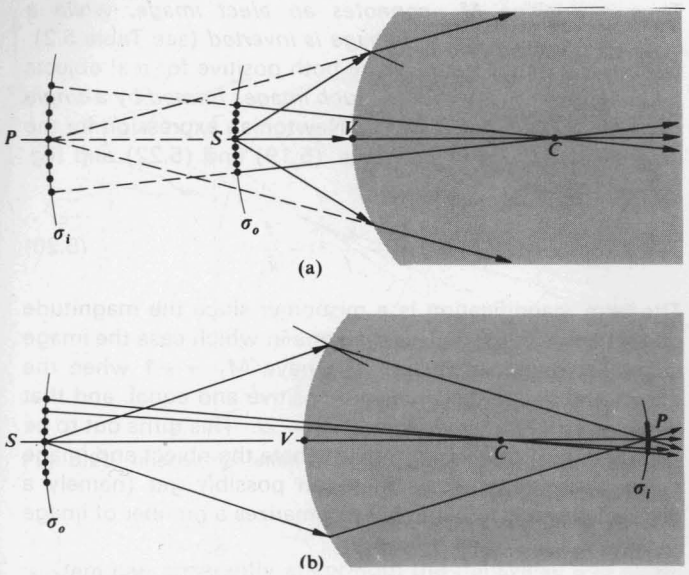


Fig. 5.22 Finite imagery.

center (Fig. 5.24). Presumably if we were to extend every incoming ray forward a little and every outgoing ray backward a bit, each pair would meet on this plane. Thus the total deviation of any ray can be envisaged as occurring all at once on that plane. This is equivalent to the actual process consisting of two separate angular shifts, one at each interface. (As we will see later, this is tantamount to saying that the two principal planes of a thin lens coincide.)

In accord with convention, transverse distances above the optical axis are taken as positive quantities while those below the axis are given negative numerical values. Therefore in Fig. 5.24 $y_o > 0$ and $y_i < 0$. Here the image is said to be *inverted* whereas if $y_i > 0$ when $y_o > 0$, it is *erect*. Observe that triangles AOF_i and $P_2P_1F_i$ are similar. Ergo

$$\frac{y_o}{|y_i|} = \frac{f}{(s_i - f)}, \tag{5.19}$$

Likewise, triangles S_2S_1O and P_2P_1O are similar and

$$\frac{y_o}{|y_i|} = \frac{s_o}{s_i}, \tag{5.20}$$

where all quantities other than y_i are positive. Hence

$$\frac{s_o}{s_i} = \frac{f}{(s_i - f)} \tag{5.21}$$

and

$$\frac{1}{f} = \frac{1}{s_o} + \frac{1}{s_i},$$

which is, of course, the Gaussian lens equation (5.17). Furthermore, triangles $S_2S_1F_o$ and BOF_o are similar and accordingly

$$\frac{f}{(s_o - f)} = \frac{|y_i|}{y_o}. \tag{5.22}$$

Using the distances measured from the focal points and combining this with Eq. (5.19) we have

$$x_o x_i = f^2. \tag{5.23}$$

This is the *Newtonian form* of the lens equation, the first statement of which appeared in Newton's *Opticks* in 1704. The signs of x_o and x_i are reckoned with respect to their concomitant foci. By convention x_o is taken to be positive

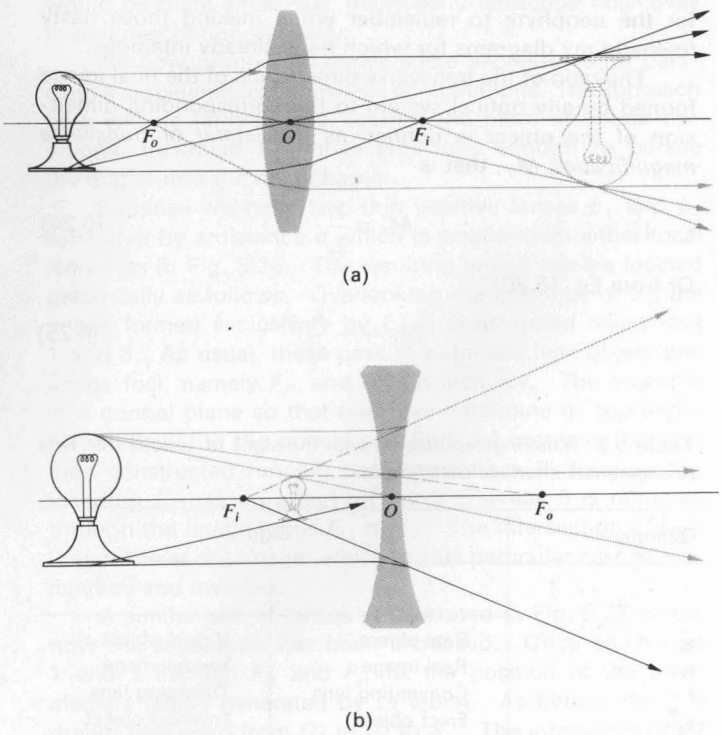


Fig. 5.23 (a) A real object and a positive lens. (b) A real object and a negative lens.

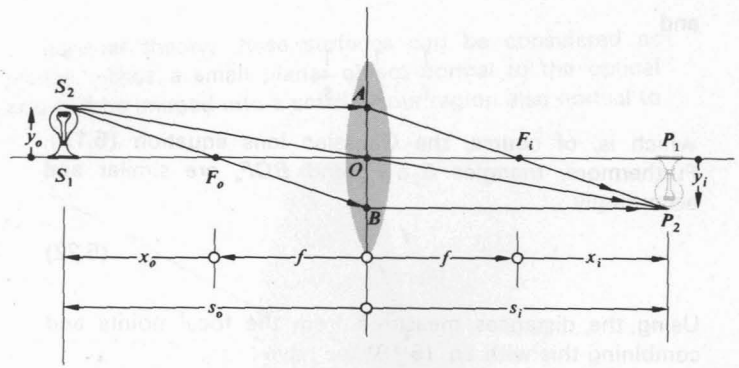


Fig. 5.24 Object and image location for a thin lens.

left of F_o whereas x_i is positive on the right of F_i. To be sure, it is evident from Eq. (5.23) that x_o and x_i have like signs and that means that *the object and image must be on opposite sides of their respective focal points*. This is a good thing for the neophyte to remember when making those hasty freehand ray diagrams for which he is already infamous.

The ratio of the transverse dimensions of the final image formed by any optical system to the corresponding dimension of the object is defined as the *lateral* or *transverse magnification* M_T, that is

$$M_T \equiv \frac{y_i}{y_o}. \tag{5.24}$$

Or from Eq. (5.20)

$$M_T = -\frac{s_i}{s_o}. \tag{5.25}$$

Table 5.2 Meanings associated with the signs of various thin lens and spherical interface parameters.

Quantity	Sign	
	+	−
s_o	Real object	Virtual object
s_i	Real image	Virtual image
f	Converging lens	Diverging lens
y_o	Erect object	Inverted object
y_i	Erect image	Inverted image
M_T	Erect image	Inverted image

Thus *a positive M_T connotes an erect image, while a negative value means the image is inverted* (see Table 5.2). Bear in mind that s_i and s_o are both positive for real objects and images. *Clearly then all such images formed by a single thin lens will be inverted.* The Newtonian expression for the magnification follows from Eqs. (5.19) and (5.22) and Fig. 5.24, whence

$$M_T = -\frac{x_i}{f} = -\frac{f}{x_o}. \tag{5.26}$$

The term magnification is a misnomer since the magnitude of M_T can certainly be less than one in which case the image is smaller than the object. We have $M_T = -1$ when the object and image distances are positive and equal, and that happens (5.17) only when $s_o = s_i = 2f$. This turns out to be (Problem 5.5) the configuration where the object and image are as close together as they can possibly get (namely a distance $4f$ apart). Table 5.3 summarizes a number of image

Table 5.3 Images of real objects formed by thin lenses.

Convex				
Object	Image			
Location	Type	Location	Orientation	Relative size
$\infty > s_o > 2f$	Real	$f < s_i < 2f$	Inverted	Minified
$s_o = 2f$	Real	$s_i = 2f$	Inverted	Same size
$f < s_o < 2f$	Real	$\infty > s_i > 2f$	Inverted	Magnified
$s_o = f$		$\pm\infty$		
$s_o < f$	Virtual	$\lvert s_i\rvert > s_o$	Erect	Magnified

Concave				
Object	Image			
Location	Type	Location	Orientation	Relative size
Anywhere	Virtual	$\lvert s_i\rvert < \lvert f\rvert$	Erect	Minified

configurations resulting from the juxtaposition of a thin lens and a real object.

Presumably the image of a three-dimensional object will itself occupy a three-dimensional region of space. The optical

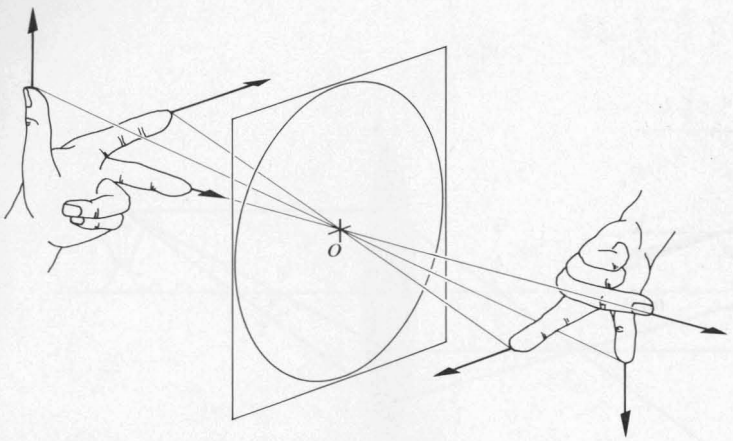


Fig. 5.25 Image orientation for a thin lens.

system can apparently affect both the transverse and longitudinal dimensions of the image. The *longitudinal magnification*, M_L, which relates to the axial direction, is defined as

$$M_L \equiv \frac{dx_i}{dx_o}. \tag{5.27}$$

This is the ratio of an infinitesimal axial length in the region of the image to the corresponding length in the region of the object. Differentiating Eq. (5.23) leads to

$$M_L = -\frac{f^2}{x_o^2} = -M_T^2 \tag{5.28}$$

for a thin lens in a single medium. Evidently, $M_L < 0$ which implies that a positive dx_o corresponds to a negative dx_i and vice versa. In other words, a finger pointing towards the lens is imaged pointing away from it (Fig. 5.25).

Form the image of a window on a sheet of paper using a simple convex lens. Assuming a lovely arboreal scene, image the distant trees on the screen. Now move the paper *away* from the lens so that it intersects a different region of the image space. The trees will fade while the nearby window itself comes into view.

iv) Thin-Lens Combinations

It is not our intent here to have the reader develop a proficiency with the subtle intricacies of modern lens design, but rather to bring him to a point where he can begin to appreciate, utilize and adapt those systems already available.

In constructing a new optical system, one generally begins by sketching out a rough arrangement using the quickest approximate calculations. Refinements are then added as the designer goes on to the prodigious and more exact ray-tracing techniques. Nowadays these computations are most often carried out by electronic digital computers. Even so, the simple thin-lens concept provides a highly useful basis for preliminary calculations in a broad range of situations.

No lens is actually a thin lens in the strict sense of having a thickness which approaches zero. Yet many simple lenses, for all practical purposes, function in a fashion equivalent to that of a thin lens. Spectacle lenses which, by the way, have been used at least since the thirteenth century are almost all in this category. When the radii of curvature are large and the lens diameter is small, the thickness will usually be small as well. A lens of this sort would generally have a large focal length compared to which the thickness would be quite small, e.g. many early telescope objectives fit that description perfectly.

We now propose to derive some expressions for parameters associated with thin-lens combinations. The approach here will be fairly simple, leaving the more elaborate traditional treatment for those tenacious enough to pursue the matter into the next chapter.

Suppose we have two thin positive lenses L_1 and L_2 separated by a distance d which is smaller than either focal length as in Fig. 5.26. The resulting image can be located graphically as follows. Overlooking the presence of L_2, the image formed exclusively by L_1 is constructed using rays 1 and 3. As usual, these pass through the lens' object and image foci, namely F_{o1} and F_{i1} respectively. The object is in a normal plane so that two rays determine its top and a perpendicular to the optical axis finds its bottom. Ray 2 is then constructed running backwards from P_1' through O_2. Inserting L_2 has no effect on ray 2 whereas 3 is refracted through the image focus F_{i2} of L_2. The intersection of rays 2 and 3 fixes the image, which in this particular case is real, minified and inverted.

A similar pair of lenses is illustrated in Fig. 5.27 where now the separation has been increased. Once again rays 1 and 3 through F_{i1} and F_{o1} fix the position of the intermediate image generated by L_1 alone. As before, ray 2 is drawn backward from O_2 to P_1' to S_1. The intersection of 2 and 3, as the latter is refracted through F_{i2}, locates the final image. This time it is real and erect. Notice that if the focal

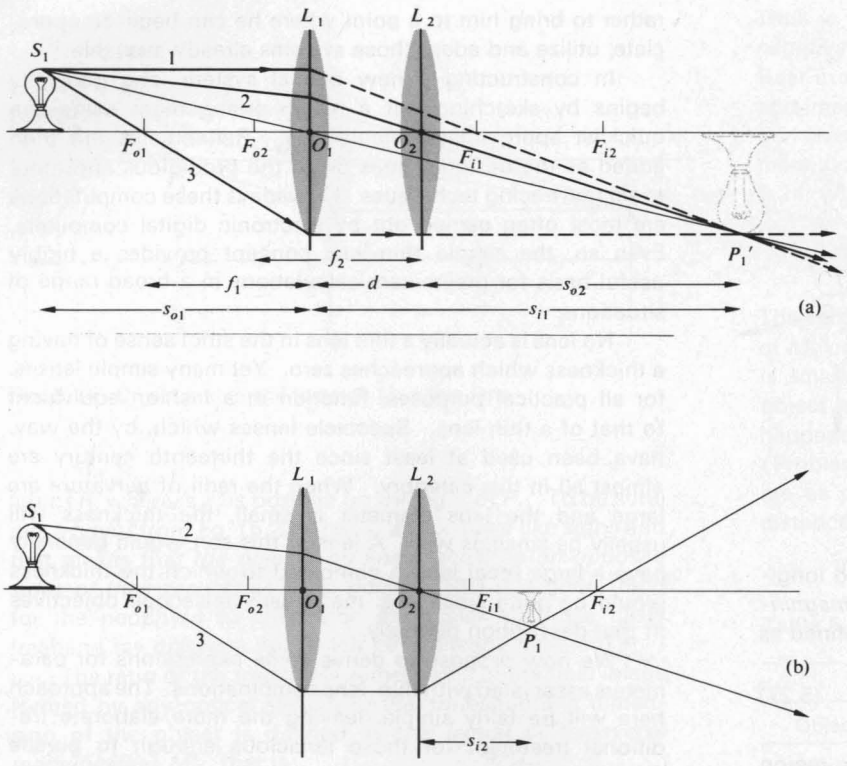


Fig. 5.26 Two thin lenses separated by a distance smaller than either focal length.

length of L_2 is increased with all else constant, the size of the image increases as well.

Analytically we have for L_1

$$\frac{1}{s_{i1}} = \frac{1}{f_1} - \frac{1}{s_{o1}} \tag{5.29}$$

or

$$s_{i1} = \frac{s_{o1} f_1}{s_{o1} - f_1}. \tag{5.30}$$

This is positive and the intermediate image is to the right of L_1, when $s_{o1} > f_1$ and $f_1 > 0$. For L_2

$$s_{o2} = d - s_{i1}, \tag{5.31}$$

and if $d > s_{i1}$ the object for L_2 is real (as in Fig. 5.27) while if $d < s_{i1}$ it is virtual ($s_{o2} < 0$ as in Fig. 5.26). In the former instance the rays approaching L_2 are diverging from P_1', while in the latter they are converging toward it. Furthermore

$$\frac{1}{s_{i2}} = \frac{1}{f_2} - \frac{1}{s_{o2}}$$

or

$$s_{i2} = \frac{s_{o2} f_2}{s_{o2} - f_2}.$$

Using Eq. (5.31) this becomes

$$s_{i2} = \frac{(d - s_{i1}) f_2}{(d - s_{i1} - f_2)}. \tag{5.32}$$

In this same way we could compute the response of any number of thin lenses. It will often be convenient to have a single expression, at least when dealing with only two lenses, and so substituting for s_{i1} from Eq. (5.29) we get

$$s_{i2} = \frac{f_2 d - f_2 s_{o1} f_1/(s_{o1} - f_1)}{d - f_2 - s_{o1} f_1/(s_{o1} - f_1)}. \tag{5.33}$$

Here s_{o1} and s_{i2} are the object and image distances of the

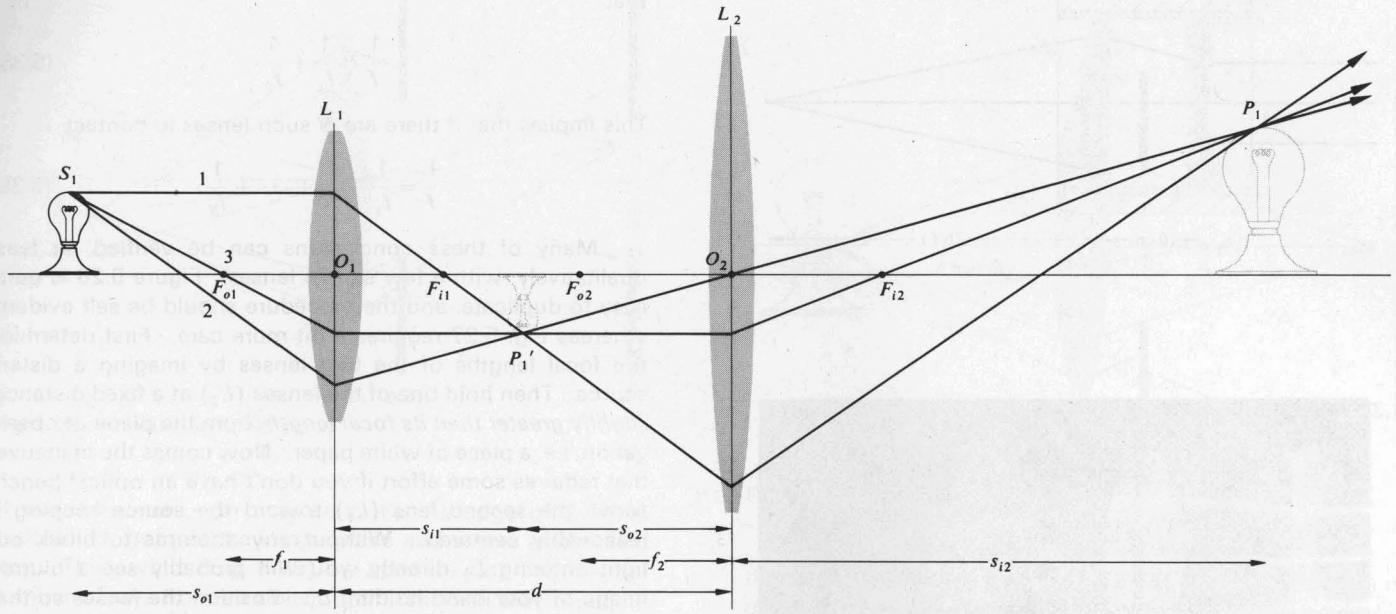


Fig. 5.27 Two thin lenses separated by a distance greater than the sum of their focal lengths.

compound lens. As an example, let's compute the image distance associated with an object placed 50 cm from the first of two positive lenses. These in turn are separated by 20 cm and have focal lengths of 30 cm and 50 cm respectively. By direct substitution (5.33)

$$s_{i2} = \frac{50(20) - 50(50)\,(30)/(50 - 30)}{20 - 50 - 50(30)/(50 - 30)} = 26.2 \text{ cm}$$

and the image is real. Inasmuch as L_2 "magnifies" the intermediate image formed by L_1, the total transverse magnification of the compound lens is the product of the individual magnifications, that is

$$M_T = M_{T1}M_{T2}.$$

It is left as a problem to show that

$$M_T = \frac{f_1 s_{i2}}{d(s_{o1} - f_1) - s_{o1}f_1}. \tag{5.34}$$

In the above example

$$M_T = \frac{30(26.2)}{20(50 - 30) - 50(30)} = -0.72$$

and just as we should have guessed from Fig. 5.26, the image is minified and inverted.

The distance from the last surface of an optical system to the second focal point of that system as a whole is known as the *back focal length* or b.f.l. Likewise the distance from the vertex of the first surface to the first or object focus is the *front focal length* or f.f.l. Consequently if we let $s_{i2} \to \infty$, s_{o2} approaches f_2, which combined with Eq. (5.31) tells us that $s_{i1} \to d - f_2$. Hence from Eq. (5.29)

$$\left.\frac{1}{s_{o1}}\right|_{s_{i2}=\infty} = \frac{1}{f_1} - \frac{1}{(d - f_2)} = \frac{d - (f_1 + f_2)}{f_1(d - f_2)}.$$

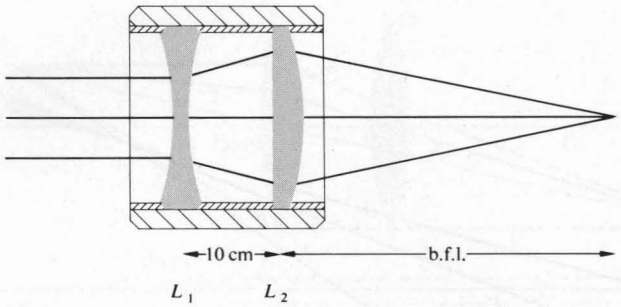
But this special value of s_{o1} is the f.f.l.:

$$\text{f.f.l.} = \frac{f_1(d - f_2)}{d - (f_1 + f_2)}. \tag{5.35}$$

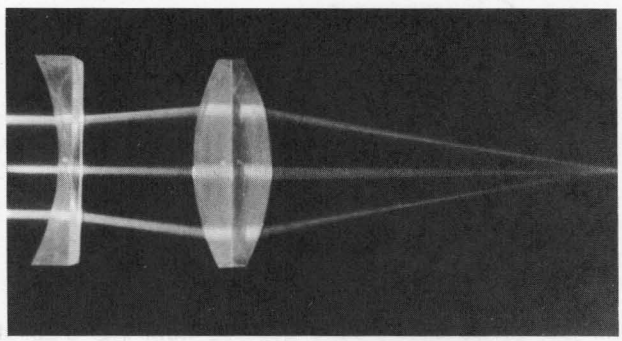
In the same way, letting $s_{o1} \to \infty$ in Eq. (5.33), $(s_{o1} - f_1) \to s_{o1}$ and since s_{i2} is then the b.f.l. we have

$$\text{b.f.l.} = \frac{f_2(d - f_1)}{d - (f_1 + f_2)}. \tag{5.36}$$

To see how this works numerically, let's find both the b.f.l.



(a)



(b)

Fig. 5.28 A positive and negative thin-lens combination. [Photo by E.H.]

and f.f.l. for the thin-lens system in Fig. 5.28(a) where $f_1 = -30$ cm and $f_2 = +20$ cm. Then

$$\text{b.f.l.} = \frac{20\,[10 - (-30)]}{10 - (-30 + 20)} = 40\,\text{cm}$$

and similarly f.f.l. = 15 cm. Incidentally, notice that if $d = f_1 + f_2$, plane waves entering the compound lens from either side will emerge as plane waves (Problem 5.18) as in telescopic systems.

Observe that if $d \to 0$, i.e. if the lenses are brought into contact, as e.g. in the case of some achromatic doublets,

$$\text{b.f.l.} = \text{f.f.l.} = \frac{f_2 f_1}{f_2 + f_1}. \tag{5.37}$$

The resultant thin lens has an *effective focal length* f such that

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}. \tag{5.38}$$

This implies that if there are N such lenses in contact

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \cdots + \frac{1}{f_N}. \tag{5.39}$$

Many of these conclusions can be verified, at least qualitatively, with a few simple lenses. Figure 5.26 is quite easy to duplicate, and the procedure should be self evident, whereas Fig. 5.27 requires a bit more care. First determine the focal lengths of the two lenses by imaging a distant source. Then hold one of the lenses (L_2) at a fixed distance, *slightly greater than its focal length*, from the plane of observation, i.e. a piece of white paper. Now comes the maneuver that requires some effort if you don't have an optical bench. Move the second lens (L_1) toward the source keeping it reasonably centered. Without any attempts to block out light entering L_2 directly, you will probably see a blurred image of your hand holding L_1. Position the lenses so that the region on the screen corresponding to L_1 is as bright as possible. The scene spread across L_1 (i.e. its image within the image) will become clear and erect as in Fig. 5.27.

5.3 STOPS

5.3.1 Aperture and Field Stops

The intrinsically finite nature of all lenses demands that they will collect only a fraction of the energy emitted by a point source. The physical limitation presented by the periphery of a simple lens therefore determines which rays shall enter the system to ultimately form an image. In that respect, the unobstructed or *clear diameter* of the lens functions as an aperture into which energy flows. Any element, be it the rim of a lens or a separate diaphragm, which determines the amount of light reaching the image is known as the *aperture stop*, abbreviated as A.S. The adjustable leaf diaphragm that is usually located behind the first few elements of a compound camera lens is just such an aperture stop. Evidently it determines the light-gathering capability of the lens as a whole. As shown in Fig. 5.29, highly oblique rays can still enter a system of this sort. These however are usually deliberately restricted in order to control the quality of the image. The element limiting the size or angular breadth of the object which can be imaged by the system is called the *field stop* or F.S.—it determines the field of view

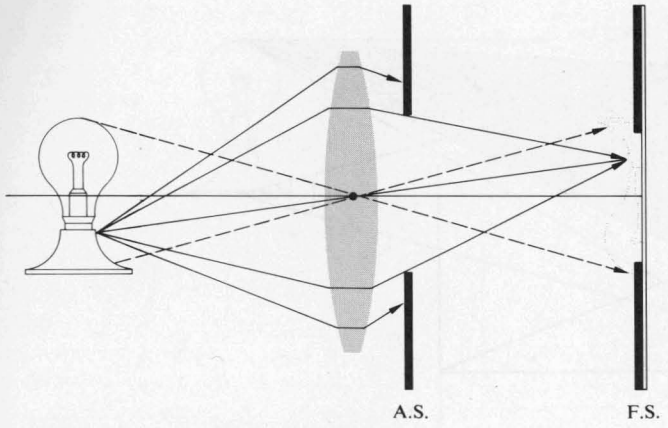


Fig. 5.29 Aperture and field stops.

of the instrument. In a camera, it's the edge of the film itself which bounds the image plane and serves as the field stop. Thus, while (Fig. 5.29) the aperture stop controls the number of rays from an object point reaching the conjugate image point, it is the field stop which will or will not obstruct those rays *in toto*. Neither the very top nor the bottom of the object in Fig. 5.29 passes the field stop. Opening the circular aperture stop would cause the system to accept a larger energy cone and in so doing increase the irradiance at each image point. In contrast, opening the field stop would allow the extremities of the object, which were previously blocked, now to be imaged.

5.3.2 Entrance and Exit Pupils

Another concept, quite useful in determining whether or not a given ray will traverse the entire optical system, is the *pupil*. This is simply an *image of the aperture stop*. The *entrance pupil* of a system is the image of the aperture stop as seen from an axial point on the object through those elements preceding the stop. If there are no lenses between the object and the A.S., the latter itself serves as the entrance pupil. To illustrate the point examine Fig. 5.30, which is a lens with a *rear aperture stop*. The image of the aperture stop in L is virtual (see Table 5.3) and magnified. It can be located by sending a few rays out from the edges of the A.S. in the usual way. In contrast, the *exit pupil* is the image of the A.S. as seen from an axial point on the image plane through the interposed lenses, if there are any. In Fig. 5.30 there are no such lenses and so the aperture stop itself serves as the exit pupil. Notice that all of this just means

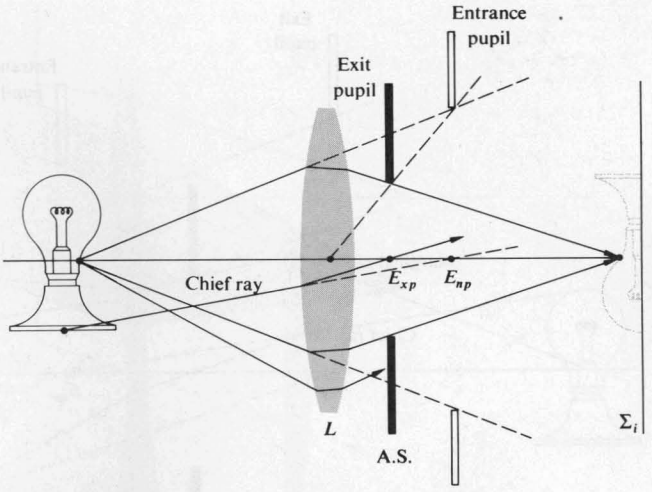


Fig. 5.30 Entrance and exit pupils.

that the cone of light actually entering the optical system is determined by the entrance pupil while the cone leaving it is controlled by the exit pupil. No rays from the source point proceeding outside of either cone will make it to the image plane.

If you wanted to use a telescope or a monocular as a camera lens, you might attach an external *front aperture stop* to control the amount of incoming light for exposure purposes. Figure 5.31 represents a similar arrangement where

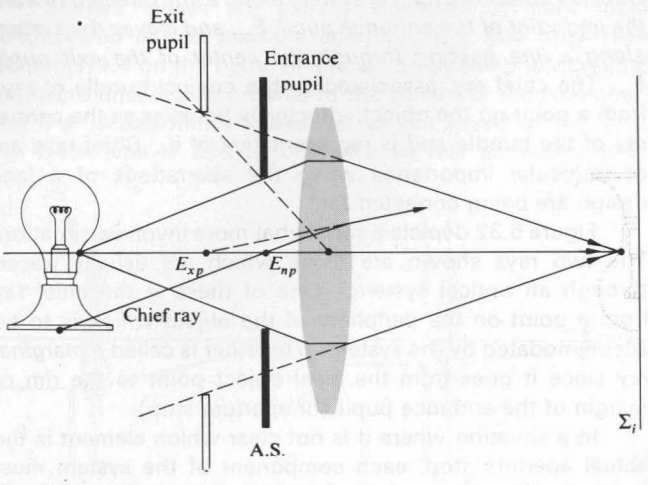


Fig. 5.31 A front aperture stop.

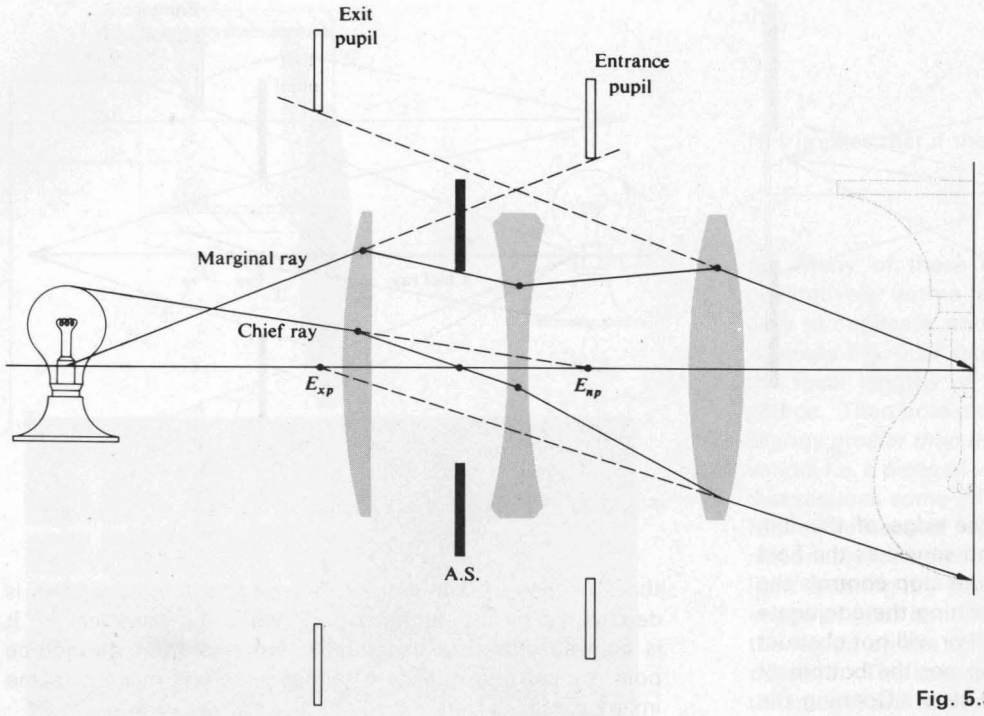


Fig. 5.32 Pupils and stops for a three-lens system.

the entrance and exit pupil locations should be self evident. The last two diagrams included a ray labeled as the *chief ray*. It is defined to be *any ray from an off-axis object point which passes through the center of the aperture stop. The chief ray enters the optical system along a line directed toward the midpoint of the entrance pupil, E_{np}, and leaves the system along a line passing through the center of the exit pupil, E_{xp}*. The chief ray, associated with a conical bundle of rays from a point on the object, effectively behaves as the central ray of the bundle and is representative of it. Chief rays are of particular importance when the aberrations of a lens design are being corrected for.

Figure 5.32 depicts a somewhat more involved situation. The two rays shown are those which are usually traced through an optical system. One of these is the chief ray from a point on the periphery of the object which is to be accommodated by the system. The other is called a *marginal ray* since it goes from the axial object point to the rim or margin of the entrance pupil (or aperture stop).

In a situation where it is not clear which element is the actual aperture stop, each component of the system must be imaged by the remaining elements to its left. *The image which subtends the smallest angle at the axial object point is the entrance pupil.* The element whose image is the entrance pupil is then the aperture stop of the system for that object point. Problem 5.21 deals with just this kind of calculation.

Notice how the cone of rays, in Fig. 5.33, which can reach the image plane becomes narrower as the object point moves off-axis. The effective aperture stop, which for the axial bundle of rays was the rim of L_1, has been markedly reduced for the off-axis bundle. The result is a gradual fading out of the image at points near its periphery, a process known as *vignetting*.

The locations and sizes of the pupils of an optical system are of considerable practical importance. In visual instruments, the observer's eye is positioned at the center of the exit pupil. The pupil of the eye itself will vary from 2 mm to about 8 mm depending on the general illumination level. Thus a telescope or binocular designed primarily for evening use might have an exit pupil of at least 8 mm (you may have heard the term *night glasses*—they were quite popular on roofs during the Second World War). In contrast, a daylight version will suffice with an exit pupil of 3 or 4 mm. The larger

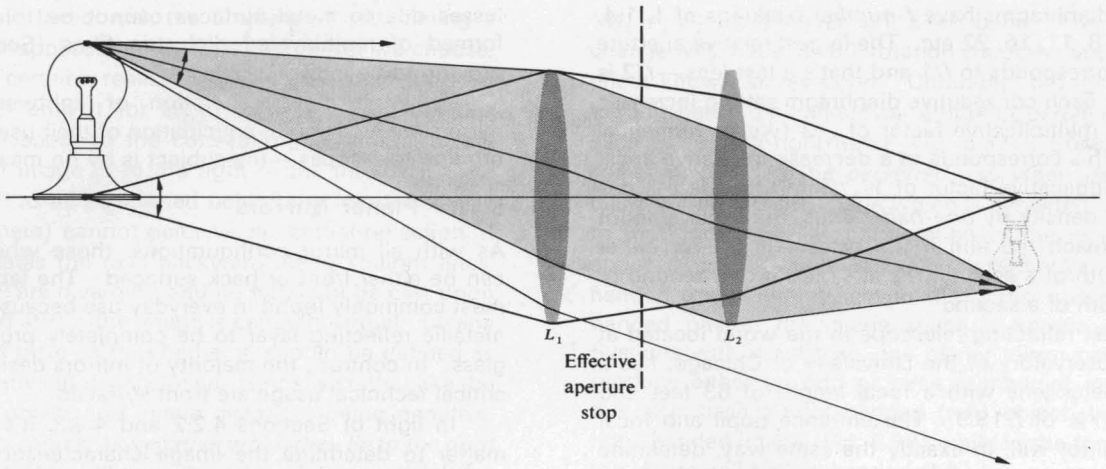


Fig. 5.33 Vignetting.

the exit pupil, the easier it will be to properly align your eye with the instrument. Obviously a telescopic sight for a high-powered rifle should have a large exit pupil located far enough behind the scope so as to avoid injury from recoil.

5.3.3 Relative Aperture and *f*-number

Suppose we wish to collect the light from an extended source and form an image of it using a lens (or mirror). The amount of energy gathered by the lens (or mirror) from some small region of a distant source will be directly proportional to the area of the lens or more generally to the area of the entrance pupil. A large *clear aperture* will intersect a large cone of rays. Obviously, if the source were a laser having a very narrow beam, this would not necessarily be true. If we neglect losses due to reflections, absorption etc., the incoming energy will be spread across a corresponding region of the image. Thus the energy per unit area per unit time, i.e. the flux density or irradiance, will be inversely proportional to the image area. The entrance pupil area, if circular, varies as the square of its radius and is therefore proportional to the square of its diameter D. Furthermore, the image area will go as the square of its lateral dimension which in turn [Eqs. (5.24) and (5.26)] is proportional to f^2. (Keep in mind that we are talking about an extended object rather than a point source. In the latter case, the image would be confined to a very small area independent of f.) Thus the flux density at the image plane varies as $(D/f)^2$. The ratio D/f is known as the *relative aperture*, while its inverse is said to be the *f-number* or $f/\#$, that is

$$f/\# \equiv \frac{f}{D} \tag{5.40}$$

where $f/\#$ should be understood as a single symbol. For example, a lens with a 25 mm aperture and a 50 mm focal length has an f-number of 2 and is usually designated as $f/2$. Figure 5.34 illustrates the point by showing a thin lens behind a variable iris diaphragm operating at either $f/2$ or $f/4$. A smaller f-number clearly permits more light to reach the image plane.

Camera lenses are usually specified by their focal lengths and largest possible apertures, e.g. you might see 50 mm, $f/1.4$ on the barrel of a lens. Since the photographic exposure time is proportional to the square of the *f-number*, the $f/\#$ is sometimes spoken of as the *speed* of the lens. An $f/1.4$ lens is said to be twice as fast as an $f/2$ lens.

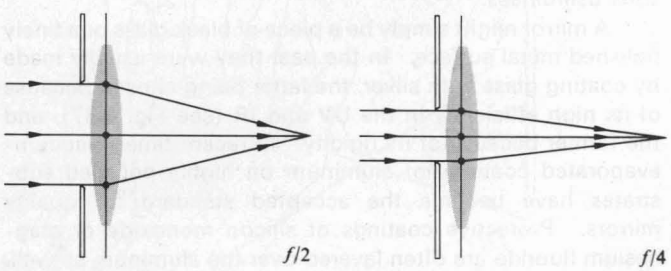


Fig. 5.34 Stopping down a lens to change the *f*-number.

Usually lens diaphragms have *f-number* markings of 1, 1.4, 2, 2.8, 4, 5.6, 8, 11, 16, 22 etc. The largest relative aperture in this case corresponds to $f/1$ and that's a fast lens—$f/2$ is more typical. Each consecutive diaphragm setting increases the $f/\#$ by a multiplicative factor of $\sqrt{2}$ (within numerical round-off). This corresponds to a decrease in relative aperture by a multiplicative factor of $1/\sqrt{2}$ and therefore a decrease in flux density by *one-half*. Thus, the same amount of light will reach the film if the camera is set for either $f/1.4$ at 1/500th of a second, $f/2$ at 1/250th of a second or $f/2.8$ at 1/125th of a second.

The largest refracting telescope in the world located at the Yerkes Observatory of the University of Chicago, has a 40 inch diameter lens with a focal length of 63 feet and therefore an $f/\#$ of $f/18.9$. The entrance pupil and focal length of a mirror will, in exactly the same way, determine its $f/\#$. Accordingly, the 200 inch diameter mirror of the Mount Palomar telescope with a prime focal length of 666 inches has an $f/\#$ of $f/3.33$.

In precise work where reflection and absorption losses in the lens itself must be taken into consideration, the *T-number* is highly useful. In effect, it is a modified (increased) *f*-number which a given real lens would actually have to have were it to transmit an amount of light corresponding to a particular value of f/D.

5.4 MIRRORS

Mirror systems are finding increasingly more extensive applications, particularly in the ultraviolet and infrared regions of the spectrum. While it is relatively simple to construct a reflecting device which will perform satisfactorily across a broad frequency bandwidth, the same cannot be said of refracting systems. For example, a silicon or germanium lens designed for the infrared will be completely opaque in the visible. As we will see later when we consider their aberrations, mirrors have other attributes which contribute to their usefulness.

A mirror might simply be a piece of black glass or a finely polished metal surface. In the past they were usually made by coating glass with silver, the latter being chosen because of its high efficiency in the UV and IR (see Fig. 4.37), and the former because of its rigidity. In recent times, vacuum-evaporated coatings of aluminum on highly polished substrates have become the accepted standard for quality mirrors. Protective coatings of silicon monoxide or magnesium fluoride are often layered over the aluminum as well. In special applications (e.g. in lasers), where even the small losses due to metal surfaces cannot be tolerated, mirrors formed of multilayered dielectric films (Section 9.9) are proving to be indispensable.

A whole new generation of lightweight precision mirrors are evolving in anticipation of their use in large-scale orbiting telescopes—the subject is by no means static.

5.4.1 Planar Mirrors

As with all mirror configurations, those which are planar can be either front or back surfaced. The latter is the kind most commonly found in everyday use because it allows the metallic reflecting layer to be completely protected behind glass. In contrast, the majority of mirrors designed for more critical technical usage are front surfaced.

In light of Sections 4.2.2 and 4.2.3, it's a rather easy matter to determine the image characteristics of a planar mirror. Examining the point source and mirror arrangement of Fig. 5.35, we can quickly show that $|s_o| = |s_i|$, i.e. the image *P* and object *S* are equidistant from the surface. To wit, $\theta_i = \theta_r$ from the law of reflection; $\theta_i + \theta_r$ is the exterior angle of triangle *SPA* and is, therefore, equal to the sum of the alternate interior angles, $\measuredangle VSA + \measuredangle VPA$. But $\measuredangle VSA = \theta_i$ and therefore $\measuredangle VSA = \measuredangle VPA$. This makes triangles *VAS* and *VPA* congruent, in which case $|s_o| = |s_i|$. (Go back and take another look at Problem 4.3 and Fig. 4.41 for the wave picture of the reflection.)

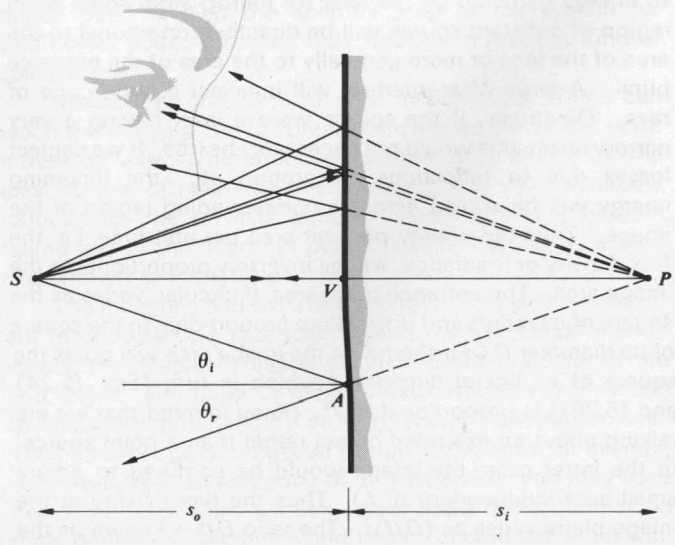


Fig. 5.35 A planar mirror.

We are now faced with the problem of determining a sign convention applicable to mirrors. Whatever we choose, and you should certainly realize that there is a choice, we need only be faithful unto it for all to be well. One obvious dilemma with respect to the convention for lenses is that now the virtual image is to the right of the interface. The observer sees P to be positioned behind the mirror because his eye (or camera) cannot perceive the actual reflection, it merely interpolates the rays backward along straight lines. The rays from P are diverging and no light can be cast upon a screen located at P—the image is certainly virtual. Clearly it is a matter of taste as to whether s_i should be defined as positive or negative in this instance. Since we rather like the idea of virtual object and image distances being negative, we shall define *s_o and s_i as negative when they lie to the right of the vertex V*. This will have the added benefit of yielding a mirror formula identical to the Gaussian lens equation (5.17). Evidently, the same definition of the transverse magnification (5.24) holds, where now as before $M_T = +1$ indicates a *life-sized*, virtual, erect image.

Each point of the extended object of Fig. 5.36, a perpendicular distance s_i from the mirror, is imaged that same distance behind the mirror. In this way, the entire image is built up point by point. This is very much different from the way a lens locates an image. The object in Fig. 5.25 was a left hand and the image formed by the lens was again a left hand; to be sure it might have been distorted ($M_L \neq M_T$), but it was still a left hand. The only evident change was that of a 180° rotation about the optical axis—an effect known as *reversion*. Contrarily, the mirror image of the left hand, determined by dropping perpendiculars from each point, is a right hand (Fig. 5.37). Such an image is sometimes said to be *perverted*, i.e. when looking into a plane mirror, it is the image which is perverted. In deference to the more usual lay connotation of the word, its use in optics is happily waning. The process that converts a right-handed coordinate system in the object space into a left-handed one in the image space is known as *inversion*. Systems with more than one planar mirror can be used to produce either an odd or even number of inversions. In the latter case a right-handed (r-h) object will generate a right-handed image (Fig. 5.38) while in the former instance, the image will be left-handed (l-h).

There are a number of practical devices which utilize rotating planar mirror systems, e.g. choppers, beam deflectors and image rotators. Quite frequently, mirrors are used to amplify and measure the slight rotations of certain laboratory apparatus, e.g. galvanometers, torsion pendulums, current balances, etc. As can be seen in Fig. 5.39, if the mirror rotates through an angle α, the reflected beam or image will move through an angle of 2α.

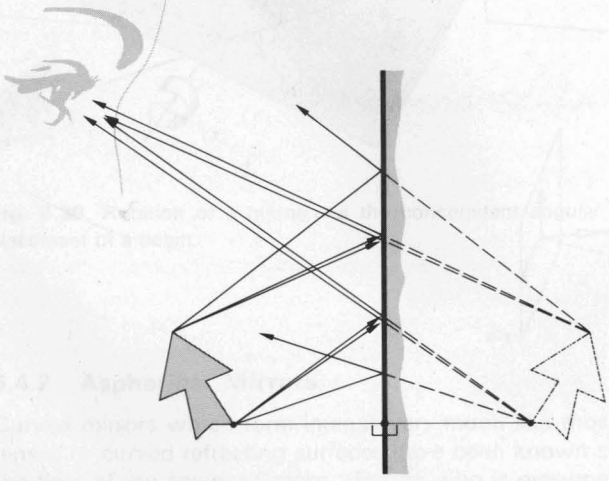


Fig. 5.36 The image of an extended object in a planar mirror.

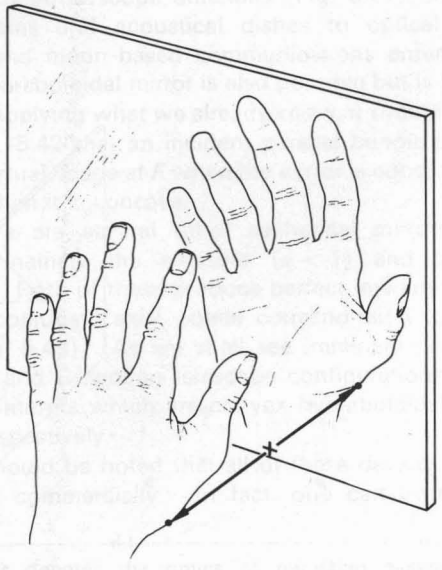


Fig. 5.37 Mirror images—inversion.

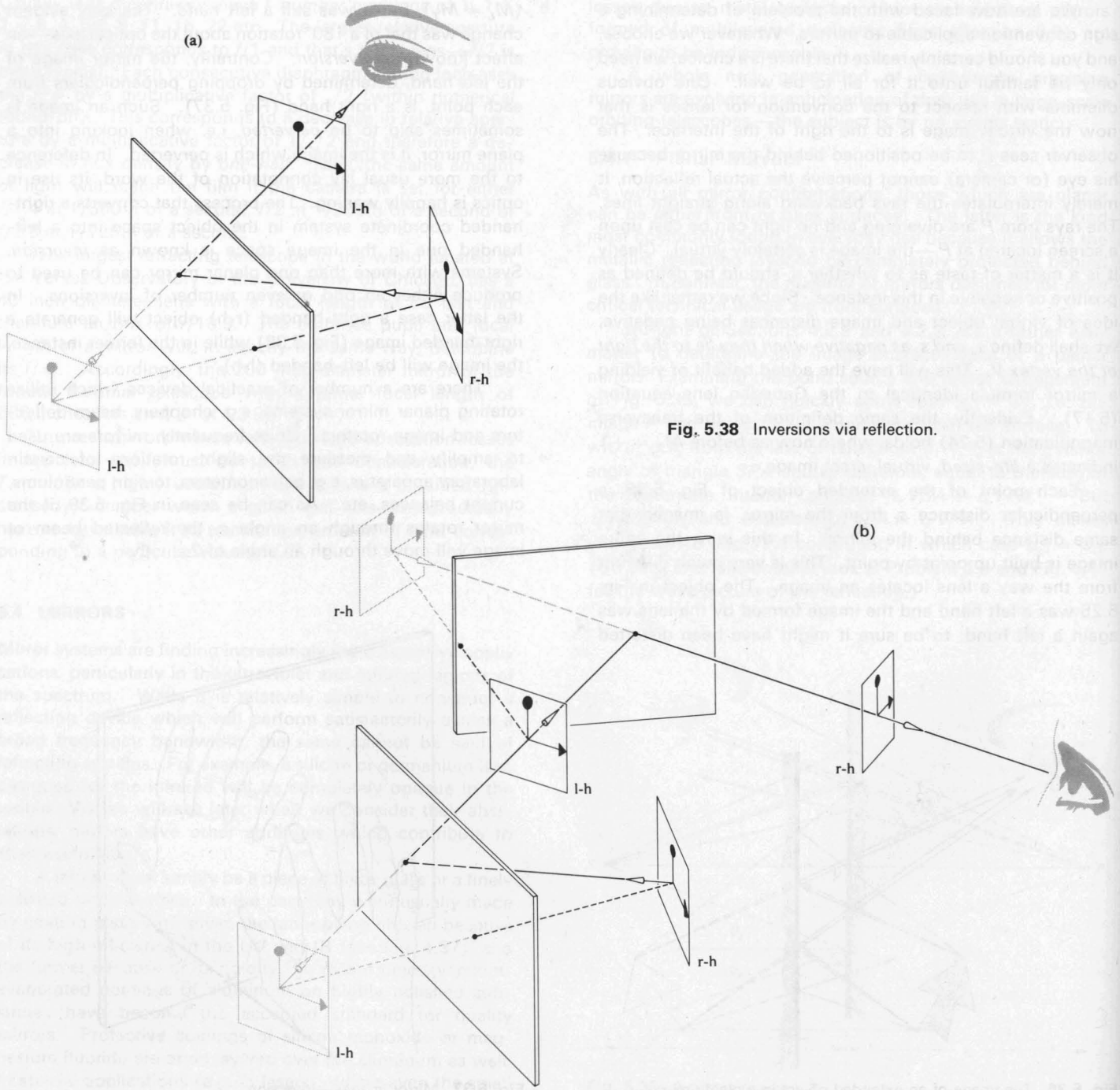


Fig. 5.38 Inversions via reflection.

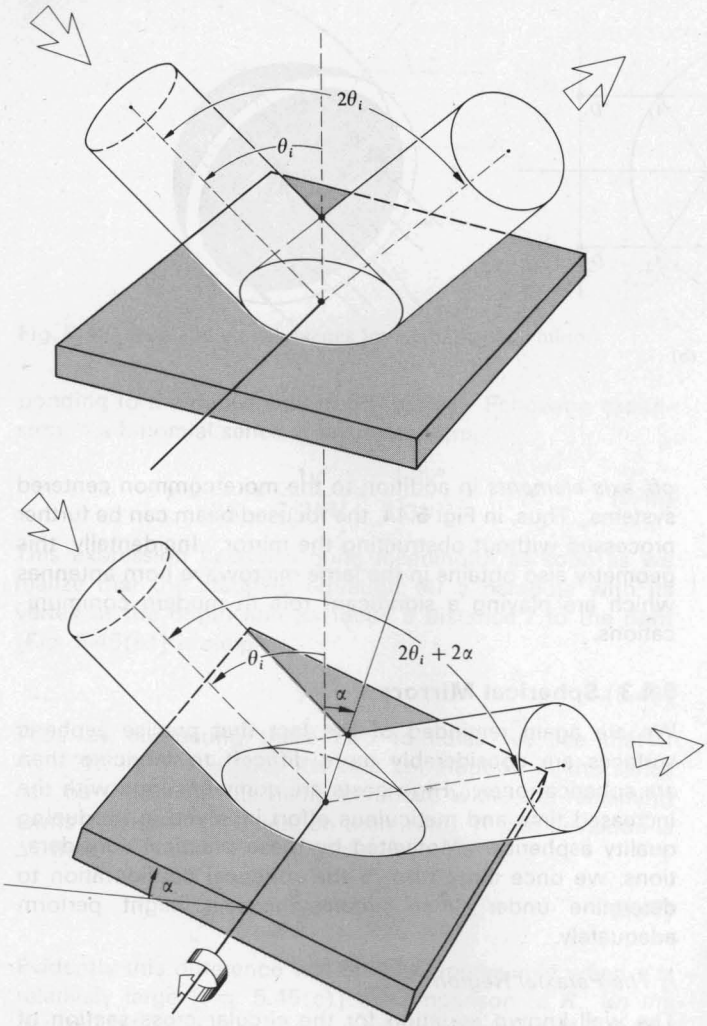


Fig. 5.39 Rotation of a mirror and the concomitant angular displacement of a beam.

5.4.2 Aspherical Mirrors

Curved mirrors which form images very much like those of lenses or curved refracting surfaces have been known since the time of the ancient Greeks. Euclid, who is presumed to have authored the book entitled *Catoptrics*, discusses in it both concave and convex mirrors.* Fortunately, we developed the conceptual basis for designing such mirrors when we spoke earlier about Fermat's principle as applied to imagery in refracting systems. Suppose then, that we would like to determine the configuration a mirror must have in order that an incident plane wave be reformed upon reflection into a converging spherical wave (Fig. 5.40). If the plane wave is to ultimately converge on some point F, the optical path lengths for all rays must be equal; accordingly, for arbitrary points A_1 and A_2

$$(\text{O.P.L.}) = \overline{W_1A_1} + \overline{A_1F} = \overline{W_2A_2} + \overline{A_2F}. \quad (5.41)$$

Since the plane Σ is parallel to the incident wavefronts,

$$\overline{W_1A_1} + \overline{A_1D_1} = \overline{W_2A_2} + \overline{A_2D_2}. \quad (5.42)$$

Equation (5.41) will therefore be satisfied for a surface for which $\overline{A_1F} = \overline{A_1D_1}$ and $\overline{A_2F} = \overline{A_2D_2}$, or more generally, one where $\overline{AF} = \overline{AD}$ for any point A on the mirror. This same condition was discussed in Section 5.2.1 in which we found $\overline{AF} = e(\overline{AD})$, where e was the eccentricity of a conic section. Here $e = 1$; in other words, the surface is a paraboloid with F as its focus and Σ its directrix. The rays could equally well be reversed, i.e. a point source at the focus of a paraboloid will result in the emission of plane waves from the system. The paraboloidal configuration ranges in its present-day applications from flashlight and auto headlight reflectors to giant radiotelescope antennas (Fig. 5.41), from microwave horns and acoustical dishes to optical telescope mirrors and moon-based communications antennas. The convex paraboloidal mirror is also possible but is less widely in use. Applying what we already know, it should be evident from Fig. 5.42 that an incident parallel bundle of rays will form a virtual image at F when the mirror is convex and a real image when it is concave.

There are several other aspherical mirrors of some interest, namely the ellipsoid ($e < 1$) and hyperboloid ($e > 1$). Both of these produce perfect imagery between a pair of conjugate axial points corresponding to their two foci (Fig. 5.43). As we shall see imminently, the Cassegrainian and Gregorian telescope configurations utilize secondary mirrors which are convex hyperboloidal and ellipsoidal respectively.

It should be noted that all of these devices are readily available commercially. In fact, one can even purchase

* *Dioptrics* denotes the optics of refracting elements whereas *catoptrics* connotes the optics of reflecting surfaces.

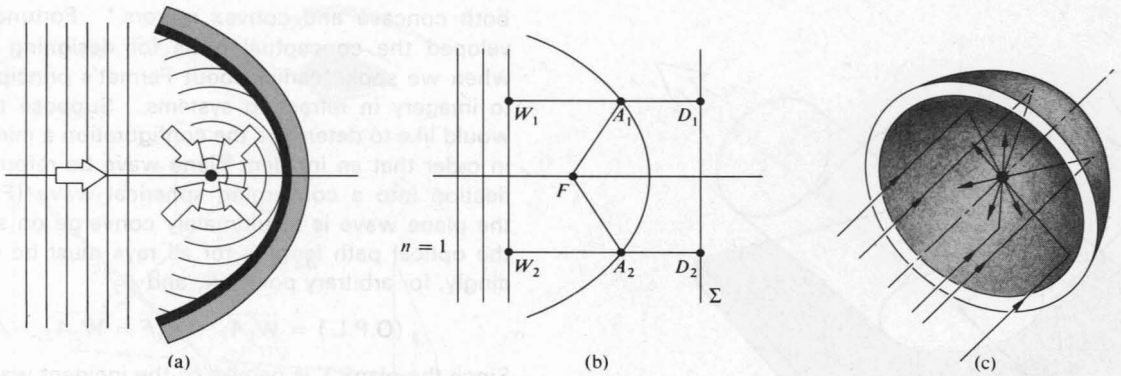


Fig. 5.40 A paraboloidal mirror.

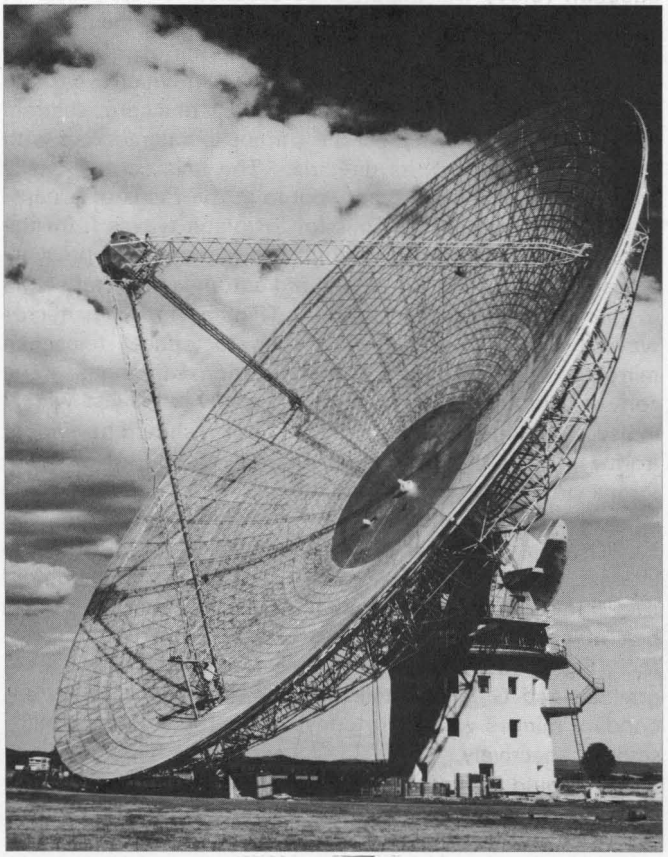


Fig. 5.41 A paraboloidal radio antenna. [Photo courtesy of the Australian News and Information Bureau.]

off-axis elements in addition to the more common centered systems. Thus, in Fig. 5.44, the focused beam can be further processed without obstructing the mirror. Incidentally, this geometry also obtains in the large microwave horn antennas which are playing a significant role in modern communications.

5.4.3 Spherical Mirrors

We are again reminded of the fact that precise aspheric surfaces are considerably more difficult to fabricate than are spherical ones. High costs are commensurate with the increased time and meticulous effort involved in producing quality aspherics. Motivated by these practical considerations, we once more turn to the spherical configuration to determine under what circumstances it might perform adequately.

i) The Paraxial Region

The well-known equation for the circular cross-section of a sphere [Fig. 5.45(a)] is

$$y^2 + (x - R)^2 = R^2, \tag{5.43}$$

where the center C is shifted from the origin O by one radius R. After writing this as

$$y^2 - 2Rx + x^2 = 0$$

we can solve for x:

$$x = R \pm (R^2 - y^2)^{1/2}. \tag{5.44}$$

Let's just concern ourselves with values of x less than R, i.e. we will study a hemisphere, open on the right, corres-

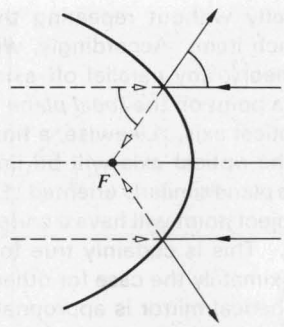


Fig. 5.42 Real and virtual images for a paraboloidal mirror.

ponding to the minus sign in Eq. (5.44). Following expansion in a binomial series, x takes the form

$$x = \frac{y^2}{2R} + \frac{1y^4}{2^2 2! R^3} + \frac{1 \cdot 3 y^6}{2^3 3! R^5} + \cdots . \tag{5.45}$$

This expression becomes quite meaningful as soon as we realize that the standard equation for a parabola with its vertex at the origin and its focus a distance f to the right [Fig. 5.45(b)] is simply

$$y^2 = 4fx. \tag{5.46}$$

Thus by comparing these two formulas we see that if $4f = 2R$, i.e. if $f = R/2$, the first contribution in the series can be thought of as being parabolic while the remaining terms represent the deviation therefrom. If that deviation is Δx then

$$\Delta x = \frac{y^4}{8R^3} + \frac{y^6}{16R^5} + \cdots . \tag{5.47}$$

Evidently this difference will only be appreciable when y is relatively large [Fig. 5.45(c)] in comparison to R. *In the paraxial region, i.e. in the immediate vicinity of the optical axis, these two configurations will be essentially indistinguishable.* Thus if we talk about the paraxial theory of spherical mirrors as a first approximation, we can again embrace the conclusions drawn from our study of the stigmatic imagery of paraboloids. In actual use, however, y will not be so limited and aberrations will appear. Moreover, aspherical surfaces produce perfect images only for pairs of axial points—they too will suffer aberrations.

ii) The Mirror Formula

The paraxial equation which relates conjugate object and image points to the physical parameters of a spherical

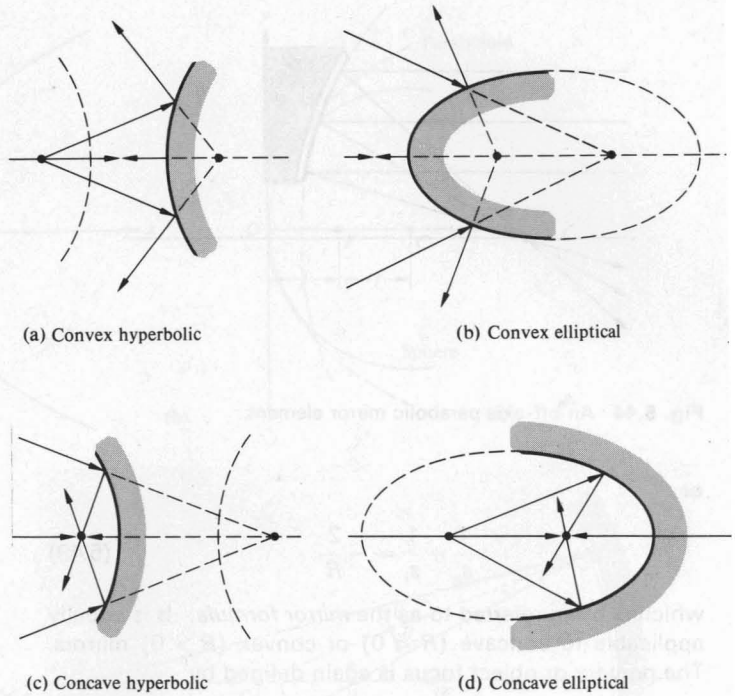


Fig. 5.43 Hyperbolic and elliptical mirrors.

mirror can be derived rather easily with the help of Fig. 5.46. To that end, observe that since $\theta_i = \theta_r$, the $\measuredangle SAP$ is bisected by $\overline{CA}$ which therefore divides the side $\overline{SP}$ of triangle SAP into segments proportional to the remaining two sides, that is

$$\frac{\overline{SC}}{\overline{SA}} = \frac{\overline{CP}}{\overline{PA}}. \tag{5.48}$$

Furthermore

$$\overline{SC} = s_o - |R| \qquad \text{and} \qquad \overline{CP} = |R| - s_i,$$

where s_o and s_i are on the left and therefore positive. If we use the same sign convention for R as we did when we dealt with refraction, it will be negative here because C is to the left of V, i.e. the surface is concave. Thus $|R| = -R$ and

$$\overline{SC} = s_o + R \qquad \text{while} \qquad \overline{CP} = -(s_i + R).$$

In the paraxial region $\overline{SA} \approx s_o$, $\overline{PA} \approx s_i$ and so Eq. (5.48) becomes

$$\frac{s_o + R}{s_o} = -\frac{s_i + R}{s_i}$$

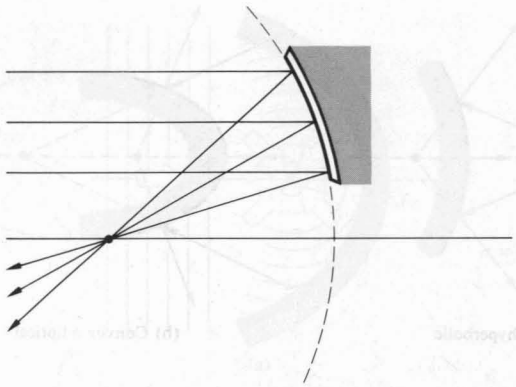


Fig. 5.44 An off-axis parabolic mirror element.

or

$$\frac{1}{s_o}+\frac{1}{s_i}=-\frac{2}{R}, \tag{5.49}$$

which is often referred to as the *mirror formula*. It is equally applicable to concave ($R<0$) or convex ($R>0$) mirrors. The primary or object focus is again defined by

$$\lim_{s_i\to\infty} s_o = f_o,$$

while the secondary or image focus corresponds to

$$\lim_{s_o\to\infty} s_i = f_i.$$

Consequently, from Eq. (5.49)

$$\frac{1}{f_o}+\frac{1}{\infty}=\frac{1}{\infty}+\frac{1}{f_i}=-\frac{2}{R},$$

to wit $f_o = f_i = -R/2$, as we knew from Fig. 5.45(c). Thus dropping the subscripts on the focal lengths, we have

$$\frac{1}{s_o}+\frac{1}{s_i}=\frac{1}{f}. \tag{5.50}$$

Observe that f will be positive for concave mirrors ($R<0$) and negative for convex mirrors ($R>0$). In the latter instance the image is formed behind the mirror and is virtual (Fig. 5.47).

iii) Finite Imagery

The remaining mirror properties are so similar to those of lenses and spherical refracting surfaces that we need only mention them briefly without repeating the entire logical development of each item. Accordingly, within the restrictions of paraxial theory, any parallel off-axis bundle of rays will be focused to a point on the *focal plane* passing through F normal to the optical axis. Likewise, a finite planar object perpendicular to the optical axis will be imaged (to a first approximation) in a plane similarly oriented. Essentially we are saying that each object point will have a corresponding image point in the plane. This is certainly true for a plane mirror but it is only approximately the case for other configurations. To be sure, if a spherical mirror is appropriately restricted in its operation, the reflected waves arising from each permitted object point will very closely approximate spherical waves. Under such circumstances good finite images of extended objects can be formed (Fig. 5.48). Just as each image point produced by a thin lens lies along a straight line through the optical center O, each image point for a spherical mirror will lie on a ray passing through both the center of curvature C and the object point. As with the thin lens (Fig. 5.23), the graphical location of the image is quite straightforward. Once more the top of the image is located at the intersection of two rays; e.g. one initially parallel to the axis and passing through F after reflection, the other going straight through C (Fig. 5.49). The ray from any off-axis object point to the vertex makes equal angles with the optical axis on reflection and is therefore particularly convenient to construct as well. So too is the ray which first passes through the focus and after reflection emerges parallel to the axis.

Notice that triangles S_1S_2V and P_1P_2V are similar and hence their sides are proportional. Taking y_i to be negative as we did in the past, since it is below the axis, $y_i/y_o = -s_i/s_o$ which of course is equal to M_T the *transverse magnification*, identical to that of the lens (5.25).

The only equation which contains information about the structure of the optical element (n, R etc.) is that for f and so, rather understandably, it differs for the thin lens and spherical mirror. The other functional expressions which relate s_o, s_i and f or y_o, y_i and M_T are, however, precisely the same. The only alteration in the previous sign convention appears in Table 5.4 where s_i on the left of V is now taken as positive. The striking similarity between the properties of a concave mirror and a convex lens on one hand and a convex mirror and a concave lens on the other are quite evident on comparing Tables 5.3 and 5.5 which are identical in all respects.

The properties summarized in Table 5.5 can easily be verified empirically. If you don't have a spherical mirror at hand, a fairly crude but functional one can be made by

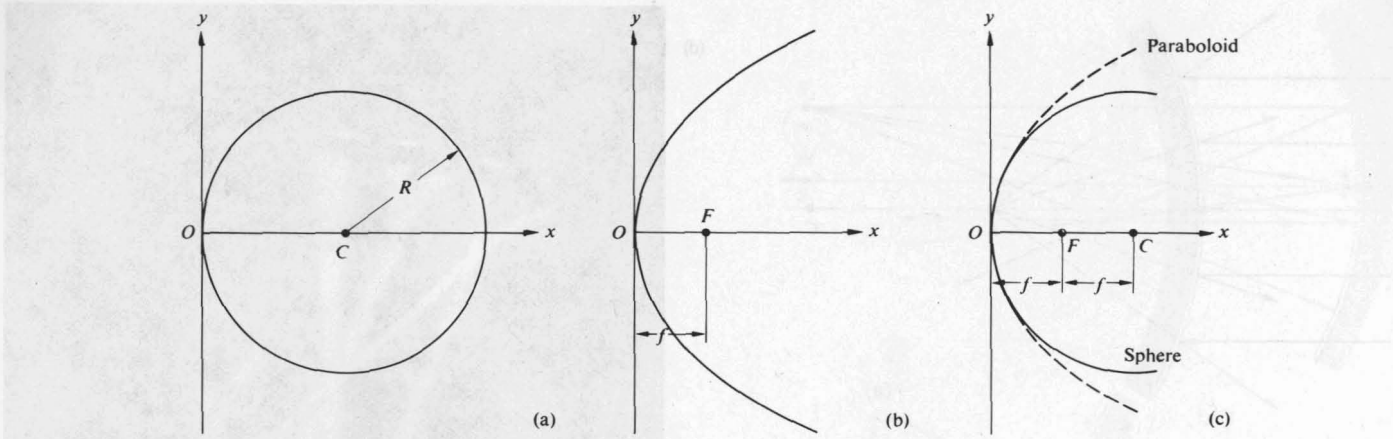


Fig. 5.45 Comparison of a spherical and paraboloidal mirror.

carefully shaping aluminum foil over a spherical form such as the end of a light bulb (in that particular case R and therefore f will be small). A rather nice qualitative experiment involves examining the image of some small object formed by a short focal length concave mirror. As you move it toward the mirror from beyond a distance of $2f = R$, the image will gradually increase until at $s_o = 2f$ it will appear inverted and life-size. Bringing it still closer will cause the image to increase even more until it fills the entire mirror with an unrecognizable blur. As s_o becomes ever smaller, the now erect, magnified image will continue to decrease until the object finally rests on the mirror where the image is again lifesize. If you are not moved by all of this to immediately jump up and make a mirror, you might try examining the image formed by a shiny spoon—either side will be interesting.

Table 5.4 Sign convention for spherical mirrors.

Quantity	Sign	
	+	−
s_o	Left of V, real object	Right of V, virtual object
s_i	Left of V, real image	Right of V, virtual image
f	Concave mirror	Convex mirror
R	C right of V, convex	C left of V, concave
y_o	Above axis, erect object	Below axis, inverted object
y_i	Above axis, erect image	Below axis, inverted image

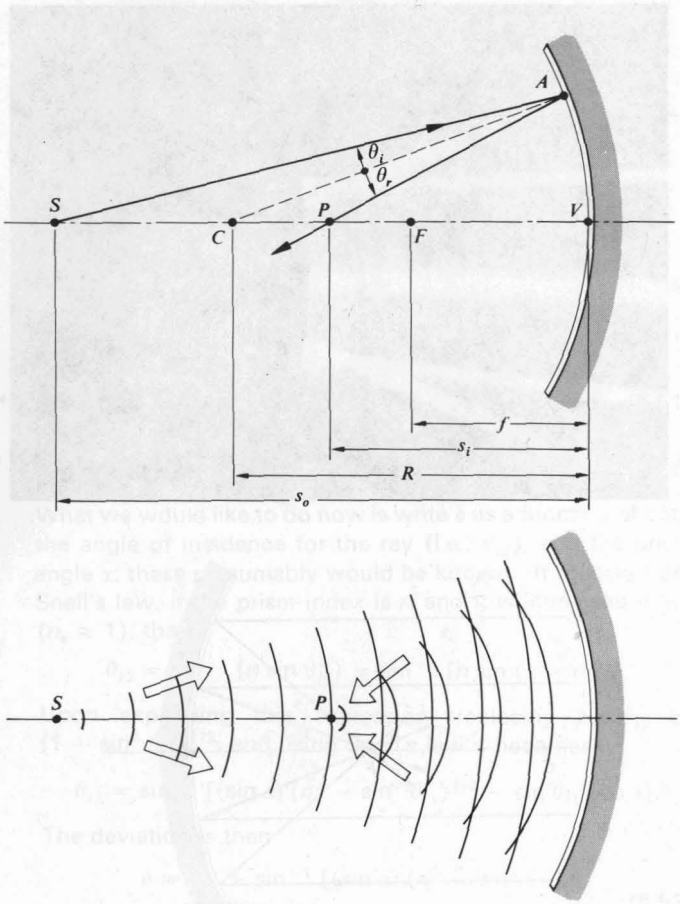
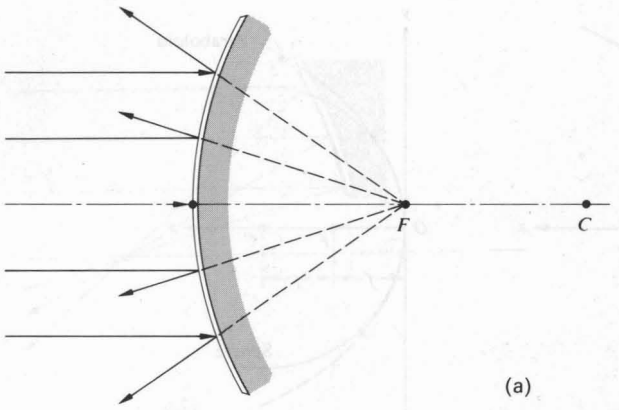
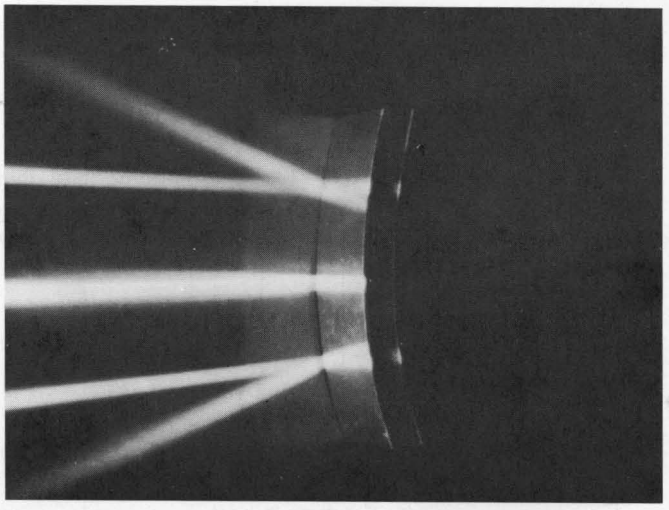


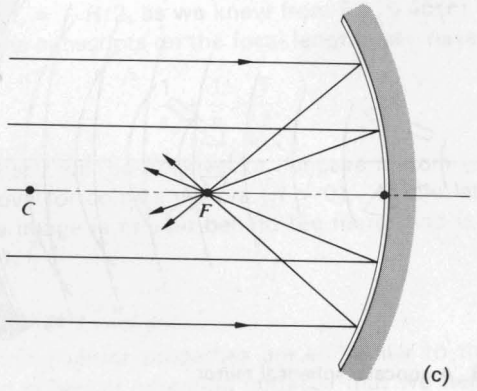
Fig. 5.46 A concave spherical mirror.



(a)



(b)



(c)

(d)

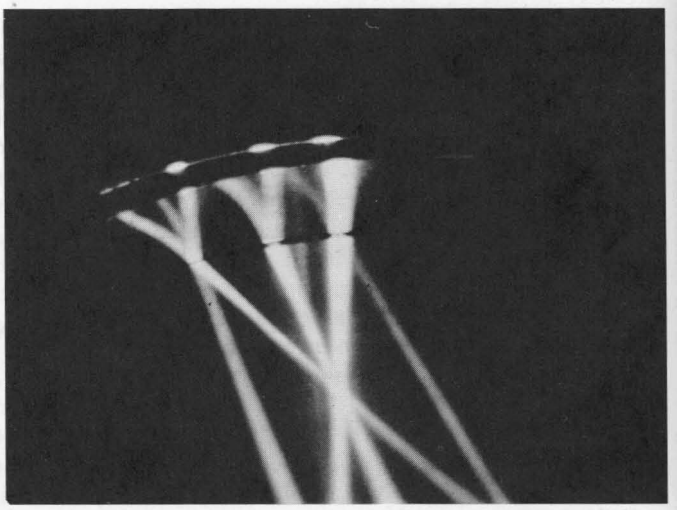


Fig. 5.47 Focusing of rays via a spherical mirror. [Photos by E.H.]

Table 5.5 Images of real objects formed by spherical mirrors.

Concave				
Object	Image			
Location	Type	Location	Orientation	Relative size
$\infty > s_o > 2f$	Real	$f < s_i < 2f$	Inverted	Minified
$s_o = 2f$	Real	$s_i = 2f$	Inverted	Same size
$f < s_o < 2f$	Real	$\infty > s_i > 2f$	Inverted	Magnified
$s_o = f$		$\pm \infty$		
$s_o < f$	Virtual	$\lvert s_i \rvert > s_o$	Erect	Magnified

Convex				
Object	Image			
Location	Type	Location	Orientation	Relative size
Anywhere	Virtual	$\lvert s_i \rvert < \lvert f \rvert$	Erect	Minified

5.5 PRISMS

Prisms play a great many different roles in optics; there are prism combinations that serve as beam-splitters (Section 4.3.4), polarizing devices (Section 8.4.3) and even interferometers. Despite this diversity, the vast majority of

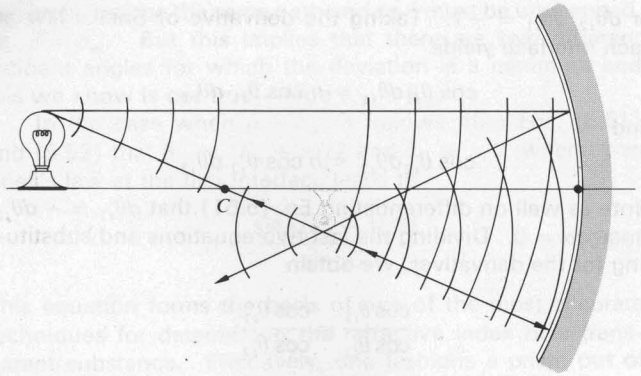


Fig. 5.48 Wavefronts reflected from a concave mirror.

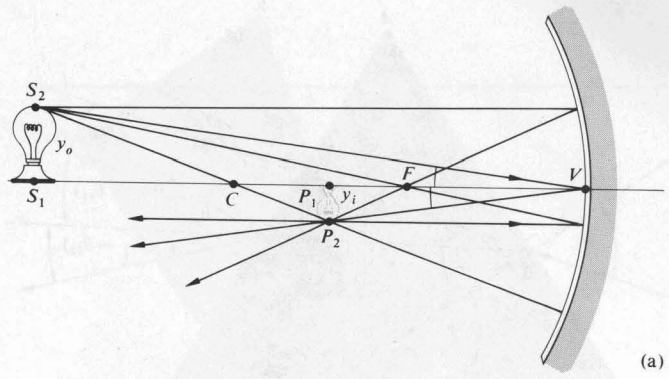
applications make use of only one of two main prism functions. Firstly, a prism can serve as a dispersive device as it does in a variety of spectrum analyzers. That is fo say, it is capable of separating, to some extent, the constituent frequency components in a polychromatic light beam. You might recall that the term *dispersion* was introduced earlier (Section 3.3.1) in connection with the frequency dependence of the index of refraction, $n(\omega)$, for dielectrics. And in fact, the prism provides a highly useful means of measuring $n(\omega)$ over a broad range of frequencies and for a wide variety of materials (including gases and liquids). Its second and most widely utilized function is to affect a change in the orientation of an image or in the direction of propagation of a beam. Prisms are incorporated in a great many optical instruments, often simply to fold the system into a confined space. There are inversion prisms, reversion prisms and prisms that deviate a beam without inversion or reversion and all of this without dispersion.

5.5.1 Dispersing Prisms

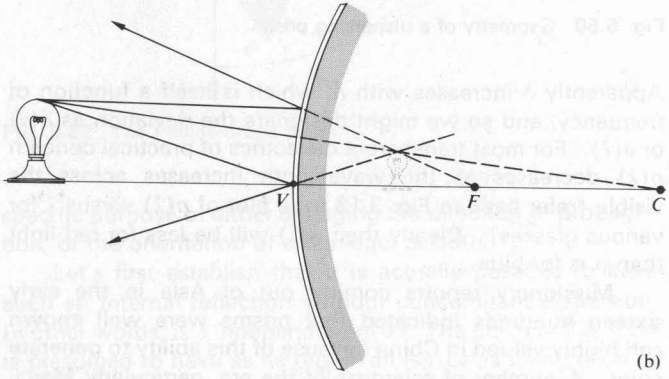
Typically, a ray entering a dispersing prism, as in Fig. 5.50, will emerge having been deflected from its original direction by an angle δ known as the *angular deviation*. At the first refraction the ray is deviated through an angle $(\theta_{i1} - \theta_{t1})$ and at the second refraction it is further deflected through $(\theta_{t2} - \theta_{i2})$. The total deviation is then

$$\delta = (\theta_{i1} - \theta_{t1}) + (\theta_{t2} - \theta_{i2}).$$

Since the polygon *ABCD* contains two right angles, $\measuredangle BCD$ must be the supplement of the *apex angle* α. Now α, being the exterior angle to triangle *BCD*, is also the sum of the alternate interior angles, that is



(a)



(b)

Fig. 5.49 Finite imagery with spherical mirrors.

$$\alpha = \theta_{t1} + \theta_{i2}. \tag{5.51}$$

Thus

$$\delta = \theta_{i1} + \theta_{t2} - \alpha. \tag{5.52}$$

What we would like to do now is write δ as a function of both the angle of incidence for the ray (i.e., θ_{i1}), and the prism angle α; these presumably would be known. It follows from Snell's law, if the prism index is n and it is immersed in air ($n_a \approx 1$), that

$$\theta_{t2} = \sin^{-1}(n \sin \theta_{i2}) = \sin^{-1}[n \sin(\alpha - \theta_{t1})].$$

Upon expanding this expression, replacing $\cos \theta_{t1}$ by $(1 - \sin^2 \theta_{t1})^{1/2}$, and using Snell's law it becomes

$$\theta_{t2} = \sin^{-1}[(\sin \alpha)(n^2 - \sin^2 \theta_{i1})^{1/2} - \sin \theta_{i1} \cos \alpha].$$

The deviation is then

$$\delta = \theta_{i1} + \sin^{-1}[(\sin \alpha)(n^2 - \sin^2 \theta_{i1})^{1/2} - \sin \theta_{i1} \cos \alpha] - \alpha. \tag{5.53}$$

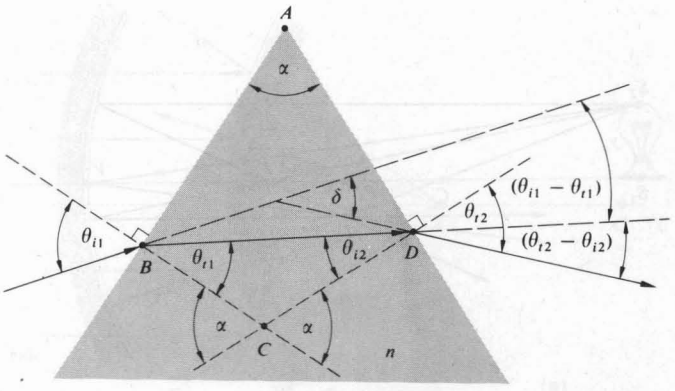


Fig. 5.50 Geometry of a dispersing prism.

Apparently δ increases with n, which is itself a function of frequency, and so we might designate the deviation as $\delta(\nu)$ or $\delta(\lambda)$. For most transparent dielectrics of practical concern $n(\lambda)$ decreases as the wavelength increases across the visible [refer back to Fig. 3.13 for a plot of $n(\lambda)$ versus λ for various glasses]. Clearly then $\delta(\lambda)$ will be less for red light than it is for blue.

Missionary reports coming out of Asia in the early sixteen hundreds indicated that prisms were well known and highly valued in China because of this ability to generate color. A number of scientists of the era, particularly Marci, Grimaldi and Boyle had made some observations using prisms but it remained for the great Sir Isaac Newton to perform the first definitive studies of dispersion. On February 6, 1672, Newton presented a classic paper to the Royal Society entitled *A New Theory about Light and Colours*. He had concluded that white light consisted of a mixture of various colors and that the process of refraction was color dependent.

Returning to Eq. (5.53), it is evident that the deviation suffered by a monochromatic beam on traversing a given prism (i.e. n and α are fixed) is a function only of the incident angle at the first face, θ_{i1}. A plot of the results of Eq. (5.53) as applied to a typical glass prism is shown in Fig. 5.51. The smallest value of δ is known as the *minimum deviation* δ_m and it is of particular interest for practical reasons. It can be determined analytically by differentiating Eq. (5.53) and then setting $d\delta/d\theta_{i1} = 0$, but a more indirect route will certainly be simpler. Differentiating Eq. (5.52) and setting it equal to zero leads to

$$\frac{d\delta}{d\theta_{i1}} = 1 + \frac{d\theta_{t2}}{d\theta_{i1}} = 0$$

or $d\theta_{t2}/d\theta_{i1} = -1$. Taking the derivative of Snell's law at each interface yields

$$\cos\theta_{i1}\, d\theta_{i1} = n\cos\theta_{t1}\, d\theta_{t1}$$

and

$$\cos\theta_{t2}\, d\theta_{t2} = n\cos\theta_{i2}\, d\theta_{i2}.$$

Note as well on differentiating Eq. (5.51) that $d\theta_{t1} = -d\theta_{i2}$ since $d\alpha = 0$. Dividing the last two equations and substituting for the derivatives, we obtain

$$\frac{\cos\theta_{i1}}{\cos\theta_{t2}} = \frac{\cos\theta_{t1}}{\cos\theta_{i2}}.$$

Making use of Snell's law once again, this can be rewritten as

$$\frac{1 - \sin^2\theta_{i1}}{1 - \sin^2\theta_{t2}} = \frac{n^2 - \sin^2\theta_{i1}}{n^2 - \sin^2\theta_{t2}}.$$

The value of θ_{i1} for which this is true is the one for which $d\delta/d\theta_{i1} = 0$. Inasmuch as $n \neq 1$, it follows that

$$\theta_{i1} = \theta_{t2}$$

and therefore

$$\theta_{t1} = \theta_{i2}.$$

This means that the ray for which the deviation is a minimum traverses the prism symmetrically, i.e. parallel to its base. Incidentally, there is a lovely argument for why θ_{i1} must equal θ_{t2} which is neither as mathematical nor as tedious as the one we have evolved. In brief, suppose a ray undergoes a minimum deviation and $\theta_{i1} \neq \theta_{t2}$. Then if we reverse the

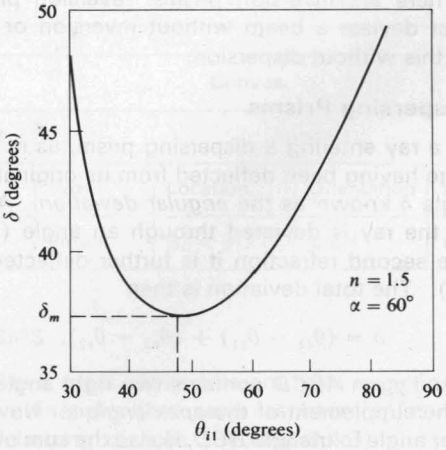


Fig. 5.51 Deviation versus incident angle.

ray, it will retrace the same path and so δ must be unchanged, i.e. $\delta = \delta_m$. But this implies that there are two different incident angles for which the deviation is a minimum and this we know is not true—ergo $\theta_{i1} = \theta_{t2}$.

In the case when $\delta = \delta_m$, it follows from Eqs. (5.51) and (5.52) that $\theta_{i1} = (\delta_m + \alpha)/2$ and $\theta_{t1} = \alpha/2$, whereupon Snell's law at the first interface leads to

$$n = \frac{\sin\left[(\delta_m + \alpha)/2\right]}{\sin \alpha/2}. \tag{5.54}$$

This equation forms the basis of one of the most accurate techniques for determining the refractive index of a transparent substance. Effectively, one fashions a prism out of the material in question and then, measuring α and $\delta_m(\lambda)$, $n(\lambda)$ is computed employing Eq. (5.54) at each wavelength of interest. Hollow prisms whose sides are fabricated of plane-parallel glass can be filled with liquids or gases under high pressure; the glass plates will not result in any deviation of their own.

Figures 5.52 and 5.53 show two examples of *constant deviation dispersing prisms* which are important primarily in spectroscopy. The *Pellin-Broca* is probably the most common of the group. Albeit a single block of glass, it can be envisaged as consisting of two 30°–60°–90° prisms and one 45°–45°–90° prism. Suppose that in the position shown a single monochromatic ray of wavelength λ traverses the component prism *DAE* symmetrically, thereafter to be reflected at 45° from face *AB*. The ray will then traverse prism *CDB* symmetrically having experienced a total deviation of 90°. Effectively, the ray can be thought of as having passed through an ordinary 60° prism (*DAE* combined with *CDB*) at minimum deviation. All other wavelengths present in the beam will emerge at other angles. If the prism is now rotated slightly about an axis normal to the paper, the incoming beam will have a new incident angle. A different wavelength component, say λ_2, will now undergo a minimum deviation which is again 90°—hence the name, *constant deviation*. With a prism of this sort, one can conveniently set up the light source and viewing system at a fixed angle (here 90°) and then simply rotate the prism in order to look at a particular wavelength. The device can be calibrated so that the prism-rotating dial reads directly in wavelength.

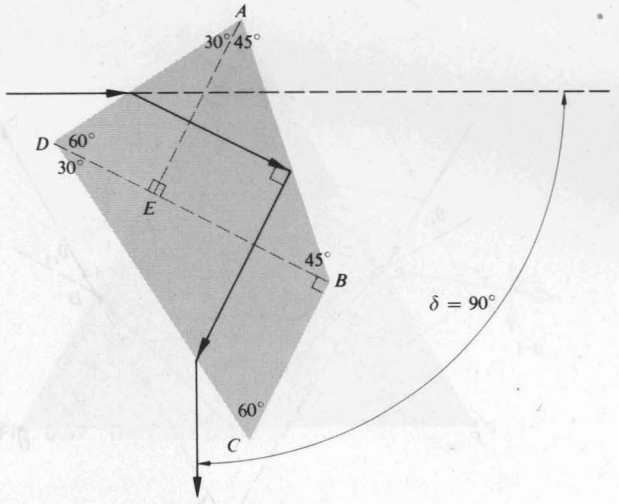


Fig. 5.52 The Pellin–Broca prism.

5.5.2 Reflecting Prisms

In contrast with the previous section, we now examine *reflecting prisms* in which dispersion is no longer desirable. In the present instance, the beam will be introduced in such a way that at least one internal reflection takes place, for the specific purpose of either changing the direction of propagation, or the orientation of the image, or both.

Let's first establish that it is actually possible to have such an internal reflection without concomitant dispersion. In other words, is δ independent of λ? The prism in Fig. 5.54 is presumed to have as its profile an isosceles triangle—this happens to be a rather common configuration in any event. The ray refracted at the first interface is later reflected from face *FG*. As we saw earlier (Section 4.3.4), this will occur

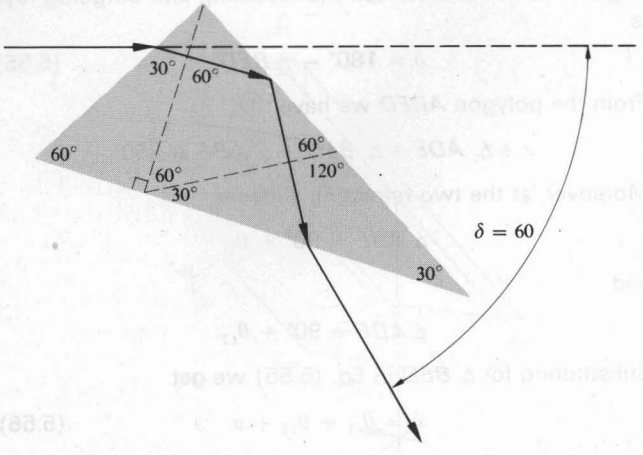
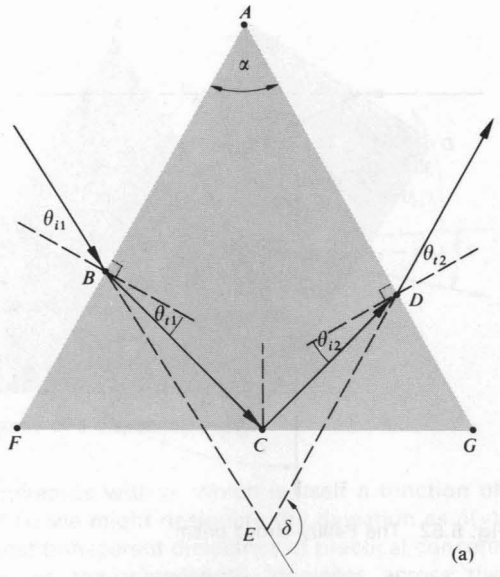


Fig. 5.53 The Abbe prism.



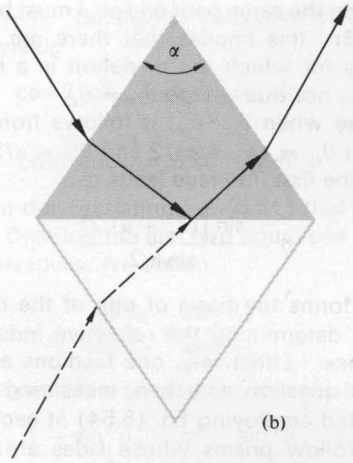


Fig. 5.54 Geometry of a reflecting prism.

when the internal incident angle is greater than the critical angle θ_c, defined by

$$\sin \theta_c = n_{ti}. \qquad [4.69]$$

For a glass–air interface, this requires that θ_i be greater than roughly 42°. To avoid any difficulties at smaller angles, let's further suppose the base of our hypothetical prism is silvered as well—certain prisms do in fact require silvered faces. The angle of deviation between the incoming and outgoing rays is

$$\delta = 180° - \measuredangle BED. \qquad (5.55)$$

From the polygon *ABED* we have that

$$\alpha + \measuredangle ADE + \measuredangle BED + \measuredangle ABE = 360°.$$

Moreover, at the two refracting surfaces

$$\measuredangle ABE = 90° + \theta_{i1}$$

and

$$\measuredangle ADE = 90° + \theta_{t2}.$$

Substituting for $\measuredangle BED$ in Eq. (5.55) we get

$$\delta = \theta_{i1} + \theta_{t2} + \alpha. \qquad (5.56)$$

Since the ray at point *C* has equal angles of incidence and reflection $\measuredangle BCF = \measuredangle DCG$. Thus, because the prism is isosceles $\measuredangle BFC = \measuredangle DGC$ and triangles *FBC* and *DGC* are similar. It follows that $\measuredangle FBC = \measuredangle CDG$ and finally, therefore, $\theta_{t1} = \theta_{i2}$. From Snell's law we know that this is equivalent to $\theta_{i1} = \theta_{t2}$, whereupon the deviation becomes

$$\delta = 2\theta_{i1} + \alpha, \qquad (5.57)$$

which is certainly independent of both λ and n. The reflection will occur without any color preferences and the prism is said to be *achromatic*. If we unfold the prism, i.e. if we draw its image in the reflecting surface *FG*, as in Fig. 5.54(b), we see that the prism is equivalent in a sense to a parallelepiped or thick planar plate. The image of the incident ray emerges parallel to itself regardless of wavelength.

A few out of the great many widely used reflecting prisms are shown in the next several figures. These are often made from BSC-2 or C-1 glass (see Table 6.2). For the most part, the illustrations are self explanatory and so the descriptive commentary will be brief.

The *right-angle* prism (Fig. 5.55) deviates rays normal to the incident face by 90°. Notice that the top and bottom of the image have been interchanged, i.e. the arrow has been flipped over while the right and left sides have not. It is therefore an inversion system with the top face acting like a plane mirror. (To see this, imagine that the arrow and

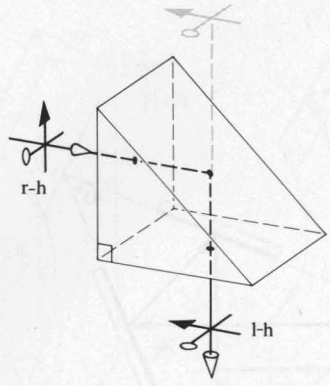


Fig. 5.55 The right-angle prism.

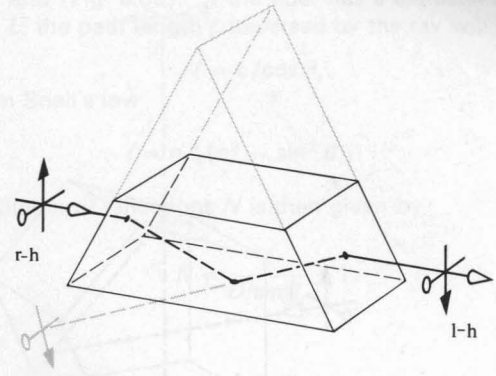


Fig. 5.57 The Dove prism.

lollypop are vectors and take their cross product. The resultant, arrow × lollypop, was initially in the propagation direction but is reversed by the prism.)

The *Porro* prism (Fig. 5.56) is physically the same as the previous one but it is used in a different orientation. After two reflections, the beam is deviated by 180°. Thus, if it enters right-handed, it leaves right-handed.

The *Dove* (Fig. 5.57) is a truncated version (to reduce size and weight) of the right-angle prism, used almost exclusively in collimated light. It has the interesting property (Problem 5.38) of rotating the image twice as fast as it is itself rotated about the longitudinal axis.

The *Amici* (Fig. 5.58) is essentially a truncated right-angle prism with a roof section added on to the hypotenuse face. In its most common usage it has the effect of splitting the image down the middle and interchanging the right and left portions.* These prisms are expensive because the 90° roof angle must be held to roughly 3 or 4 seconds of arc or a troublesome double image will result. They are often used in simple telescope systems to correct for the reversion introduced by the lenses.

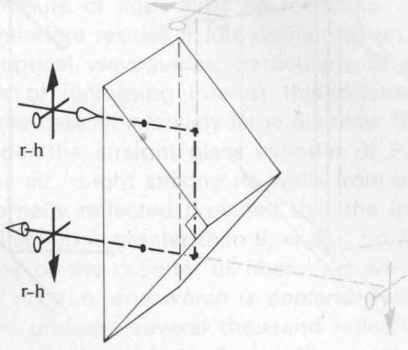


Fig. 5.56 The Porro prism.

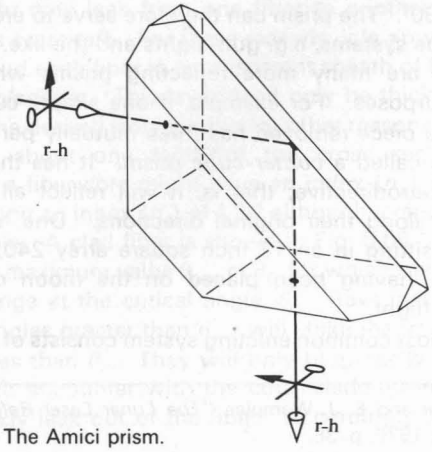


Fig. 5.58 The Amici prism.

* You can see how it actually works by placing two plane mirrors at right angles and looking directly into the combination. If you wink your *right* eye, the image will wink its *right* eye. Incidentally, if your eyes are equally strong, you will see two seams (images of where the mirrors met), one running down the middle of each eye with your nose presumably between them. If either eye is stronger, there will be only one seam down the middle of that eye. If you close it, the seam will jump over to the other eye. This must be tried to be appreciated.

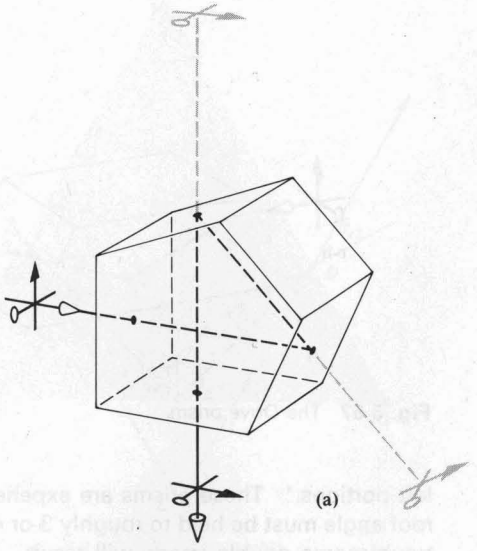


Fig. 5.59 The penta prism and its mirror equivalent.

The *Penta* prism (Fig. 5.59) will deviate the beam by 90° without affecting the orientation of the image. Note that two of its surfaces must be silvered. These prisms are often used as end reflectors in small rangefinders.

The *Rhomboid* prism (Fig. 5.60) displaces the line of sight without producing any angular deviation or changes in the orientation of the image.

The *Leman–Springer* prism (Fig. 5.61) also has a 90° roof. Here the line of sight is displaced without being deviated but the emerging image is right-handed and rotated through 180°. The prism can therefore serve to erect images in telescope systems, e.g. gun-sights and the like.

There are many more reflecting prisms which serve specific purposes. For example, if one simply cuts a cube so that the piece removed has three mutually perpendicular faces, it is called a *corner-cube* prism. It has the property of being retrodirective; that is, it will reflect all incoming rays back along their original directions. One hundred of these are sitting in an 18 inch square array 240,000 miles from here, having been placed on the moon during the Apollo 11 flight.*

The most common erecting system consists of two Porro prisms as illustrated in Fig. 5.62. These are relatively easy to manufacture and are shown here with rounded corners to reduce weight and size. Since there are four reflections, the exiting image will be right-handed. A small slot is often cut in the hypotenuse face to obstruct rays which are internally reflected at glancing angles. Finding these slots after dismantling the family's binoculars is all too often an inexplicable surprise.

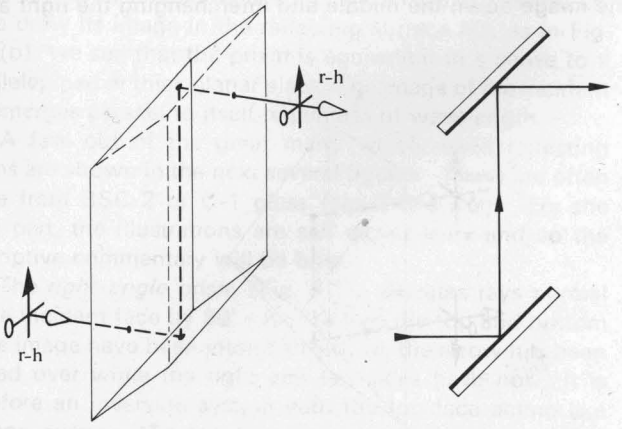


Fig. 5.60 The rhomboid prism and its mirror equivalent.

* J. E. Foller and E. J. Wampler, "*The Lunar Laser Reflector*," *Sci. Am.*, March 1970, p. 38.

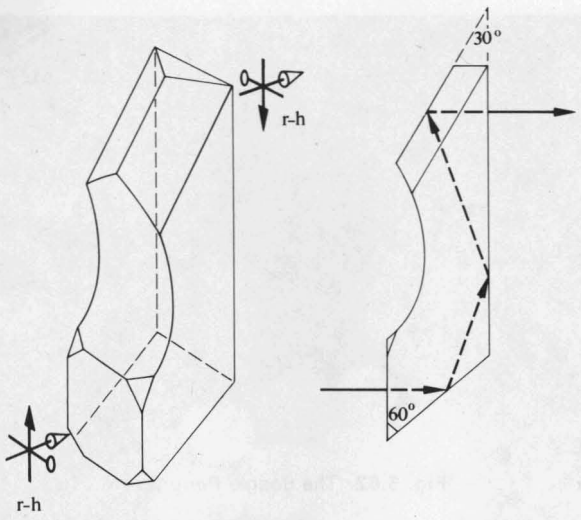


Fig. 5.61 The Leman–Springer prism.

5.6 FIBER OPTICS

In recent times, techniques have evolved for efficiently conducting light from one point in space to another via transparent, dielectric fibers. As long as the diameter of these fibers is large compared to the wavelength of the radiant energy, the inherent wave nature of the propagation mechanism is of little importance and the process obeys the familiar laws of geometrical optics. On the other hand, if the diameter is of the order of λ, the transmission closely resembles the manner in which microwaves advance along waveguides. Some of the propagation modes are evident in the photomicrographic end views of fibers shown in Fig. 5.63. Here the wave nature of light must be reckoned with and this behavior therefore resides in the domain of physical optics. Although optical waveguides, particularly of the thin film variety, are of increasing interest this discussion will be limited to the case of relatively large diameter fibers.

Consider the straight glass cylinder of Fig. 5.64 surrounded by air. Light striking its walls from within will be totally internally reflected provided that the incident angle at each reflection is greater than $\theta_c = \sin^{-1} n_a/n_f$, where n_f is the index of the cylinder or fiber. As we will show, a *meridional ray* (i.e. *one which is coplanar with the optical axis*) might undergo several thousand reflections per foot as it bounces back and forth along a fiber until it emerges at the far end (Fig. 5.65). If the fiber has a diameter D and a length L, the path length ℓ traversed by the ray will be

$$\ell = L/\cos\theta_t,$$

or from Snell's law

$$\ell = n_f L(n_f^2 - \sin^2\theta_i)^{-1/2}. \tag{5.58}$$

The number of reflections N is then given by

$$N = \frac{\ell}{D/\sin\theta_t} \pm 1$$

or

$$N = \frac{L\sin\theta_i}{D(n_f^2 - \sin^2\theta_i)^{1/2}} \pm 1 \tag{5.59}$$

rounded off to the nearest whole number. The ± 1, which depends on where the ray strikes the end face, is of no significance when N is large, as it is in practice. Thus if D is 50 μ, (i.e. 50 microns where $1\ \mu = 10^{-6}$ m $= 39.37 \times 10^{-6}$ in) which is about 2×10^{-3} in, (a hair from the head of a human is roughly 50 μ in diameter) and if $n_f = 1.6$ and $\theta_i = 30°$, N turns out to be approximately 2000 reflections per foot. Fibers are available in diameters from about 2 μ up to $\frac{1}{4}$ inch or so but are seldom used in sizes much less than about 10 μ. The large diameter rods are generally called *light pipes*. Extremely thin glass (or plastic) filaments are quite flexible as witnessed by the well-known fact that glass fibers can even be woven into fabric.

The smooth surface of a single fiber must be kept clean of moisture, dust, oil, etc., if there is to be no leakage of light (via the mechanism of frustrated total internal reflection). Similarly if large numbers of fibers are packed in close proximity, light may leak from one fiber to another in what is known as *cross talk*. For these reasons, it is now customary to enshroud each fiber in a transparent sheath of lower index called a *cladding*. This layer need only be thick enough to provide the desired isolation but for other reasons it generally occupies about one tenth of the cross sectional area. Typically a fiber core might have an index (n_f) of 1.62 and the cladding an index (n_c) of 1.52 although a range of values is available. A clad fiber is shown in Fig. 5.66. Notice that there is a maximum value θ_{max} of θ_i, for which the internal ray will impinge at the critical angle, θ_c. Rays incident on the face at angles greater than θ_{max} will strike the interior wall at angles less than θ_c. They will only be partially reflected at each such encounter with the core–cladding interface and will quickly leak out of the fiber. Accordingly, θ_{max} defines

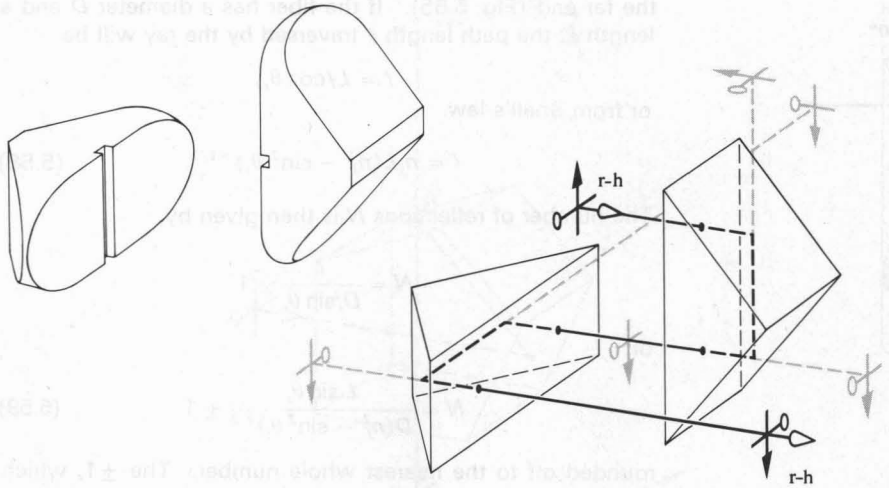


Fig. 5.62 The double Porro prism.

the half-angle of the acceptance cone of the fiber. To determine it we write

$$\sin \theta_c = n_c/n_f = \sin (90 - \theta_t)$$

Thus

$$n_c/n_f = \cos \theta_t$$

or

$$n_c/n_f = (1 - \sin^2 \theta_t)^{1/2}.$$

Making use of Snell's law and rearranging matters, we have

$$\sin \theta_{\max} = \frac{1}{n_o}(n_f^2 - n_c^2)^{1/2}. \tag{5.60}$$

The quantity $n_o \sin \theta_{\max}$ is defined as the *numerical aperture* or N.A. Its square is a measure of the light gathering power of the system. The term originates in microscopy where the equivalent expression characterizes the corresponding capabilities of the objective lens. It should clearly relate to the *speed* of the system and, in fact,

$$f/\# = \frac{1}{2(\text{N.A.})}.$$

Thus for a fiber

$$\text{N.A.} = (n_f^2 - n_c^2)^{1/2}. \tag{5.61}$$

The left-hand side of Eq. (5.60) cannot exceed one and in air ($n_o = 1.00028 \approx 1$) that means that the largest value of N.A. is one. In this case, the half-angle $\theta_{\max}$ equals 90° and the fiber totally internally reflects all light entering its face (Problem 5.39). Fibers with a wide variety of numerical apertures running from about 0.2 up to and including 1.0 are commercially obtainable.

Bundles of free fibers whose ends are bound together (e.g. with epoxy) and ground and polished form flexible light guides. If no attempt is made to align the fibers in an ordered array, they form an *incoherent bundle*. This (unfortunate use of the term which ought not to be confused with coherence theory) just means e.g. that the first fiber in the top row at the entrace face may have its terminus anywhere in the bundle at the exit face. These *flexible light carriers* are, for that reason, relatively easy to make and inexpensive. Their primary function is simply to conduct light from one region to another. Conversely, when the fibers are carefully arranged so that their terminations occupy the same relative positions in both of the bound ends of the bundle, the bundle is said to be *coherent*. Such an arrangement is capable of transmitting images and is consequently known as a *flexible image carrier* (Fig. 5.67). Not all fiber optics arrays are made flexible, for example, fused, rigid, coherent fiber faceplates or *mosaics* are used to replace homogeneous low-resolution sheet glass on cathode-ray tubes, vidicons, image intensifiers, etc. Mosaics consisting of literally millions of fibers with their claddings fused together have mechanical properties almost identical to homogeneous glass. Similarly a sheet of fused *tapered* fibers can either magnify or minify an image depending on whether the light enters the smaller or larger end of the fiber. The compound

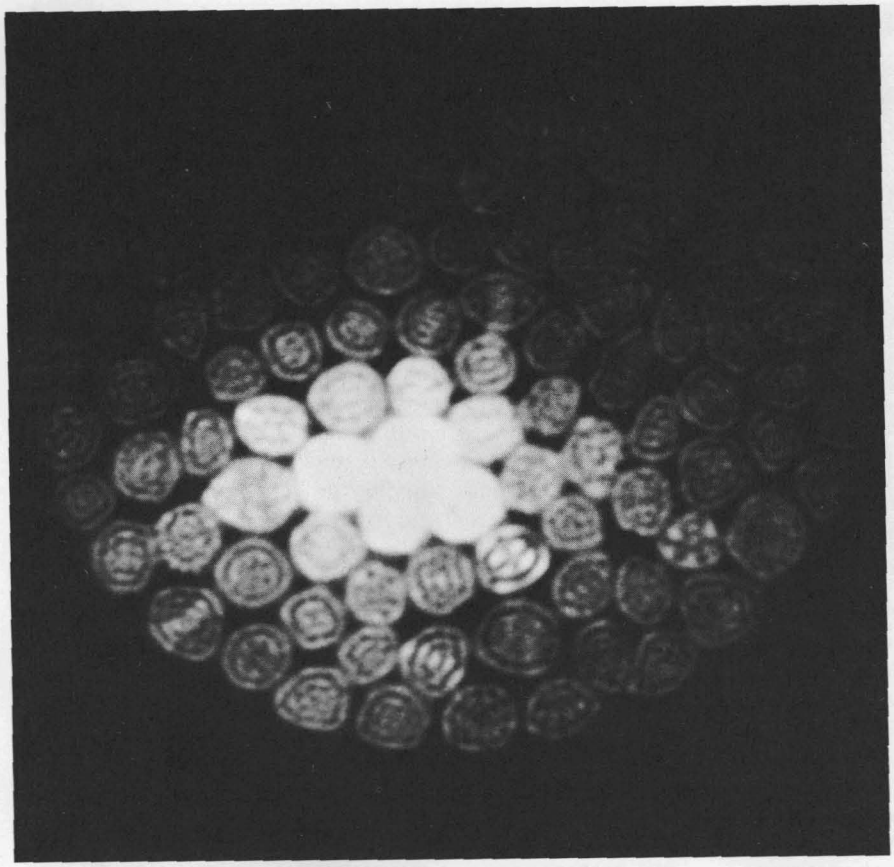


Fig. 5.63 Optical waveguide mode patterns seen in the end faces of small-diameter fibers. [Photo courtesy of Narinder S. Kapany.]

eye of an insect like the housefly is effectively a bundle of tapered fiber optical filaments. The rods and cones that make up the human retina may also channel light via total internal reflection. Another common application of mosaics involving imaging is the *field flattener*. If the image formed by a lens system resides on a curved surface, it is often desirable to reshape it into a plane so as, e.g. to match a film plate. A mosaic can be ground and polished on one of its end surfaces to correspond to the contour of the image and on the other to match the detector. Incidentally, there is a naturally occurring fibrous crystal known as *ulexite* which when polished responds surprisingly like a fiber optical mosaic. (Hobby shops often sell it for use in making jewelry.)

If you have never seen the kind of light conduction we've been talking about, try looking down the edges of a stack of microscope slides. Even better are the much thinner (0.18 mm) cover glass slides. Figure 5.68 shows the way light is conveyed to the upper surface of a stack of a few hundred of these held together by a rubber band.

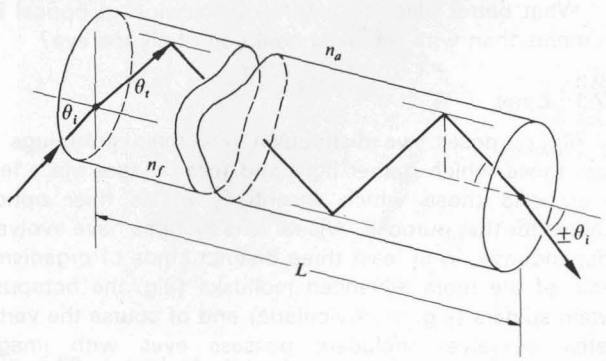


Fig. 5.64 Rays reflected within a dielectric cylinder.

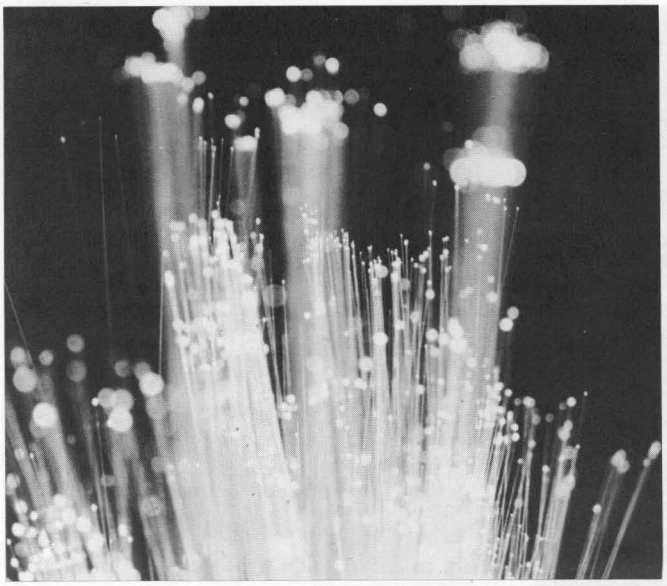


Fig. 5.65 Light emerging from the ends of a loose bundle of glass fibers. [Photo by E.H.]

5.7 OPTICAL SYSTEMS

We have thus far developed paraxial theory to a point where it is now possible to appreciate the underlying principles operative in the majority of practical optical systems. To be sure, the subtleties involved in controlling aberrations are extremely important and still quite beyond this discussion. Even so, for example, one could build a telescope (admittedly not a very good one, but a telescope nonetheless) using the conclusions already drawn from first-order theory.

What better place to begin a discussion of optical instruments than with the most common of all, the eye?

5.7.1 Eyes

For our purposes, we distinguish two main groupings of eyes; those which gather light and form images via a lens system and those which essentially utilize fiber optical bundles for that purpose. Visual lens systems have evolved, independently, in at least three distinct kinds of organisms. Some of the more advanced mollusks (e.g. the octopus), certain spiders (e.g. the Avicularia) and of course the vertebrates, ourselves included, possess eyes with image-forming lenses.

The manner in which these eyes function was, for centuries, a point of great misconception until in 1625 the German Jesuit Christoph Scheiner (1575–1650) performed a classic and irrefutible experiment. He removed the coating on the back of an animal's eye, and peering through its transparent retina from behind, was able to perceive a small inverted image of the scene beyond the eye. At just about the same time Descartes performed similar experiments.

i) Structure of the Human Eye

In essence the human eye can be thought of as a positive lens system which casts a real image on a light-sensitive surface. Figure 5.69(a) shows the arrangement of its basic components. The eye is an almost spherical jelly-like mass contained within a tough shell, the *sclera*. Except for the front portion or *cornea* which is transparent, the sclera is white and opaque. Bulging upward from the body of the sphere, the cornea's curved surface (which is slightly flattened, thereby cutting down on spherical aberration) serves as the first and strongest convex element of the lens system. Indeed most of the bending imparted to a bundle of rays takes place at the air–cornea interface. Incidentally, one of the reasons you can't see very well under water ($n_W \approx 1.33$) is that its index is too close to that of the cornea ($n_C \approx 1.376$) to allow for adequate refraction. Light emerging from the cornea passes through a chamber filled with a clear watery fluid called the *aqueous humor* ($n_{ah} \approx 1.336$). A ray which was strongly refracted toward the optical axis at the air–cornea interface will only be slightly redirected at the cornea–aqueous humor interface because of the similarity of their indices. Immersed in the aqueous is a diaphragm known as the *iris* which serves as the aperture stop controlling the amount of light entering the eye by way of the hole

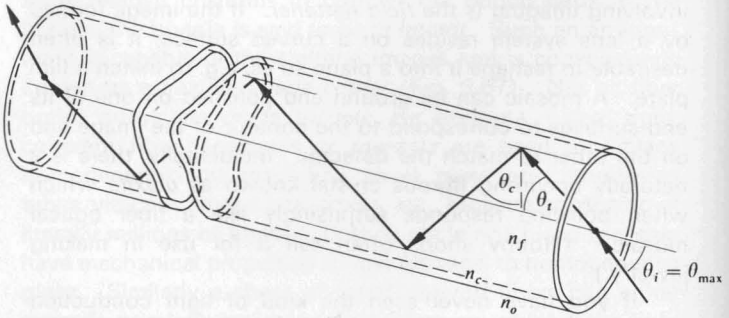


Fig. 5.66 Rays in a clad optical fiber.

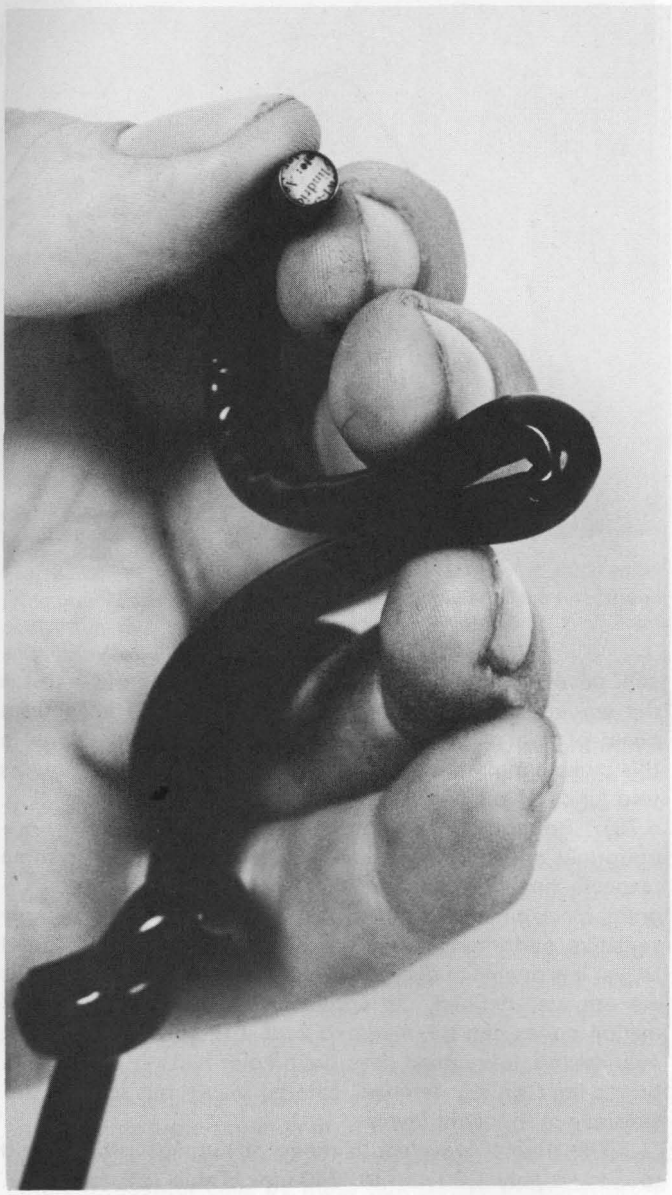


Fig. 5.67 A coherent bundle of 10 μ glass fibers transmitting an image even though knotted and sharply bent. [Photo courtesy of American Cystoscope Makers, Inc.]

or *pupil*. It is the iris (from the Greek word for rainbow) which gives the eye its characteristic blue, brown, gray, green, or hazel color. Made up of circular and radial muscles, the iris can expand or contract the pupil over a range from about 2 mm in bright light, to roughly 8 mm in darkness. In addition to this function, it is also linked to the focusing response and will contract to increase image sharpness when doing close work. Immediately behind the iris is the *crystalline lens*. The name, which is somewhat misleading, dates back to about 1000 A.D. and the work of Abû 'Alî al Hasan ibn al Hasan ibn al Haitham alias Alhazen of Cairo who described the eye as partitioned into three regions which were watery, crystalline and glassy, respectively. The lens, which has both the size and shape of a small bean, is a complex layered fibrous mass surrounded by an elastic membrane. In structure it is somewhat like a transparent onion, formed of roughly 22,000 very fine layers. It has some remarkable characteristics which distinguish it from man-made lenses in use today, in addition to the fact that it continues to grow in size. Because of its laminar structure, rays traversing it will follow paths made up of minute, discontinuous segments. The lens as a whole is quite pliable, albeit less so with age. Moreover, its index of refraction varies from about 1.406 at the inner core to approximately 1.386 at the less dense cortex. (Glass lenses with inhomogeneous indices are now receiving serious study.) The crystalline lens provides the needed fine-focusing mechanism through changes in its shape, i.e. it has a variable focal length—a feature we'll come back to presently.

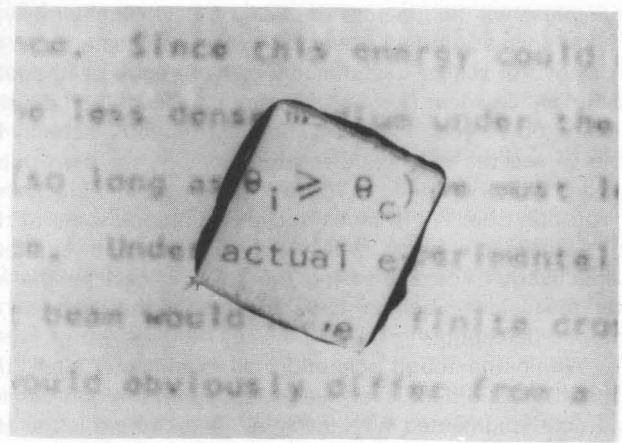
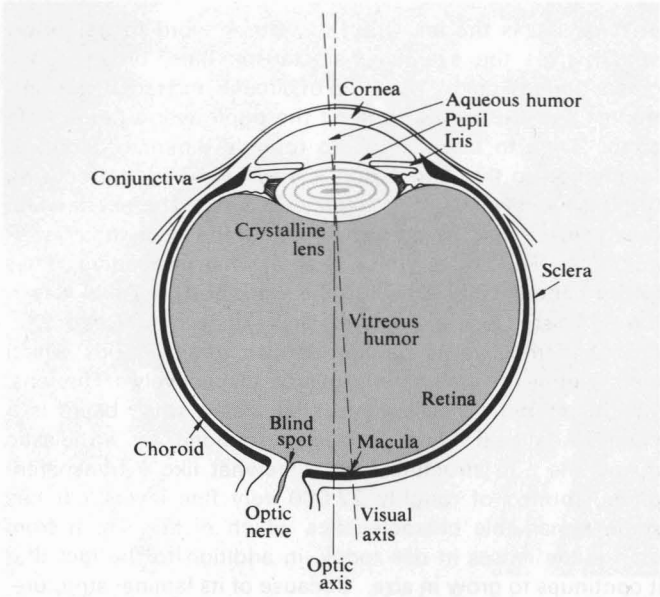


Fig. 5.68 A stack of cover glass slides held together by a rubber band serves as a coherent light guide. [Photo by E.H.]



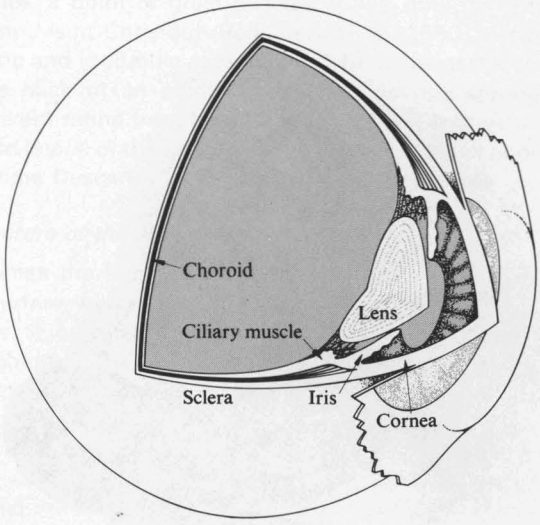


Fig. 5.69 The human eye.

Behind the lens is another chamber filled with a transparent gelatinous substance known as the *vitreous humor* ($n_{vh} \approx 1.337$). As an aside, it should be noted that the vitreous humor contains microscopic particles of cellular debris floating freely about. Their shadows, outlined with diffraction fringes, are easily seen within your own eye by just squinting at a light source or looking at the sky through a pinhole—strange little amoeba-like objects (*muscae volitantes*) will float across the field of view. Incidentally, a marked increase in one's perception of these floaters may be indicative of retinal detachment. While you're at it, squint at the source again (a broad diffuse fluorescent works well). Closing down your lids almost completely, you'll actually be able to see the near circular periphery of your own pupil, beyond which the glare of light will disappear into blackness. If you don't believe it, block and then unblock some of the light; the glare circle will visibly expand and contract, respectively. You are seeing the shadow cast by the iris from the inside! Seeing internal objects like this is known as entoptic perception.

Within the tough sclerotic wall is an inner shell, the choroid It is a dark layer, well supplied with blood vessels and richly pigmented with melanin. The choroid is the absorber of stray light, as is the coat of black paint on the inside of a camera. A thin (about 0.02 in) layer of light receptor cells covers much of the inner surface of the choroid—this is the *retina* (from the Latin *rete* meaning net). The focused beam of light is absorbed via electrochemical reactions in this pinkish multilayered structure. The human eye contains two kinds of photoreceptor cells, the *rods* and *cones* (Fig. 5.70). Roughly 125 million of them are intermingled non-uniformly over the retina. The ensemble of rods in some respects has the characteristics of a high-speed, coarse-grain, black and white film (such as Tri-X). It is exceedingly sensitive, performing in light too dim for the cones to respond to, yet it is unable to distinguish color and the images it relays are not well defined. In contrast, the ensemble of 6 or 7 million cones can be imagined as if it were a separate, but overlapping, low-speed, fine-grain color film. It performs in bright light giving detailed, colored views but is fairly insensitive at low light levels.

The normal wavelength range of human vision is said to be roughly 390 nm to 780 nm (Table 3.2, p. 56). Despite this, studies have extended these limits down to about 310 nm in the ultraviolet and up to roughly 1050 nm in the infrared—indeed people have reported "seeing" X-radiation. The limitation on ultraviolet transmission in the eye is set by the crystalline lens which absorbs in the UV. People who have had a lens removed surgically have greatly improved UV sensitivity.

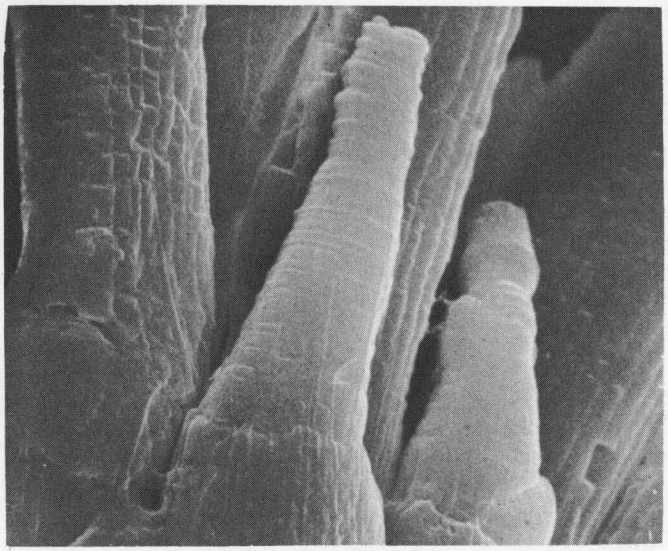


Fig. 5.70 An electron micrograph of the retina of a salamander (Necturus Maculosus). Two visual cones appear in the foreground and several rods behind them. [Photo from E. R. Lewis, Y. Y. Zeevi and F. S. Werblin, *Brain Research* **15**, 559 (1969).]

The point of exit of the optic nerve from the eye contains no receptors and that area is insensitive to light; accordingly it is known as the *blind spot* (see Fig. 5.71). The optic nerve spreads out over the back of the interior of the eye in the form of the retina.

Just about at the center of the retina is a small depression from 2.5 to 3 mm in diameter known as the yellow spot or *macula*. There is a tiny rod-free region about 0.3 mm in diameter at its center, the *fovea centralis*. (In comparison the image of the full moon on the retina is about 0.2 mm in diameter—Problem 5.40.) Here the cones are thinner and more densely packed than anywhere else in the retina. That region provides the sharpest and most detailed information. For that reason, the eyeball is continuously moving so that light coming from the area on the object of primary interest falls on the fovea. An image is constantly shifted across different receptor cells by these normal eye movements. If

X 1 2

Fig. 5.71 To verify the existence of the blind spot, close one eye and, at about a distance of 10 inches, look directly at the X—the 2 will disappear. Moving closer will cause the 2 to reappear while the 1 vanishes.

such movements are negated so that the image is kept stationary on a given set of photoreceptors, it would actually tend to fade out. Another fact which indicates the complexity of the sensing system is that the rods are multiply connected to nerve fibers and a single such fiber can be activated by any one of about a hundred rods. By contrast cones in the fovea are individually connected to nerve fibers. The actual perception of a scene is constructed by the eye–brain system in a continuous analysis of the time-varying retinal image. Just think how little trouble the blind spot causes even with one eye closed.

Between the nerve-fiber layer of the retina and the humor is a network of large retinal blood vessels which can be observed entoptically. One way is to close your eye and place a bright *small* source against the lid. You'll "see" a pattern of shadows (*Purkinje Figures*) cast by the blood vessels on the sensitive retinal layer.

ii) Accommodation

The fine focusing or *accommodation* of the human eye is a function performed by the crystalline lens. The lens is suspended in position behind the iris by ligaments which are connected to the *ciliary muscles*. Ordinarily, these muscles are relaxed and in that state they pull back on the network of fine fibers holding the rim of the lens. This draws the pliable lens into a fairly flat configuration increasing its radii which, in turn, increases its focal length (5.16). With the muscles completely relaxed, the light from an object at infinity will be focused on the retina (Fig. 5.72). As the object moves closer to the eye, the ciliary muscles contract, relieving the external tension on the periphery of the lens which then bulges slightly under its own elastic forces. In so doing, the focal length decreases such that s_i is kept constant. As the object comes still closer, the ciliary muscles become more tensely contracted and the lens surfaces take on even smaller radii. The closest point on which the eye can focus is known as the *near point*. In a normal eye it might be about 7 cm for a teenager, 25 cm or so for a young adult, and roughly 100 cm in the middle aged. Visual instruments are designed with this in mind so that the eye need not strain unnecessarily. Clearly, the eye cannot focus on two different objects at once. This will be made obvious if, while looking through a piece of glass, you try to focus on it and the scene beyond at the same time.

Mammals, like man, generally accommodate by varying the lens curvature, but there are other means. Fish, in contrast, move only the lens itself toward or away from the retina, just as the camera lens is moved to focus. Some mollusks

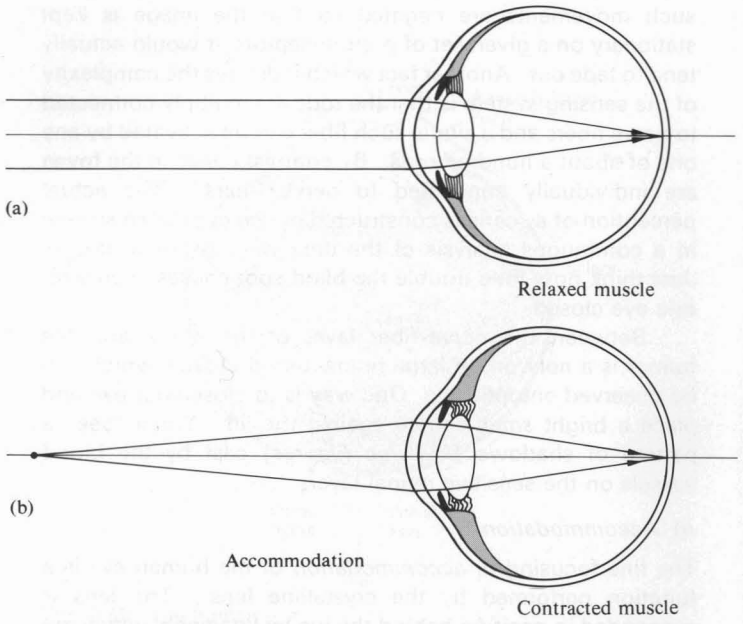


Fig. 5.72 Accommodation—changes in the lens configuration.

accomplish the same thing by contracting or expanding the whole eye, thus altering the relative distance between lens and retina. For birds of prey, where keeping a rapidly moving object in constant focus over a wide range of distances is a matter of survival, the accommodation mechanism is quite different. They accommodate by greatly changing the curvature of the cornea.

5.7.2 Eyeglasses

Spectacles were probably invented some time in the 13th century, in northern Italy. And polished gem stones were no doubt employed as lorgnettes long before that.

It is customary and quite convenient in physiological optics to speak about the *dioptric power* $\mathscr{D}$ *of a lens, which is simply the reciprocal of the focal length.* When f is in meters, the unit of power is the inverse meter or *diopter*. For example, if a converging lens has a focal length of $+1$ meter, its power is $+1$ diopter; with a focal length of -2 meters (a diverging lens) $\mathscr{D} = -\frac{1}{2}$ diopter; for $f = +10$ cm, $\mathscr{D} = 10$ diopters. Since a thin lens of index n_l in air has a focal length given by

$$\frac{1}{f} = (n_l - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right), \qquad [5.16]$$

its power is

$$\mathscr{D} = (n_l - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right). \qquad (5.62)$$

You can get a sense of the direction in which we are moving by considering, in rather loose terms, that each surface of a lens bends the incoming rays—the more the bending, the stronger the surface. A convex lens which strongly bends the rays at both surfaces has a short focal length and a large dioptric power. We already know that the focal length for two thin lenses in contact is given by

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}. \qquad [5.38]$$

This means that the combined power is the sum of the individual powers, that is

$$\mathscr{D} = \mathscr{D}_1 + \mathscr{D}_2.$$

Thus a convex lens with $\mathscr{D}_1 = +10$ diopters in contact with a negative lens of $\mathscr{D}_2 = -10$ diopters results in $\mathscr{D} = 0$; the combination behaves like a parallel sheet of glass. Furthermore, we can imagine a lens, e.g. a double convex lens, as being composed of two planar-convex lenses in intimate contact, back to back. The power of each of these follows from Eq. (5.62), thus for the first planar-convex lens ($R_2 = \infty$)

$$\mathscr{D}_1 = \frac{(n_l - 1)}{R_1}, \qquad (5.63)$$

while for the second

$$\mathscr{D}_2 = \frac{(n_l - 1)}{-R_2}. \qquad (5.64)$$

These expressions may equally well be defined as giving the *powers of the respective surfaces* of the initial double convex lens. In other words, *the power of any thin lens is equal to the sum of the powers of its surfaces.* Because R_2 for a convex lens is a negative number, both $\mathscr{D}_1$ and $\mathscr{D}_2$ will be positive in that case. The power of a *surface*, defined in this way, is not generally the reciprocal of its focal length, although it is when immersed in air. Relating this terminology to the generally used model for the human eye we note that the power of the crystalline lens *surrounded by air* is about $+19$ diopters. The cornea provides roughly $+43$ of the total $+58.6$ diopters of the intact eye.

A normal eye, despite the connotation of the word, is not really as common an occurrence as one might expect. By the term normal, or its synonym *emmetropic*, we mean an

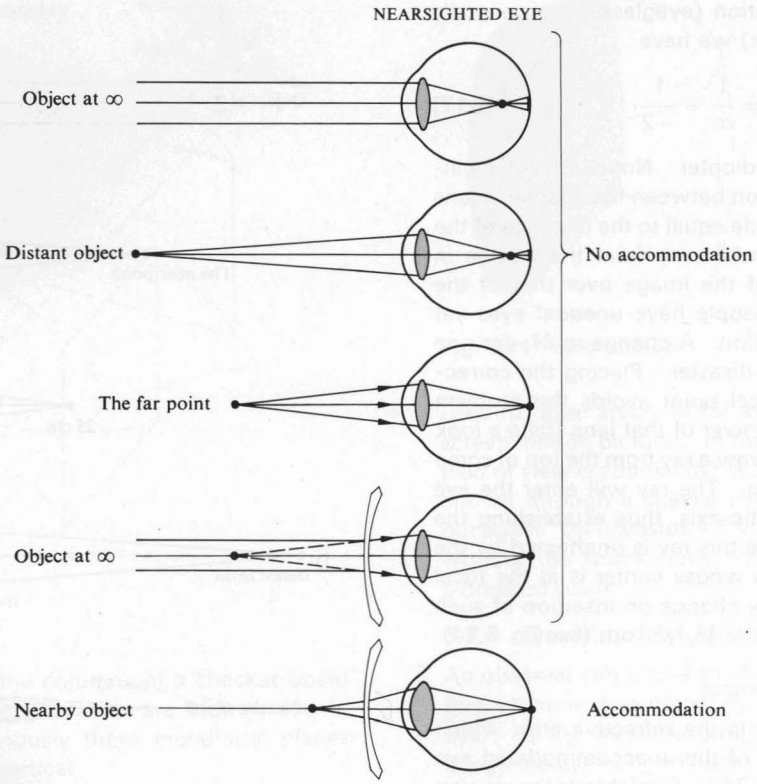


Fig. 5.73 Correction of the nearsighted eye.

eye which is capable of focusing parallel rays on the retina while in a relaxed condition, i.e. one whose second focal point lies on the retina. The most distant point which can be brought to focus, the *far point*, is therefore located at infinity. In contrast, when the second focal point does not lie on the retina the eye is *ametropic* (it suffers e.g. hyperopia, myopia or astigmatism).

i) Nearsightedness—Negative Lenses

Myopia is the condition where parallel rays are brought to focus in front of the retina; the power of the lens system as configured is too large for the anterior–posterior axial length of the eye. This can happen in a number of ways, e.g. the eye may elongate even though its power remains normal; the cornea may experience an increase in curvature; there is also a form which results from abnormal refractivity of the eye's optical media. In any event, images of distant objects fall in front of the retina, the far point is closer in than infinity, and all points beyond it will appear blurred. This is why myopia is often called *nearsightedness*—an eye with this defect sees nearby objects clearly (Fig. 5.73). To correct the condition, or at least its symptoms, we place an additional lens in front of the eye such that the combined spectacle–eye lens system has its second focal point on the retina. Since the myopic eye can clearly see objects closer than the far point, the spectacle lens must cast relatively nearby images of distant objects. Hence we introduce a negative lens which will diverge the rays a bit. Resist the temptation to suppose that we are merely reducing the power of the system. In point of fact, the power of the lens–eye combination is most often made to equal that of the unaided eye. If you are wearing glasses to correct myopia take them off; the world gets blurry but it doesn't change size. Try casting a real image on a piece of paper using your glasses—it can't be done.

Suppose an eye has a far point of 2 m, all would be well if the spectacle lens were to appear to bring more distant objects in closer than 2 m. If the virtual image of an object at infinity is formed by a concave lens at 2 m, the eye will see the object clearly with an unaccommodated lens. Thus

using the thin-lens approximation (eyeglasses are generally thin to reduce weight and bulk) we have

$$\frac{1}{f} = \frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{\infty} + \frac{1}{-2}, \qquad [5.17]$$

and $f = -2$ m while $\mathscr{D} = -\frac{1}{2}$ diopter. Notice that this calculation overlooks the separation between the spectacle lens and the eye. This is usually made equal to the distance of the first focal point of the eye ($\approx$16 mm) from the cornea in order that no magnification of the image over that of the unaided eye occurs. Many people have unequal eyes yet both yield the same magnification. A change in M_T for one and not the other would be a disaster. Placing the correcting lens at the eye's first focal point avoids the problem completely, regardless of the power of that lens [take a look at Eq. (6.8)]. To see this, just draw a ray from the top of some object through that focal point. The ray will enter the eye traversing it parallel to the optic axis, thus establishing the height of the image. Yet, since this ray is unaffected by the presence of the spectacle lens whose center is at the focal point, the image's location may change on insertion of such a lens, but its height and therefore M_T will not (see Eq. 5.24).

ii) Farsightedness—Positive Lenses

Hyperopia (or *hypermetropia*) is the refractive error which causes the second focal point of the unaccommodated eye to lie behind the retina (Fig. 5.74). *Farsightedness*, as you might have guessed it would be called, is often due to a shortening of the anteroposterior axis of the eye—the lens is too close to the retina. To increase the bending of the rays, a positive spectacle lens is placed in front of the eye. The hyperopic eye can and must accommodate to see distant objects distinctly but it will be at its limit to do so for a near point which is much farther away than it would be normally (this we take as 25 cm). It will consequently be unable to see clearly. A converging corrective lens with positive power will effectively move a close object out beyond the near point where the eye has adequate acuity, i.e. it will form a distant virtual image which the eye can then see clearly. Suppose that a hyperopic eye has a near point of 125 cm. For an object at +25 cm to have its image at $s_i = -125$ cm, so that it can be seen as if through a normal eye, the focal length must be

$$\frac{1}{f} = \frac{1}{(-1.25)} + \frac{1}{0.25} = \frac{1}{0.31}$$

or $f = 0.31$ m and $\mathscr{D} = +3.2$ diopters. This is in accord with

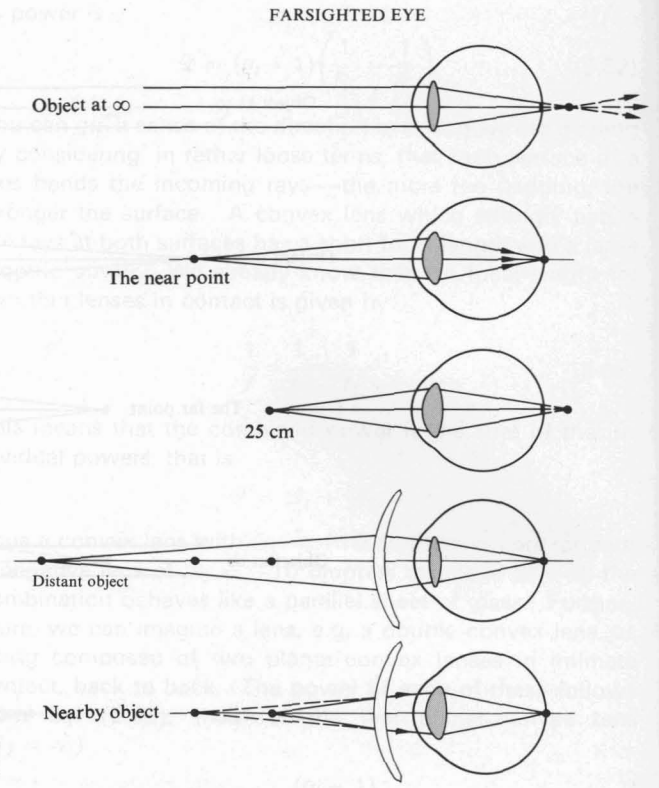


Fig. 5.74 Correction of the farsighted eye.

Table 5.3 where $s_o < f$. These spectacles will cast real images—try it if you're hyperopic.

Very gentle finger pressure on the lids above and below the cornea will temporarily distort it changing your vision from blurred to clear and vice versa.

iii) Astigmatism—Anamorphic Lenses

Another and perhaps the most common eye defect is *astigmatism*. It arises from an uneven curvature of the cornea. In other words, the cornea is asymmetric. Suppose we passed two meridional planes (ones containing the optic axis) through the eye such that the (curvature or) power is maximal on one and minimal on the other. If these planes are perpendicular, the *astigmatism* is *regular* and correctible; if not, it is *irregular* and not easily corrected. Regular astigmatism can take different forms where the eye is emmetropic, myopic or hyperopic in various combinations and degrees on the two perpendicular meridional planes.

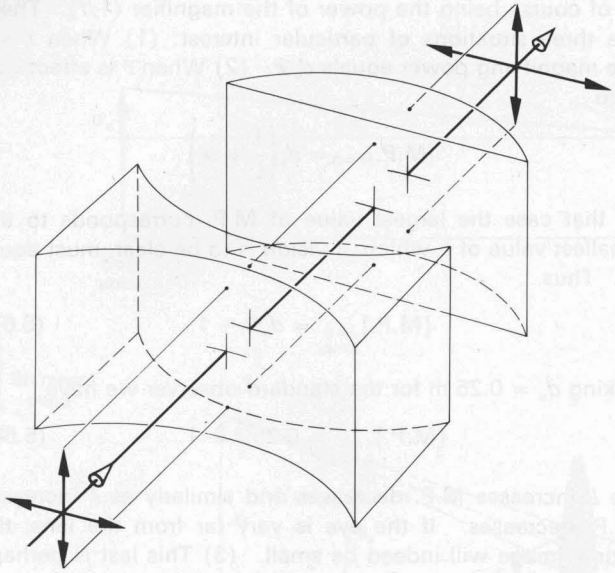


Fig. 5.75 An anamorphic system.

Thus, as a simple example, the columns of a checker board might be well focused while the rows are blurred due to myopia or hyperopia. Obviously these meridional planes need not be horizontal and vertical.

The great astronomer Sir George B. Airy used a concave sphero-cylindrical lens to ameliorate his own myopic astigmatism in 1825. This was probably the first time astigmatism had been corrected. But it was not until the publication in 1862 of a treatise on cylindrical lenses and astigmatism by the Dutchman Franciscus Cornelius Donders (1818–89) that opthalmologists were moved to adopt the method on a large scale.

Any optical system which has a different value of M_T or $\mathscr{D}$ in two principal meridians is said to be *anamorphic*. Thus, for example, if we rebuilt the system depicted in Fig. 5.28, this time using cylindrical lenses (Fig. 5.75), the image would be distorted, having been magnified in only one plane. This is just the sort of distortion needed to correct for astigmatism when a defect exists in only one meridian. An appropriate planar cylindrical spectacle lens, either positive or negative, would restore essentially normal vision. When both perpendicular meridians require correction, the lens may, for example, be *sphero-cylindrical*, or even *toric* as in Fig. (5.76).

Just as an aside, we mention that anamorphic lenses are

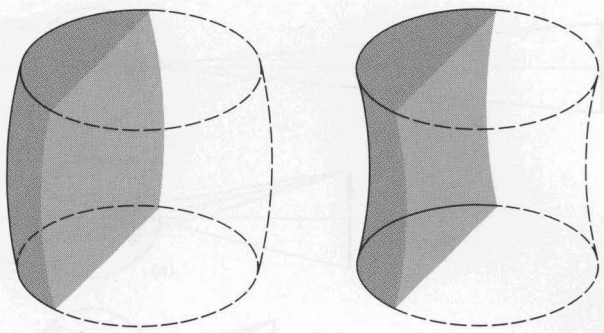


Fig. 5.76 Toric surfaces.

used in other areas, as for example in the making of wide-screen motion pictures. In that way, an extra-large horizontal field of view is compacted onto the regular film format. When shown through a special lens the distorted picture spreads out again. On occasion a T.V. station will show short excerpts without the special lens—you may have seen the weirdly elongated result.

5.7.3 The Magnifying Glass

An observer can cause an object to appear larger, and therefore examine it in detail, by simply bringing it closer to her eye. As the object is brought nearer and nearer its retinal image increases, remaining in focus until the crystalline lens can no longer provide adequate accommodation. Should the object come closer than this *near point*, the image will blur (Fig. 5.77). A single positive lens can be used, in effect to add refractive power to the eye, so that the object can be brought still closer and yet be in focus. The lens so used is referred to variously as a *magnifying glass*, a *simple magnifier* or a *simple microscope*. In any event, its function is *to provide an image of a nearby object which is larger than that seen by the unaided eye*. Devices of this sort have been around for a long time. In fact, a quartz convex lens ($f \approx 10$ cm), which may have served as a magnifier, was unearthed in 1885 amongst the ruins of the palace of King Sennacherib (705–681 B.C.) of Assyria.

Evidently, it would be desirable for the lens to form a magnified, erect image. Furthermore, the rays entering the normal eye should not be converging. Table 5.3 (p. 112) immediately suggests placing the object within the focal length, i.e. $s_o < f$. The result is shown in Fig. 5.78. Because of the relatively tiny size of the eye's pupil, it will almost certainly always be the aperture stop, and as in Fig. 5.30 (p. 117), it will also be the exit pupil.

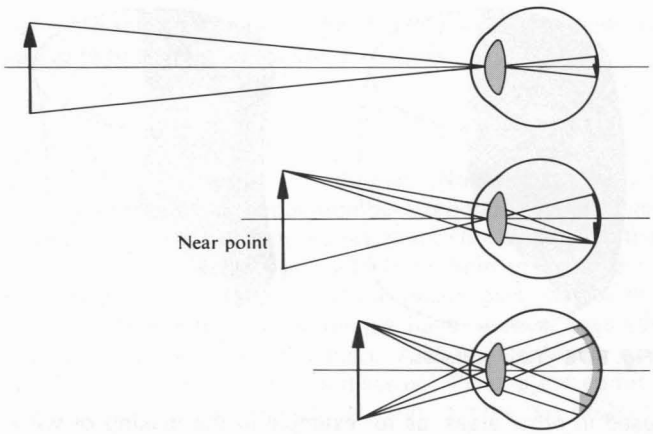


Fig. 5.77 Images in relation to the near point.

The *magnifying power, M.P.*, or equivalently the *angular magnification*, M_A, of a visual instrument is defined as *the ratio of the size of the retinal image as seen through the instrument* over *the size of the retinal image as seen by the unaided eye at normal viewing distance*. The latter is generally taken as the distance to the near point, d_o. The ratio of angles α_a and α_u (which are made by chief rays from the top of the object in the instance of the aided and unaided eye, respectively) is equivalent to M.P., that is

$$\text{M.P.} = \frac{\alpha_a}{\alpha_u}. \tag{5.65}$$

Keeping in mind that we are restricted to the paraxial region $\tan \alpha_a = y_i/L \approx \alpha_a$ and $\tan \alpha_u = y_o/d_o \approx \alpha_u$ and so

$$\text{M.P.} = \frac{y_i d_o}{y_o L},$$

wherein y_i and y_o are above axis and positive. Making d_o and L positive quantities will yield a positive M.P. which is quite reasonable. Using Eqs. (5.24) and (5.25) for M_T along with the thin-lens equation, the expression becomes

$$\text{M.P.} = -\frac{s_i d_o}{s_o L} = \left(1 - \frac{s_i}{f}\right)\frac{d_o}{L}.$$

Inasmuch as the image distance is negative $s_i = -(L - \ell)$ and consequently

$$\text{M.P.} = \frac{d_o}{L}[1 + \mathscr{D}(L - \ell)], \tag{5.66}$$

$\mathscr{D}$ of course being the power of the magnifier $(1/f)$. There are three situations of particular interest: (1) When $\ell = f$ the magnifying power equals $d_o\mathscr{D}$. (2) When ℓ is effectively zero

$$[\text{M.P.}]_{\ell=0} = d_o\left(\frac{1}{L} + \mathscr{D}\right).$$

In that case the largest value of M.P. corresponds to the smallest value of L which, if vision is to be clear, must equal d_o. Thus

$$[\text{M.P.}]_{\substack{\ell=0 \\ L=d_o}} = d_o\mathscr{D} + 1. \tag{5.67}$$

Taking $d_o = 0.25$ m for the standard observer we have

$$[\text{M.P.}]_{\substack{\ell=0 \\ L=d_o}} = 0.25\mathscr{D} + 1. \tag{5.68}$$

As L increases M.P. decreases and similarly as ℓ increases M.P. decreases. If the eye is very far from the lens, the retinal image will indeed be small. (3) This last is perhaps the most common situation. Here we position the object at the focal point ($s_o = f$) in which case the virtual image is at infinity ($L = \infty$). Thus from Eq. (5.66)

$$[\text{M.P.}]_{L=\infty} = d_o\mathscr{D} \tag{5.69}$$

for all practical values of ℓ. Because the rays are parallel, the eye views the scene in a relaxed, unaccommodated configuration, a highly desirable feature. Notice that $M_T = -s_i/s_o$ approaches infinity as $s_o \to f$, while in marked contrast M_A merely decreases by 1 under the same circumstances.

A magnifier with a power of 10 diopters has a focal length $(1/\mathscr{D})$ of 0.1 m and a M.P. equal to 2.5 when $L = \infty$. This is conventionally denoted as 2.5× which means that the retinal image is 2.5 times larger with the object at the focal length of the lens than it would be were the object at the near point of the unaided eye (where the largest clear image is possible). The simplest single-lens magnifiers are limited by aberrations to roughly 2× or 3×. A large field of view generally implies a large lens and that, for practical reasons, most often dictates fairly small curvature on the surfaces. The radii are large, as is f, and therefore M.P. is small. The reading glass, the kind Sherlock Holmes made famous, is a typical example. The watchmaker's eye loupe is frequently a single element lens, also of about 2× or 3×. Figure 5.79 shows a few more complicated magnifiers designed to operate in the range from roughly 10× to 20×. The double lens is quite common in a number of configurations. Although not particularly good, they perform satis-

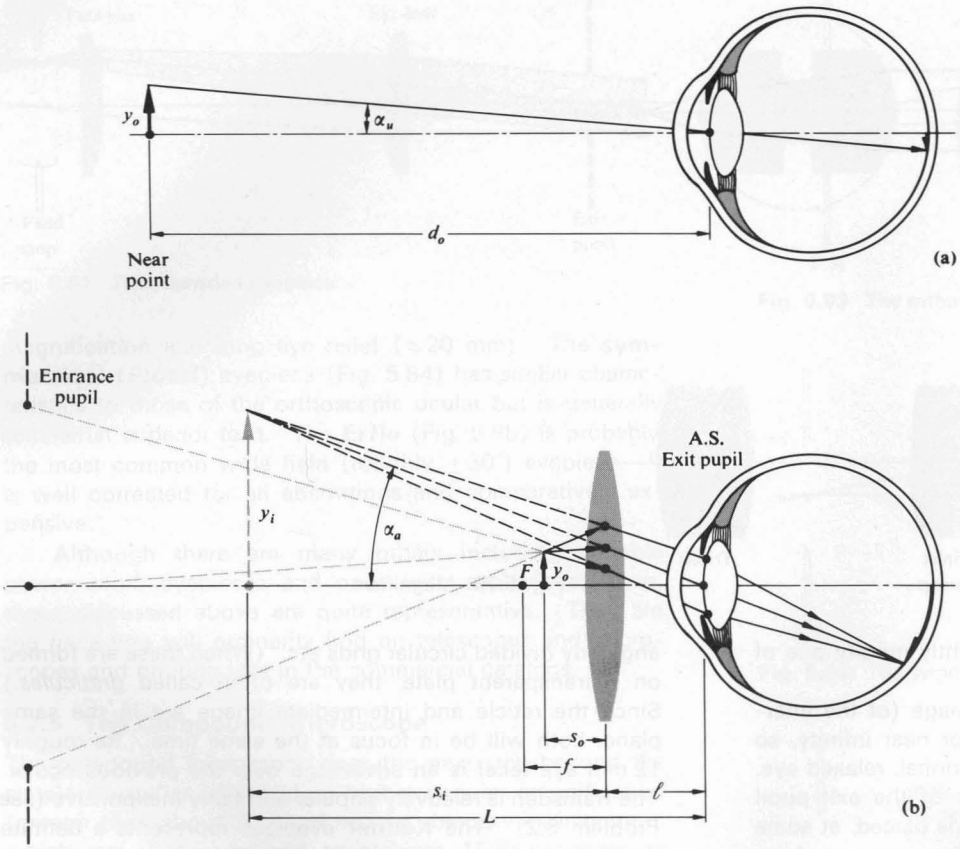


Fig. 5.78 A magnifying glass.

factorily, as for example, in high-powered loupes. The Coddington is essentially a sphere with a slot cut in it to allow an aperture smaller than the pupil of the eye. A clear marble (any small sphere of glass qualifies) will also greatly magnify—but not without a good deal of distortion.

The relative refractive index of a lens and the medium in which it is immersed, n_{lm}, is wavelength dependent. But since the focal length of a simple lens varies with $n_{lm}(\lambda)$, this means that f is a function of wavelength and the constituent colors of white light will focus at different points in space. The resultant defect is known as *chromatic aberration*. In order that the image be free of this coloration, positive and negative lenses made of different glasses are combined to form *achromates* (see Section 6.3.2). Achromatic, cemented, doublet and triplet lenses are comparatively expensive and are usually found in small, highly corrected, high-powered magnifiers.

5.7.4 Eyepieces

The *eyepiece* or *ocular* is a visual optical instrument. Fundamentally a magnifier, its function is to view not an actual object, but the intermediate image of that object as formed by a preceding lens system. In effect, the eye looks into the ocular and the ocular looks into the optical system—be it a spotting scope, compound microscope, telescope or binocular. A single lens could serve the purpose, but poorly. If the retinal image is to be more satisfactory, the ocular cannot generally suffer extensive aberrations. The eyepiece of a special instrument, however, might be designed as part of the complete system so that its lenses can be utilized in the overall scheme to balance out aberrations. Even so, standard eyepieces are used interchangeably on most telescopes and compound microscopes. Moreover, eyepieces are quite difficult to design and the usual, and perhaps most

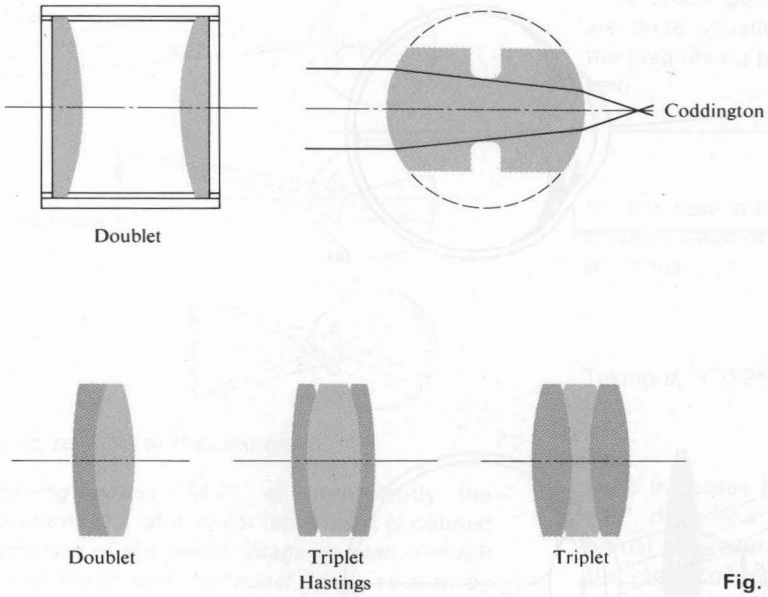


Fig. 5.79 Magnifiers.

fruitful approach is to incorporate or slightly modify one of the existing designs.

The ocular must provide a virtual image (of the intermediate image), most often located at or near infinity, so that it can be comfortably viewed by a normal, relaxed eye. Furthermore, it must position the center of the exit pupil or *eye-point* at which the observer's eye is placed, at some convenient location preferably at least 10 mm or so from the last surface. As before, ocular magnification is the product $d_o\mathscr{D}$ or as is often written M.P. = (250 mm)/f.

The **Huygens** ocular, which dates back over 250 years, is still in wide use today (Fig. 5.80) particularly in microscopy. The lens adjacent to the eye is known as the *eye-lens* while the first lens in the ocular is the *field-lens*. The distance from the eye-lens to the eye-point is known as the *eye relief* and for the Huygens ocular, it's only an uncomfortable 3 mm or so. Notice that this ocular requires the incoming rays to be converging so as to form a virtual object for the eye-lens. Clearly then, the Huygens eyepiece cannot be used as an ordinary magnifier. Its contemporary appeal rests in its low purchase price [see Section 6.3.2(ii)]. Another old standby is the **Ramsden** eyepiece (Fig. 5.81). This time the principal focus is in front of the field-lens and so the intermediate image will appear there in easy access. This is where you would place a *reticle* (or *reticule*) which might contain a set of cross hairs, precision scales, or angularly divided circular grids etc. (When these are formed on a transparent plate, they are often called *graticules*.) Since the reticle and intermediate image are in the same plane, both will be in focus at the same time. Its roughly 12 mm eye relief is an advantage over the previous ocular. The Ramsden is relatively popular and fairly inexpensive (see Problem 6.2). The **Kellner** eyepiece represents a definite increase in image quality although eye relief is between that of the previous two devices. The Kellner is essentially an achromatized Ramsden (Fig. 5.82). It is most commonly used in moderately wide-field telescopic instruments. The **orthoscopic** eyepiece (Fig. 5.83) has a wide field, high

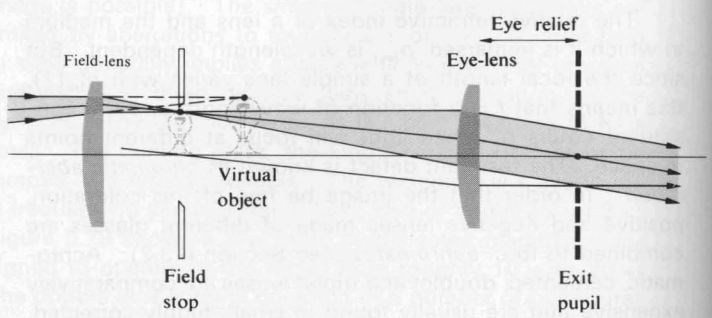


Fig. 5.80 The Huygens eyepiece.

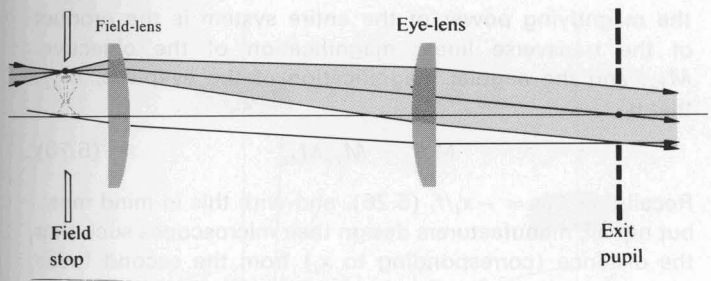


Fig. 5.81 The Ramsden eyepiece.

magnification and long eye relief ($\approx$20 mm). The **symmetrical (Plössl)** eyepiece (Fig. 5.84) has similar characteristics to those of the orthoscopic ocular but is generally somewhat superior to it. The **Erfle** (Fig. 5.85) is probably the most common wide field (roughly $\pm 30°$) eyepiece. It is well corrected for all aberrations and comparatively expensive.*

Although there are many others including variable power *zoom* eyepieces and ones with aspheric surfaces, those discussed above are quite representative. They are the ones you will ordinarily find on telescopes and microscopes and on long lists in the commercial catalogs.

5.7.5 The Compound Microscope

The compound microscope goes the next step beyond the simple magnifier in providing higher angular magnification (greater than about 30×) of *nearby* objects. Its invention, which may have occurred as early as 1590, is generally accredited to a Dutch spectacle-maker Zacharias Janssen of Middleburg. Galileo runs a close second, having announced his invention of a compound microscope in 1610.

* Detailed designs of these and other oculars can be found in the *Military Standardization Handbook—Optical Design*, MIL-HDBK-141.

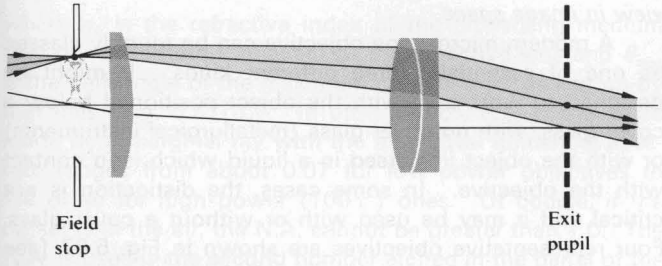


Fig. 5.82 The Kellner eyepiece.

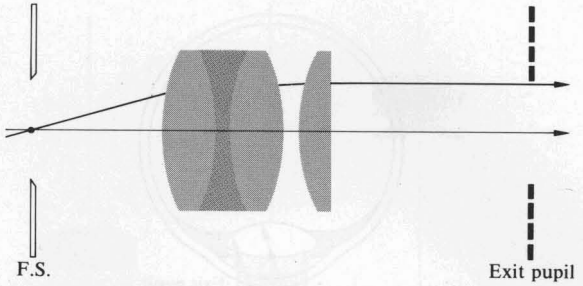


Fig. 5.83 The orthoscopic eyepiece.

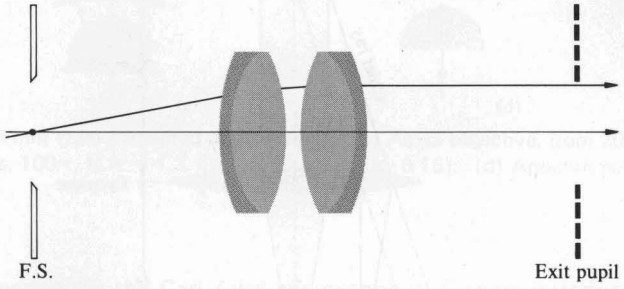


Fig. 5.84 The symmetrical (Plössl) eyepiece.

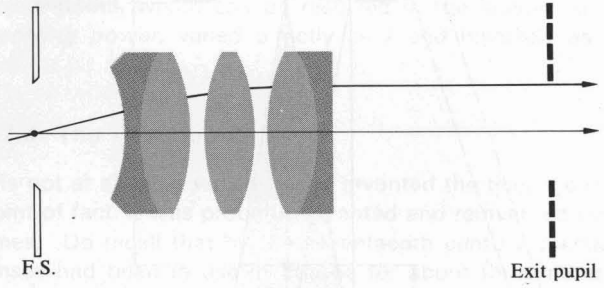


Fig. 5.85 The Erfle eyepiece.

A simple version, which is closer to these earliest devices than it is to a modern laboratory microscope, is depicted in Fig. 5.86. The lens system, here a singlet, closest to the object is referred to as the *objective*. It forms a real, inverted and, usually magnified, image of the object. This image resides in space on the plane of the field stop of the eyepiece. Rays diverging from each point of this image will emerge from the eye-lens (which in this simple case is the eyepiece itself) parallel to each other as in the previous section. The ocular magnifies this intermediate image still further. Thus

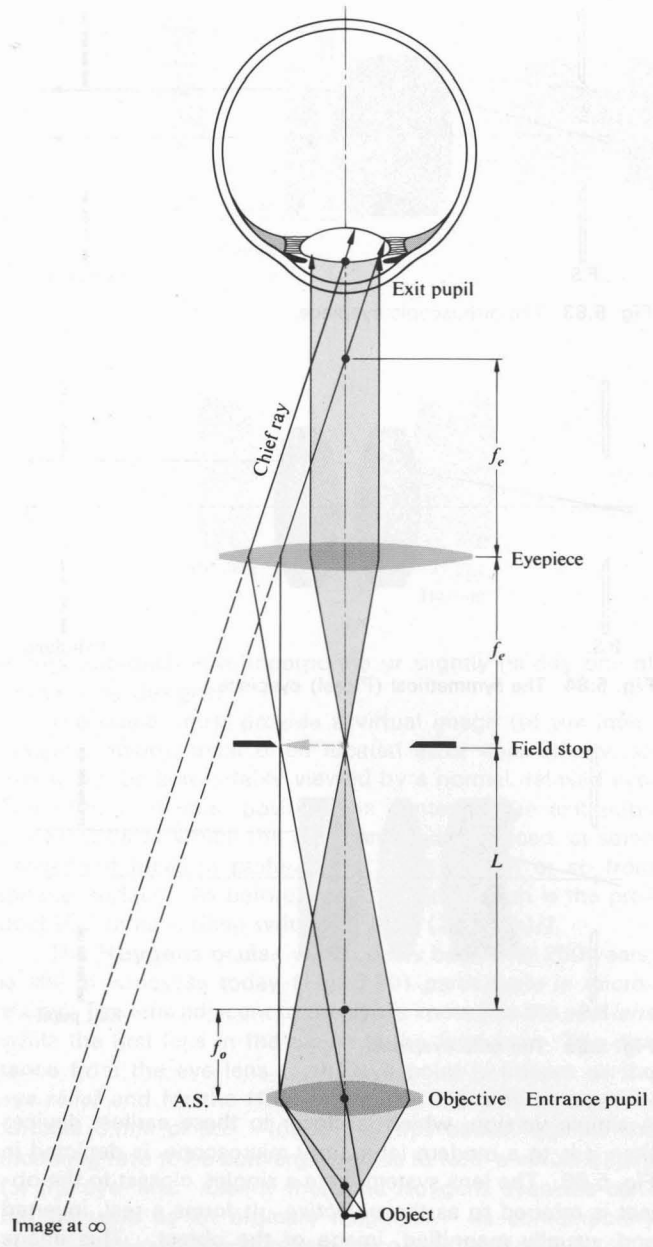


Fig. 5.86 A rudimentary compound microscope.

the magnifying power of the entire system is the product of the transverse linear magnification of the objective, M_{To}, and the angular magnification of the eyepiece, M_{Ae}, that is

$$\text{M.P.} = M_{To}M_{Ae}. \tag{5.70}$$

Recall that $M_T = -x_i/f$, (5.26), and with this in mind most, but not all, manufacturers design their microscopes such that the distance (corresponding to x_i) from the second focus of the objective to the first focus of the eyepiece is standardized at 160 mm. This distance, known as the *tube length*, is denoted by L in the figure. (Some authors define tube length as the image distance of the objective.) Hence, with the final image at infinity and the standard near point taken as 10 inches or 254 mm

$$\text{M.P.} = \left(-\frac{160}{f_o}\right)\left(\frac{254}{f_e}\right) \tag{5.71}$$

and the image is inverted (M.P. < 0). Accordingly, the barrel of an objective with a focal length f_o of say 32 mm will be engraved with the markings $5\times$ (or $\times 5$) indicating a power of 5. Combined with a $10\times$ eyepiece ($f_e = 1$ inch) the microscope M.P. would then be $50\times$.

To maintain the distance relationships between the objective, field stop and ocular, while positioning a focused intermediate image of the object in the first focal plane of the eyepiece, all three elements are moved as a single unit.

The objective itself functions as the aperture stop and entrance pupil. Its image, formed by the eyepiece, is the exit pupil into which the eye is positioned. The field stop, which limits the extent of the largest object that can be viewed is fabricated as part of the ocular. The image of the field stop formed by the optical elements following it is called the *exit window*, and the image formed by the optical elements preceding it is the *entrance window*. The cone angle subtended at the center of the exit pupil by the periphery of the exit window is said to be the *angular field of view in image space*.

A modern microscope objective can be roughly classed as one of essentially three different kinds. It might be designed to work best with the object positioned below a cover glass, with no cover glass (metallurgical instruments) or with the object immersed in a liquid which is in contact with the objective. In some cases, the distinction is not critical and it may be used with or without a cover glass. Four representative objectives are shown in Fig. 5.87 [see Section 6.3.1(i)]. In addition, the ordinary low power

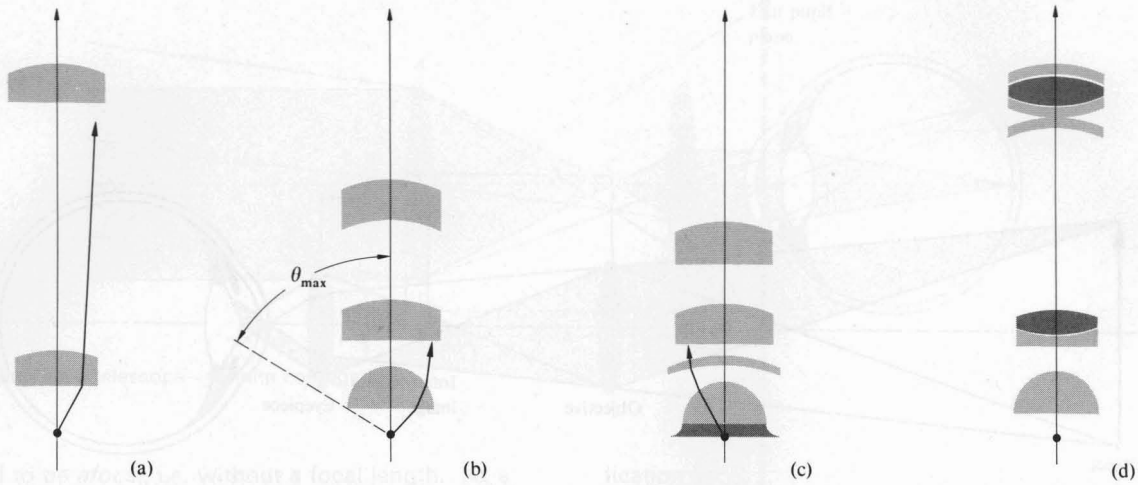


Fig. 5.87 Microscope objectives: (a) Lister objective, 10×, N.A. = 0.25, f = 16 mm (two cemented achromates). (b) Amici objective, from 20×, N.A. = 0.5, f = 8 mm to 40×, N.A. = 0.8, f = 4 mm. (c) Oil-immersion objective, 100×, N.A. = 1.3, f = 1.6 mm (see Fig. 6.16). (d) Apochromatic objective, 55 ×, N.A. = 0.95, f = 3.2 (contains two flourite lenses).

(about 5×) cemented doublet achromate is quite common. [Relatively inexpensive medium power (10× or 20×) achromatic objectives, because of their short focal lengths, can also conveniently be used when expanding and spatially filtering laser beams.]

There is one other characteristic quantity of importance which must be mentioned here even if only briefly. The brightness of the image is, in part, dependent on the amount of light gathered in by the objective. The f-number is a useful parameter for describing this quantity, particularly when the object is a distant one (Section 5.3.3). However, for an instrument working at *finite conjugates* (s_i and s_o both finite), the numerical aperture N.A. is more appropriate (Section 5.6). In the present instance

$$\text{N.A.} = n_o \sin \theta_{max} \tag{5.72}$$

where n_o is the refractive index of the immersing medium (air, oil, water, etc.) adjacent to the objective lens and θ_{max} is the half-angle of the maximum cone of light picked up by that lens [Fig. 5.87(b)]. In other words, θ_{max} is the angle made by a marginal ray with the axis. The numerical aperture ranges from about 0.07 for low-power objectives to 1.4 or so for high-power (100×) ones. Of course, if the object is in the air, the N.A. cannot be greater than 1.0. The N.A. is usually the second number etched in the barrel of the objective. Incidentally, Ernst Abbe (1840–1905), while working in the Carl Zeiss microscope workshop, introduced the concept of the numerical aperture. It was he who recognized that the minimum transverse distance between two object points which can be resolved in the image, i.e. the *resolving power*, varied directly as λ and inversely as the N.A.

5.7.6 The Telescope

It is not at all clear who actually invented the telescope. In point of fact, it was probably invented and reinvented many times. Do recall that by the seventeenth century spectacle lenses had been in use in Europe for about three hundred years. During that long span of time, the fortuitous juxtapositioning of two appropriate lens to form a telescope seems almost unavoidable. In any event, it is most likely that a Dutch optician, possibly even the ubiquitous Zacharias Jenssen of microscope fame, first constructed a telescope and in addition had inklings of the value of what he was peering into. The earliest indisputable evidence of the discovery, however, dates to October 2, 1608 when Hans Lippershey petitioned the States-General of Holland for a patent on a device for seeing at a distance (which is what *teleskopos* means in Greek). Incidentally, as you might have guessed, its military possibilities were immediately recognized. His patent was therefore not granted; instead the

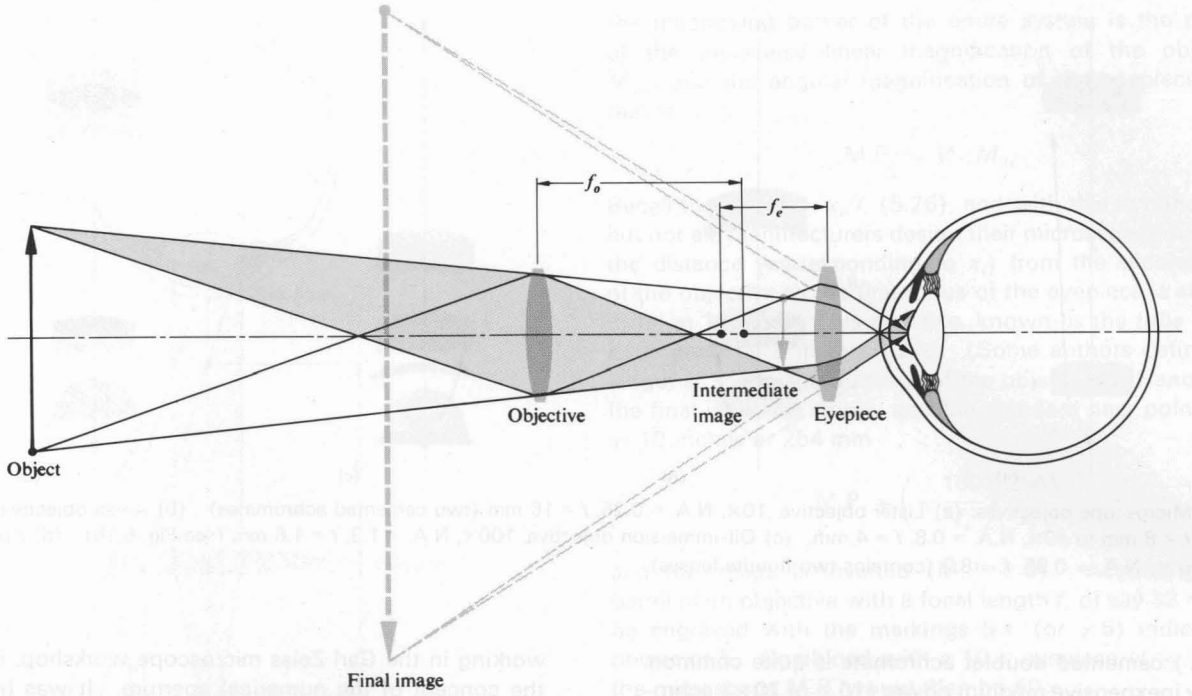


Fig. 5.88 Keplerian astronomical telescope (accommodating eye).

government purchased the rights to the instrument and he received a commission to continue research. Galileo heard of this work and by 1609 he had fashioned a telescope of his own using two lenses and an organ pipe as a tube. It was not long before he constructed a number of greatly improved instruments and began to astound the world with the forthcoming astronomical discoveries for which he is so justly famous.

i) Refracting Telescopes

A simple *astronomical* telescope is shown in Fig. 5.88. Unlike the compound microscope, which it so closely resembles, its primary function is to enlarge the retinal image of a *distant* object. In the illustration, the object is at a finite far distance from the objective so that the real intermediate image is formed just beyond its second focal point. This image will be the object for the next lens system, i.e. the ocular. It follows from Table 5.3 (p. 112) that if the eyepiece is to form a virtual magnified final image (within the range of normal accommodation), the object distance must be less than or equal to the focal length, f_e. In practice, *the position of the intermediate image is fixed and only the eyepiece is moved to focus the instrument.* Notice that *the final image is inverted,* but as long as the scope is used for astronomical observations, this is of little consequence especially since most work is photographic.

At very great object distances the incident rays are effectively parallel—the intermediate image resides at the second focus of the objective. Usually the eyepiece is located so that its first focus overlaps the second focus of the objective in which case rays diverging from a point on the intermediate image will leave the ocular parallel to each other. A normal viewing eye can then focus the rays in a relaxed configuration. Certainly if the eye is nearsighted or farsighted, the ocular can be moved in or out so that the rays diverge or converge a bit to compensate. (If you are astigmatic, you'll have to keep your glasses on when using ordinary visual instruments.) We saw earlier [Section 5.2.3, (iv)] that both the back and front focal lengths of a thin lens combination go to infinity when the two lenses are separated by a distance d equal to the sum of their focal lengths (Fig. 5.89). The astronomical telescope in this configuration of *infinite con-*

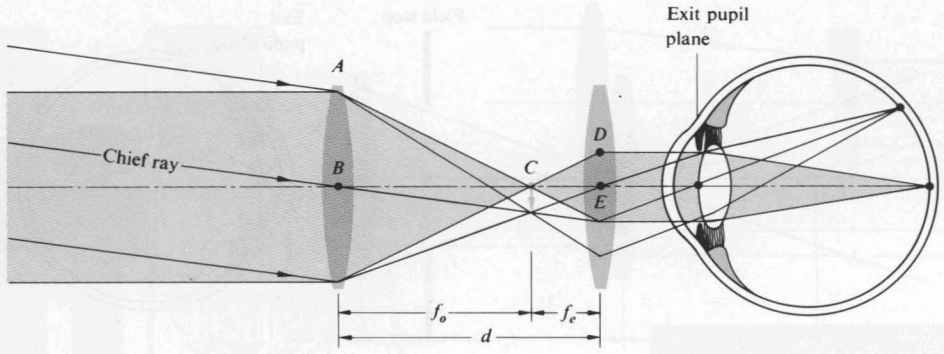


Fig. 5.89 Astronomical telescope—infinite conjugates.

jugates is said to be *afocal*, i.e. without a focal length. As a side note, if you shine a collimated (parallel rays, i.e. plane waves) narrow laser beam into the back end of a scope focused at infinity, it will emerge still collimated, but with an increased cross-section. It is often desirable to have a broad, quasimonochromatic, plane-wave beam, and specific devices of this sort are now available commercially.

The periphery of the objective is the aperture stop and it encompasses the entrance pupil as well, there being no lenses to the left of it. If the telescope is trained directly on some distant galaxy, the visual axis of the eye will presumably be collinear with the central axis of the scope. The entrance pupil of the eye should then coincide in space with the exit pupil of the scope. However, the eye is not immobile. It will move about scanning the entire field of view, which quite often contains many points of interest. In effect, the eye examines different regions of the field by rotating so that rays from a particular area fall on the fovea centralis. The direction established by the chief ray through the center of the entrance pupil to the fovea centralis is the *primary line of sight*. The axial point, fixed in reference to the head, through which the primary line of sight always passes, regardless of the orientation of the eyeball, is called the *sighting intersect*. When it is desirable to have the eye surveying the field, the sighting intersect should be positioned at the center of the telescope's exit pupil. In that case, the primary line of sight will always correspond to a chief ray through the center of the exit pupil, however the eye moves.

Suppose that the margin of the visible object subtends a half-angle of α at the objective (Fig. 5.90). This is essentially the same as the angle α_u which would be subtended at the unaided eye. As in previous sections, the angular magnification is

$$\text{M.P.} = \frac{\alpha_a}{\alpha_u} \qquad [5.65]$$

Here α_u and α_a are measures of the field of view in object and image space, respectively. The first is the half-angle of the actual cone of rays collected while the second relates to the apparent cone of rays. If a ray arrives at the objective with a negative slope, it will enter the eye with a positive slope and vice versa. Thus to make the sign of M.P. positive for erect images, and therefore consistent with previous usage (Fig. 5.78), either α_u or α_a must be taken to be negative—we choose the former because the ray has a negative slope. Observe that the ray passing through the first focus of the objective passes through the second focus of the eyepiece, i.e. F_{o1} and F_{e2} are conjugate points. In the paraxial approximation $\alpha \approx \alpha_u \approx \tan \alpha_u$ and $\alpha_a \approx \tan \alpha_a$. The image fills the region of the field stop and so half its extent equals the distance $\overline{BC} = \overline{DE}$. Thus, from triangles $F_{o1}BC$ and $F_{e2}DE$, the ratio of the tangents yields

$$\text{M.P.} = -\frac{f_o}{f_e}. \qquad (5.73)$$

Another convenient expression for the M.P. comes from considering the transverse magnification of the ocular. Inasmuch as the exit pupil is the image of the objective (Fig. 5.90) we have

$$M_{Te} = -\frac{f_e}{x_o} = -\frac{f_e}{f_o}.$$

Furthermore if D_o is the *diameter of the objective* and D_{ep} is

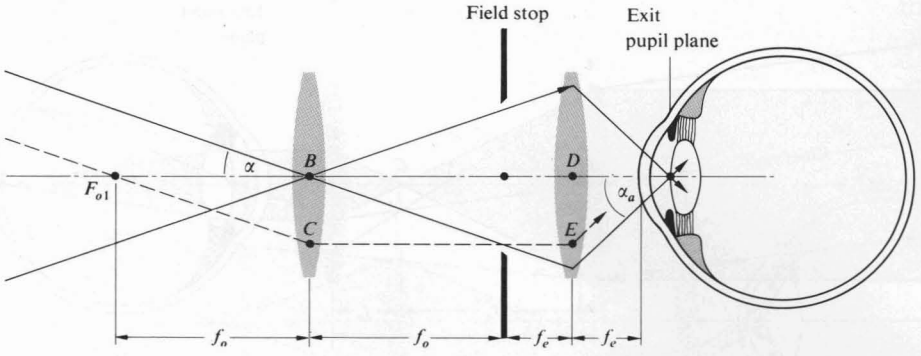


Fig. 5.90 Ray angles for a telescope.

the *diameter of its image, the exit pupil,* then $M_{Te} = D_{ep}/D_o$. These two expressions for M_{Te} compared with Eq. (5.73) yield

$$\text{M.P.} = \frac{D_o}{D_{ep}}. \tag{5.74}$$

Here D_{ep} is actually a negative quantity since the image is inverted. It is an easy matter to build a simple refracting scope by just holding a long focal length lens in front of one having a short focal length and making sure that $d = f_o + f_e$. But again, well-corrected telescopic instruments generally have multielement objectives, usually doublets or triplets.

To be useful when the orientation of the object is of importance, a scope must contain an additional *erecting system*—such an arrangement is known as a *terrestrial telescope*. Most often a single erecting lens or lens system is located between the ocular and objective with the result that the image is right side up. Figure 5.91 shows one with a cemented doublet objective and a Kellner eyepiece. It will obviously have to have a long draw tube, the picturesque kind that comes to mind when you think of wooden ships and cannonballs.

For that reason, *binoculars* (binocular telescopes) generally utilize erecting prisms which accomplish the same thing in less space, and also increase the separation of the objectives, thereby enhancing the stereoscopic effect. Most often these are double Porro prisms, as in Fig. 5.92 (notice the involved modified Erfle eyepiece, the wide field stop and achromatic doublet objective). Binoculars customarily bear several numerical markings, e.g. 6 × 30, 7 × 50, or 20 × 50, etc. The initial number is the magnification, here 6×, 7×, or 20×. The second number is the entrance pupil diameter, or equivalently the clear aperture of the objective, expressed in millimeters. It follows from Eq. (5.74) that the exit-pupil diameters will be the second number divided by the first or in this case 5, 7.1 and 2.5, all in millimeters. You can hold the instrument away from your eye and see the bright circular exit pupil surrounded by blackness. To measure it, focus the device at infinity, point it at the sky, and observe the emerging well-defined sharp disk of light

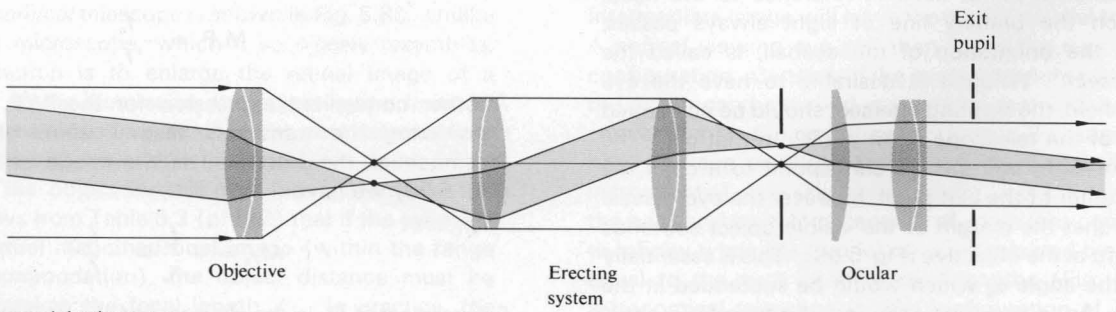


Fig. 5.91 A terrestrial telescope.

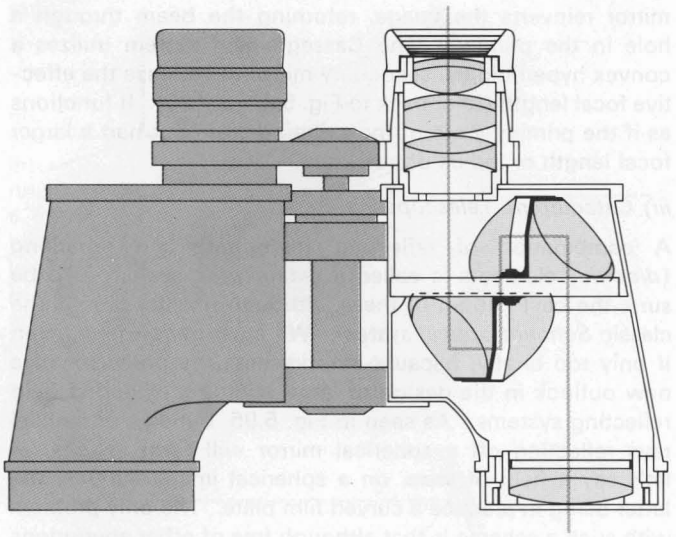


Fig. 5.92 A binocular.

using a piece of paper as a screen. Determine the eye relief while you're at it.

By the way, as long as $d = f_o + f_e$, the scope will be afocal, even if the eyepiece is negative (i.e. $f_e < 0$). The telescope built by Galileo (Fig. 5.93) had just such a negative lens as an eyepiece and therefore formed an erect image [$f_e < 0$ and M.P. > 0 in Eq. (5.73)]. As a telescope the system is now mainly of historical and pedagogical interest although one can still purchase two such scopes mounted side by side to form a Galilean fieldglass. It is quite useful, however, as a laser-beam expander because it has no internal focal points where a high power beam would otherwise ionize the surrounding air.

ii) Reflecting Telescopes

The difficulties inherent in making large lenses become underscored when we note that the largest refracting instrument is the 40-inch Yerkes telescope in Williams Bay, Wisconsin, while the reflector on Palomar Mountain, U.S.A. is 200 inches in diameter and the Soviet Union is constructing a 236 inch reflector at their Crimea Observatory. The problems are evident ones; a lens must be transparent and free of internal bubbles, etc. A front-surfaced mirror obviously need not be so, indeed it need not even be transparent. A lens can only be supported by its rim and may sag under its own weight; a mirror can be supported by its rim and back as well. Furthermore, since there is no refraction, and therefore no

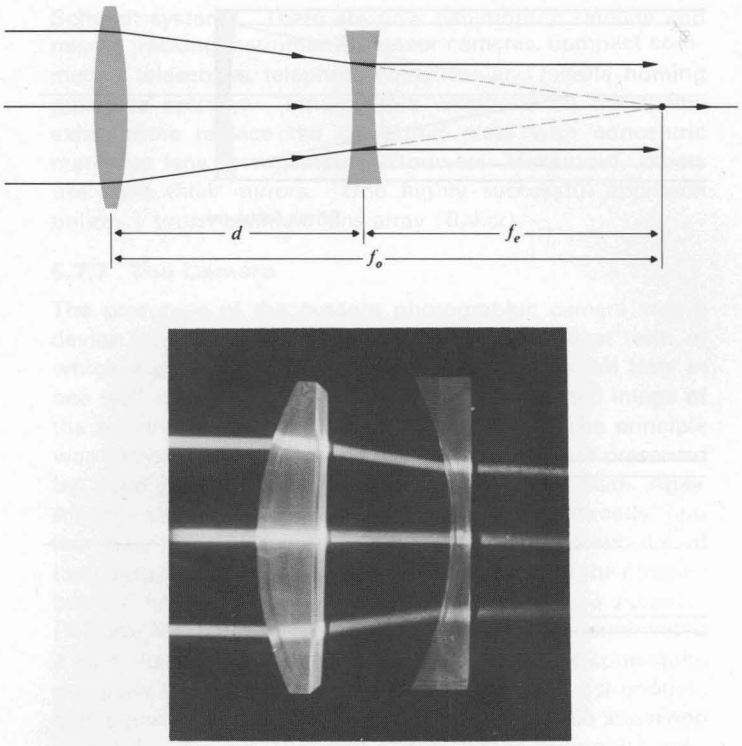
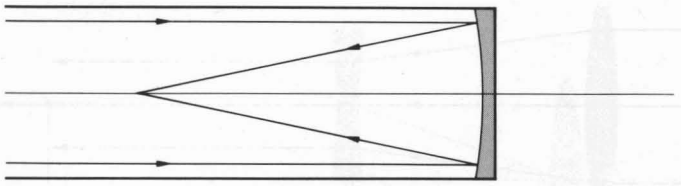


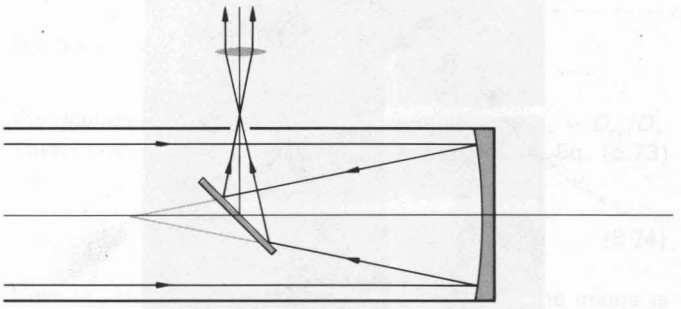
Fig. 5.93 The Galilean telescope. Galileo's first scope had a planar-convex objective (5.6 cm in diameter, $f = 1.7$ m, $R = 93.5$ cm) and a planar-concave eyepiece, both of which he ground himself. It was 3× in contrast to his last scope which was 32×. [Photo by E.H.]

effect on the focal length due to the wavelength dependence of the index, mirrors suffer no chromatic aberration. For these and other reasons (e.g. their frequency response) reflectors predominate in large telescopes.

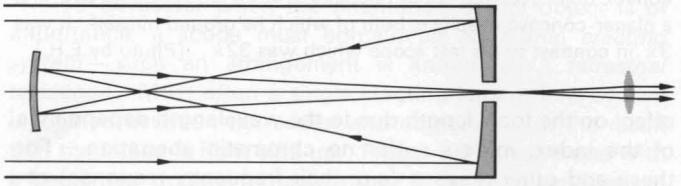
Invented by the Scotsman James Gregory (1638–75) in 1661, the reflecting telescope was first successfully constructed by Newton in 1668, and only became an important research tool in the hands of William Herschel a century later. Figure 5.94 depicts a number of reflector arrangements, each having concave paraboloidal primary mirrors. The 200 inch Hale telescope is so large that a little enclosure, where an observer can sit, is positioned at the prime focus. In the Newtonian version, a plane mirror or prism brings the beam out at right angles to the axis of the scope where it can be photographed, viewed, spectrally analyzed or photoelectrically processed. In the Gregorian arrangement, which is not particularly popular, a concave ellipsoidal secondary



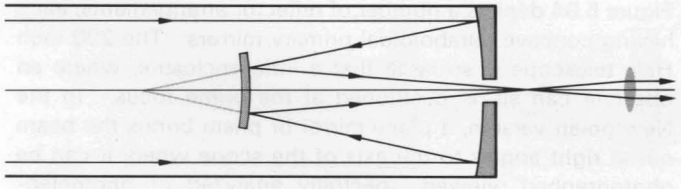
Prime focus (a)



Newtonian (b)



Gregorian (c)



Cassegrainian (d)

Fig. 5.94 Reflecting telescopes.

mirror reinverts the image, returning the beam through a hole in the primary. The Cassegrainian system utilizes a convex hyperboloidal secondary mirror to increase the effective focal length (refer back to Fig. 5.43, p. 125). It functions as if the primary were of the same aperture but had a larger focal length or radius of curvature.

iii) Catadioptric Telescopes

A combination of reflecting (*catoptric*) and refracting (*dioptric*) elements is called a *catadioptric* system. To be sure, the best known of these, although not the first, is the classic *Schmidt optical system*. We must treat it here, even if only too briefly, because it represents the precursor of a new outlook in the design of large-aperture, extended-field reflecting systems. As seen in Fig. 5.95, bundles of parallel rays reflecting off a spherical mirror will form images, of let's say a field of stars, on a spherical image surface, the latter being in practice a curved film plate. The only problem with such a scheme is that although free of other aberrations [see Section 6.3.1 (iv)] we know that rays reflected from the outer regions of the mirror will not arrive at the same focus as those from the paraxial region. In other words, the mirror is a sphere, not a paraboloid and it suffers *spherical aberration* [Fig. 5.95(b)]. If this could be corrected, the system (in theory at least) would be capable of perfect imagery over a wide field of view. Since there is no one central axis, there are, in effect, no off-axis points. Recall that the paraboloid forms perfect images only at axial points, the image deteriorating rapidly off axis. One evening in 1929, while sailing on the Indian ocean (returning from an eclipse expedition to the Philippines), Bernhard Voldemar Schmidt (1879–1935) showed a colleague a sketch of a system he had designed to cope with the spherical aberration of a spherical mirror. He would use a thin glass corrector plate on whose surface would be ground a very shallow toroidal curve [Fig. 5.95(c)]. Light rays traversing the outer regions would be deviated by just the amount needed to be sharply focused on the image sphere. The corrector must overcome one defect without introducing appreciable amounts of other aberrations. This first system was built in 1930 and in 1949 the famous 48 inch Schmidt telescope of the Palomar Observatory was completed. It is a fast ($f/2.5$), wide-field device, ideal for surveying the night sky. A single photograph could encompass a region the size of the bowl of the Big Dipper—this compared with roughly 400 photos by the 200 inch reflector to cover the same area.

Major advances in the design of catadioptric instrumentation have occurred since the introduction of the original

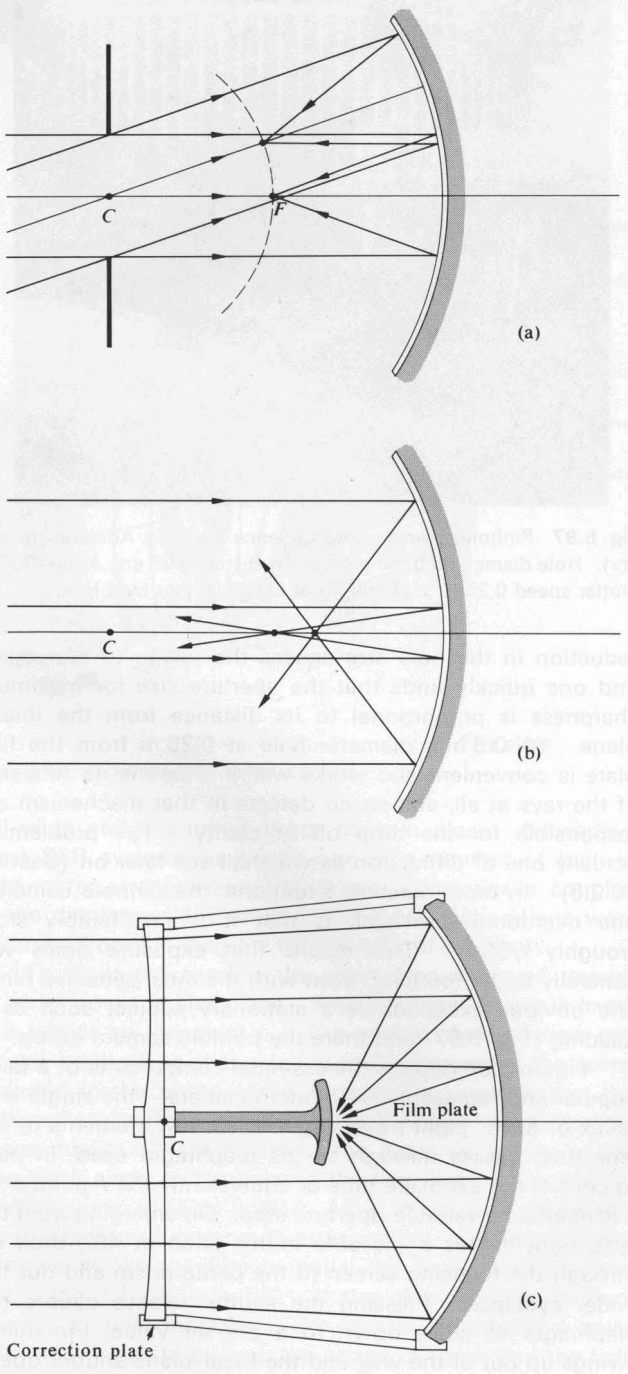


Fig. 5.95 The Schmidt optical system.

Schmidt system.* There are now catadioptric satellite and missile tracking instruments, meteor cameras, compact commercial telescopes, telephoto objectives and missile-homing guidance systems. Innumerable variations on the theme exist; some replace the correcting plate with concentric meniscus lens arrangements (Bouwers–Maksutov), others use solid thick mirrors. One highly successful approach utilizes a triplet aspheric lens array (Baker).

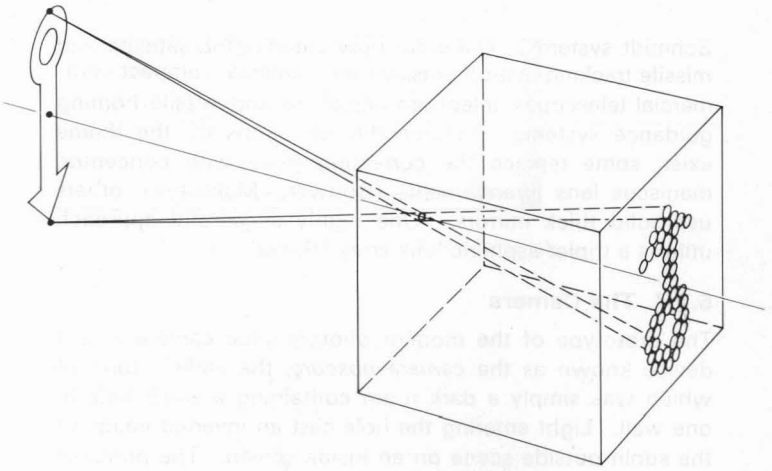
5.7.7 The Camera

The prototype of the modern photographic camera was a device known as the *camera obscura*, the earliest form of which was simply a dark room containing a small hole in one wall. Light entering the hole cast an inverted image of the sunlit outside scene on an inside screen. The principle was known to Aristotle and his observations were preserved by Arab scholars throughout Europe's long Dark Ages. Alhazen utilized it to examine solar eclipses indirectly, and this over eight hundred years ago. The notebooks of Leonardo da Vinci contain several descriptions of the obscura but the first detailed treatment appears in *Magia naturalis* (Natural Magic) by Giovanni della Porta. He recommended it as a drawing aid, a function to which it was soon quite popularly put. Johannes Kepler, the renowned astronomer, had a portable tent version which he used while surveying in Austria. By the latter part of the sixteen hundreds, small hand-held *camera obscuras* were commonplace. We note parenthetically that the eye of the Nautilus, a little cuttlefish, is literally an open pinhole obscura which simply fills with sea water on immersion.

By replacing the viewing screen with a photosensitive surface, such as a film plate, the obscura becomes a camera in the modern sense of the word. The very first permanent photograph was made in 1826 by Joseph Nicéphore Niépce (1765–1833) using a box camera with a small convex lens, a sensitized pewter plate and a roughly eight hour exposure. It is a roof-top scene, taken from the workroom window of his estate near Châlon-sur-Saône in France. Although blurry and spotty (in its unretouched form), the large slanting roof of a barn, a pigeon-house, and a distant tree are still discernible.

The lensless pinhole camera (Fig. 5.96) is by far the least complicated device for the purpose, and yet it has several endearing and, indeed, remarkable virtues. It can form a very well-defined, practically undistorted, image of

* For further reading see e.g. J. J. Villa, "Catadioptric Lenses," *Optical Spectra* (March/April, 1968), p. 57.



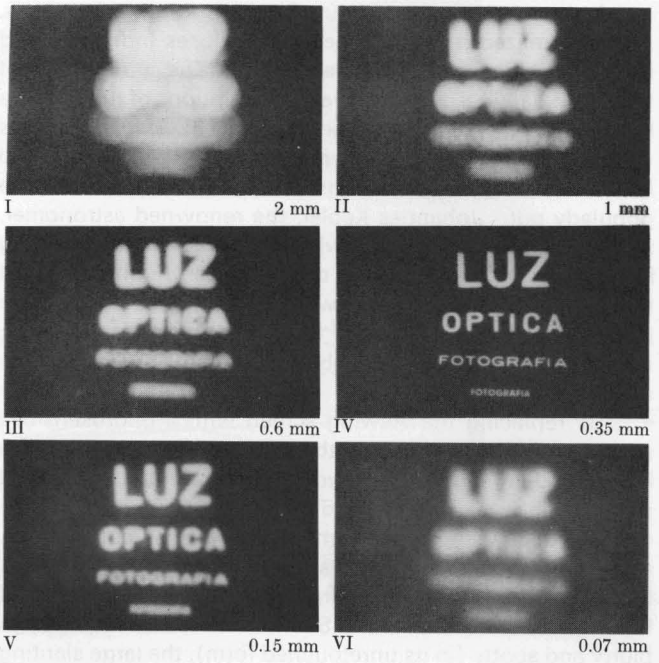


Fig. 5.96 The pinhole camera. Note the variation in image clarity as the hole diameter decreases. [Photos courtesy Dr. N. Joel, UNESCO.]

objects across an extremely wide angular field (due to great depth of focus) and over a large range of distances (great depth of field). If initially the entrance pupil is very large, no image results. As it is decreased in diameter, the image forms and grows ever sharper. After a point, further

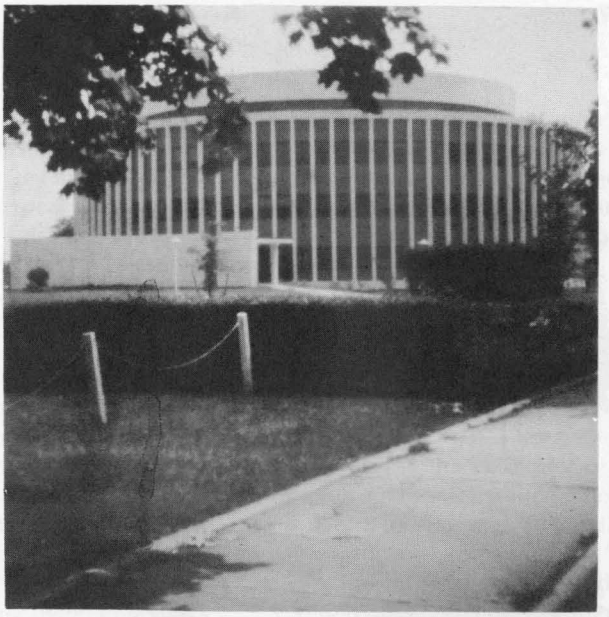


Fig. 5.97 Pinhole camera photo (Science Building, Adelphi University). Hole diameter 0.5 mm, film plane distance 25 cm, A.S.A. 3000, shutter speed 0.25 s. Note depth of field. [Photo by E.H.]

reduction in the hole size causes the image to blur again and one quickly finds that the aperture size for maximum sharpness is proportional to its distance from the image plane. (A 0.5 mm diameter hole at 0.25 m from the film plate is convenient and works well.) There is no focusing of the rays at all, and so no defects in that mechanism are responsible for the drop off in clarity. The problem is actually one of diffraction as we shall see later on (Section 10.2.5). In most practical situations, the pinhole camera's one overriding drawback is that it is insufferably slow (roughly $f/500$). This means that exposure times will generally be far too long, even with the most sensitive films. The obvious exception is a stationary subject such as a building (Fig. 5.97) and there the pinhole camera excels.

Figure 5.98 depicts the essential components of a fairly popular and representative modern camera—the single lens reflex or SLR. Light traversing the first few elements of the lens then passes through an iris diaphragm used, in part, to control the exposure time or equivalently the f-number—it is in effect a variable-aperture stop. On emerging from the lens, light strikes a movable mirror tilted at 45°, then up through the focusing screen to the penta prism and out the finder eyepiece. Pressing the shutter release causes the diaphragm to close down to a pre-set value, the mirror swings up out of the way and the focal-plane shutter opens exposing the film. The shutter closes, the diaphragm opens

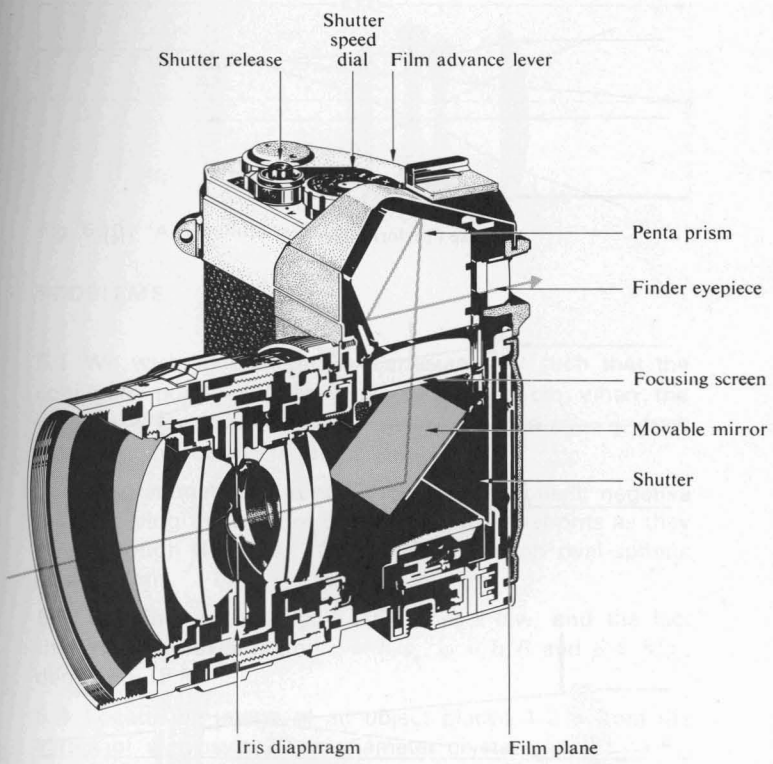


Fig. 5.98 A single-lens reflex camera.

full wide and the mirror drops back in place. Nowadays most SLR systems have any one of a number of built-in light-meter arrangements which are automatically coupled to the diaphragm and shutter, but those components were excluded from the diagram for the sake of simplicity.

To focus the camera, the entire lens is moved toward or away from the film plane. Since its focal length is fixed, as s_o varies so too must s_i. The *angular field of view* can loosely be thought of as relating to the fraction of the scene included in the photo. It is furthermore required that the entire photo surface correspond to a region of satisfactory image quality. More precisely, the angle subtended at the lens, by a circle encompassing the film area, is the angular field of view φ (Fig. 5.99). As a rough but reasonable approximation of a common arrangement, take the diagonal distance across the film to equal the focal length. Thus $\varphi/2 \approx \tan^{-1} \frac{1}{2}$, i.e. $\varphi \approx 53°$. If the object comes in from infinity, s_i must increase. The lens is then backed away from the film plate to keep the image in focus and the field of view, as recorded on the film whose periphery is the field

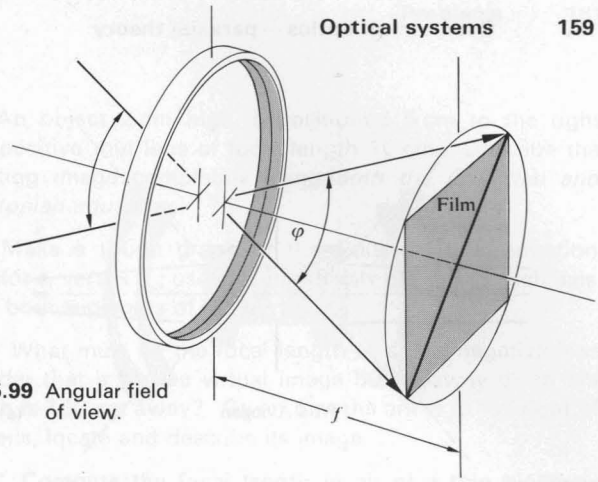
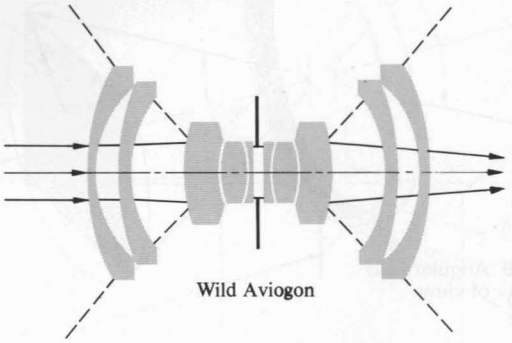
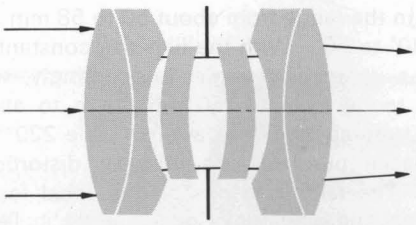


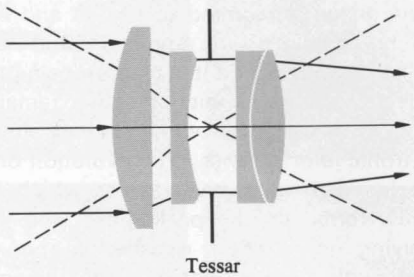
Fig. 5.99 Angular field of view.

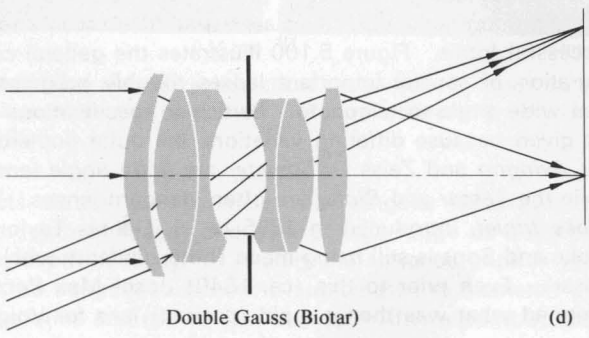
stop, decreases. A *standard* SLR lens has a focal length somewhere in the range from about 50 to 58 mm and a field of view of 40° to 50°. With the film size constant, reducing f results in a wider field angle. Accordingly, *wide-angle* SLR lenses range from $f \approx 40$ mm down to about 6 mm and φ goes from about 50° to a remarkable 220° (the latter being a special purpose lens wherein distortion is unavoidable). The *telephoto* has a long focal length, from roughly 80 mm and upwards. Consequently, its field of view drops off rapidly until it is only a few degrees at $f \approx 1000$ mm.

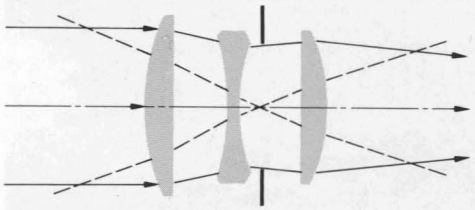
The standard photographic objective must have a large relative aperture, $1/(f/\#)$, to keep exposure times short. Moreover, the image is required to be flat and undistorted and the lens should have a wide angular field of view as well. All of this is no mean task and it is not surprising that a high quality innovative photographic objective remains particularly difficult to design even with our marvelous, mathematical, electronic *idiot savants*. The evolution of a modern lens still begins with a creative insight which leads to a promising new form. In the past, these were laboriously perfected relying on intuition, experience and, of course, trial and error with a succession of developmental lenses. Today, for the most part, the computer serves this function without the need of numerous prototypes. Many contemporary photographic objectives are variations of well-known successful forms. Figure 5.100 illustrates the general configuration of several important lenses roughly progressing from wide angle to telephoto. Particular specifications are not given because differing variations are quite numerous. The *Aviogon* and Zeiss *Orthometer* are wide angle lenses, while the *Tessar* and *Biotar* are often standard lenses. The *Cooke triplet*, introduced in 1895 by H. Dennis Taylor of Cooke and Sons is still being made (note similarity with the Tessar). Even prior to this (ca. 1840) Josef Max Petzval designed what was then a rapid (portrait) lens for Voight-

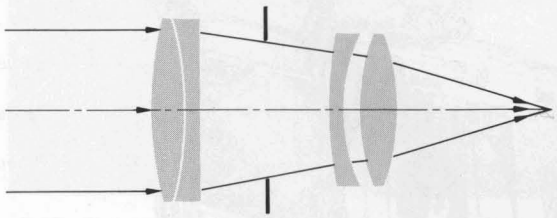












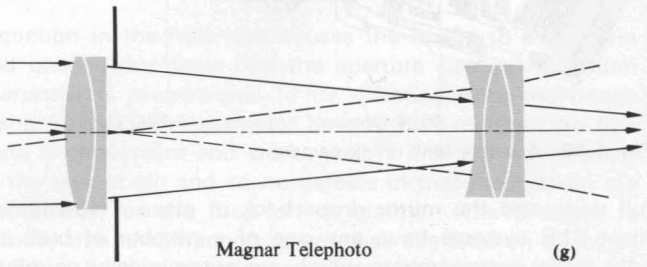


Fig. 5.100 Camera lenses.

länder and Son. Its modern offshoots are myriad. In general, a telephoto objective has a positive front grouping and a distant negative rear grouping. It often resembles the Galilean scope except that the lenses are shifted a bit so that the system is not afocal. These are usually rather large and heavy at the longer focal lengths, although calcium fluoride elements have begun to help in both respects. As can be seen in Fig. 5.101, the telephoto has a large effective focal length, e.f.l., i.e. it behaves as if it were a long focal length positive lens located a large distance in front of the focal plane. Thus while the image size is large, the back focal length is conveniently short, allowing it to be handily slipped into a standard camera body.

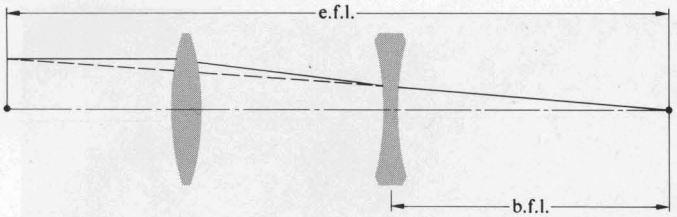


Fig. 5.101 A telephoto lens.

PROBLEMS

5.1 We wish to construct a Cartesian oval such that the conjugate points will be separated by 11 cm when the object is 5 cm from the vertex. If $n_1 = 1$ and $n_2 = \frac{3}{2}$, draw several points on the required surface.

5.2 Diagrammatically construct an ellipto-spheric negative lens showing the form of both rays and wavefronts as they pass through the lens. Do the same for an oval-spheric positive lens.

5.3* Making use of Fig. 5.102, Snell's law, and the fact that in the paraxial region $\alpha = h/s_o$, $\varphi \approx h/R$ and $\beta \approx h/s_i$, derive Eq. (5.8).

5.4 Locate the image of an object placed 1.2 m from the vertex of a gypsy's 20 cm diameter crystal ball ($n = 1.5$). Make a sketch of the thing (not the gypsy, the rays).

5.5 Prove that the minimum separation between conjugate *real* object and image points for a thin positive lens is $4f$.

5.6 A biconcave lens ($n_l = 1.5$) has radii of 20 cm and 10 cm and an axial thickness of 5 cm. Describe the image of a one inch tall object placed 8 cm from the first vertex.

5.7* Use the thin-lens equation on the previous problem to see how far off it is in determining the final-image location.

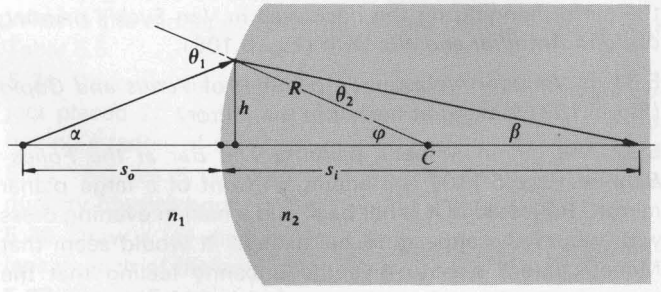


Fig. 5.102

5.8 An object 2 cm high is positioned 5 cm to the right of a positive thin lens of focal length 10 cm. Describe the resulting image completely using *both the Gaussian and Newtonian equations*.

5.9 Make a rough graph of the Gaussian lens equation, i.e. plot s_i versus s_o, using unit intervals of f along each axis. (Get both segments of the curve.)

5.10 What must be the focal length of a thin negative lens in order that it form a virtual image 50 cm away of an ant which is 100 cm away? Given that the ant is to the right of the lens, locate and describe its image.

5.11* Compute the focal length in air of a thin biconvex lens ($n_l = 1.5$) having radii 20 cm and 40 cm. Locate and describe the image of an object 40 cm from the lens.

5.12 Determine the focal length of a planar-concave lens ($n_l = 1.5$) having a radius of curvature of 10 cm. What is its power in diopters?

5.13 Write an expression for the focal length (f_w) of a thin lens immersed in water ($n_w = \frac{4}{3}$) in terms of its focal length when it's in air (f_a).

5.14* A convenient way to measure the focal length of a positive lens makes use of the following fact. If a pair of conjugate object and (real) image points (S and P) are separated by a distance $L > 4f$, there will be two locations of the lens, a distance d apart, for which the same pair of conjugates obtain. Show that

$$f = \frac{L^2 - d^2}{4L}.$$

Note that this avoids measurements made specifically from the vertex which are generally not easy to do.

5.15 An equiconvex thin lens L_1 is cemented in intimate contact with a thin negative lens, L_2, such that the combination has a focal length of 50 cm in air. If their indices are 1.50 and 1.55 respectively and if the focal length of L_2 is -50 cm, determine all of the radii of curvature.

5.16 Verify Eq. (5.34) which gives M_T for a combination of two thin lenses.

5.17 Compute the image location and magnification of an object 30 cm from the front doublet of the thin-lens combination in Fig. 5.103. Do the calculation by finding the effect of each lens separately. Make a sketch of appropriate rays.

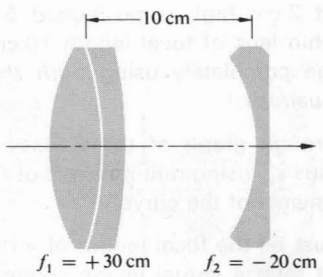


Fig. 5.103

5.18* Draw a ray diagram for the combination of two positive lenses wherein their separation equals the sum of their respective focal lengths. Do the same thing for the case where one of the lenses is negative.

5.19 Redraw the ray diagram for a compound microscope (Fig. 5.86) but this time treat the intermediate image as if it were a real object—this approach should be a bit simpler.

5.20* Redraw the telescope of Fig. 5.89 taking advantage of the fact that the intermediate image can be thought of as a real object (as in the previous problem).

5.21 Consider the case of two positive thin lenses L_1 and L_2 separated by 5 cm. Their diameters are 6 cm and 4 cm respectively and their focal lengths are $f_1 = 9$ cm and $f_2 = 3$ cm. If a diaphragm with a 1 cm dia. hole is located between them, 2 cm from L_2, find (a) the A.S. and (b) the locations and sizes of the pupils for an axial point, S, 12 cm in front of (to the left of) L_1.

5.22 Make a sketch roughly locating the aperture stop and entrance and exit pupils for the lens in Fig. 5.104.

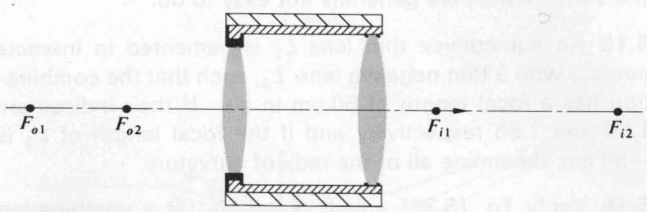


Fig. 5.104

5.23 Make a sketch roughly locating the aperture stop and entrance and exit pupils for the lens in Fig. 5.105 assuming the object point to be beyond (to the left of) F_{o1}.

5.24 Draw a ray diagram locating the images of a point source as formed by a pair of mirrors at 90° (Fig. 5.106).

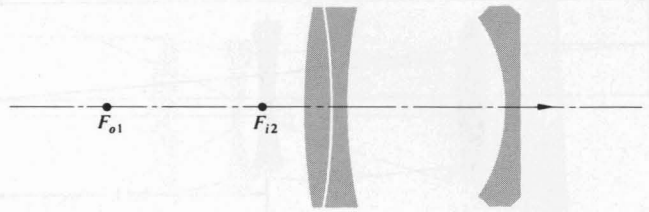


Fig. 5.105

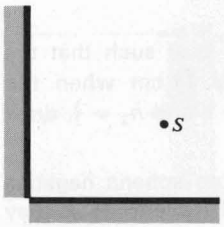


Fig. 5.106

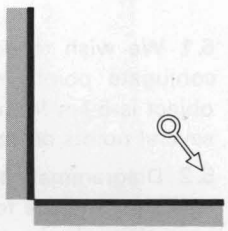


Fig. 5.107

5.25* Make a sketch of a ray diagram locating the images of the arrow shown in Fig. 5.107.

5.26 Show that Eq. (5.49) for a spherical surface is equally applicable to a plane mirror.

5.27 Locate the image of a paperclip 100 cm away from a convex spherical mirror having a radius of curvature of 80 cm.

5.28* Describe the image you would see standing 5 feet from, and looking directly toward, a 1 ft diameter brass ball hanging in front of a pawn shop.

5.29 The image of a red rose is formed by a concave spherical mirror on a screen 100 cm away. If the rose is 25 cm from the mirror, determine its radius of curvature.

5.30 From the image configuration determine the shape of the mirror hanging on the back wall in Van Eyck's painting of *John Arnolfini and His Wife* (Fig. 5.108).

5.31 Is Venus in Velasquez's painting of *Venus and Cupid* (Fig. 5.109) looking at herself in the mirror?

5.32 The girl in Manet's painting *The Bar at the Folies-Bergère* (Fig. 5.110) is standing in front of a large planar mirror. Reflected in it is her back and a man in evening dress with whom she appears to be talking. It would seem that Manet's intent was to give the uncanny feeling that the viewer is standing where that gentleman must be. From the laws of geometrical optics what is amiss?

Fig. 5.108 Detail of *John Arnolfini and His Wife* by Jan Van Eyck—National Gallery, London.

5.33* Suppose you have a concave spherical mirror of focal length 10 cm. At what distance must an object be placed if its image is to be erect and one and a half times as large? What is the radius of curvature of the mirror? Check with Table 5.5.

5.34 Describe the image that would result for a 3 in tall object placed 20 cm from a spherical concave shaving mirror having a radius of curvature of -60 cm.

5.35* Figures 5.111 and 5.112 are taken from an introductory physics book. What's wrong with them?

5.36 Draw a ray diagram for a finite object and a positive lens in each of the configurations treated in Table 5.3.

5.37 Figure 5.113 shows a lens system, an object and the appropriate pupils. Diagrammatically locate the image.

5.38 Referring to the dove prism of Fig. 5.57 rotate it through 90° about an axis along the ray direction. Sketch the new configuration and determine the angle through which the image is rotated.

5.39 Determine the numerical aperture of a single clad optical fiber given that the core has an index of 1.62 and the clad 1.52. When immersed in air, what is its maximum acceptance angle? What would happen to a ray incident at say 45°?

5.40 Using the information on the eye in Section 5.7.2 compute the approximate size (in millimeters) of the image of the moon as cast on the retina. The moon has a diameter of 2160 miles and is roughly 230,000 miles from here although this, of course, varies.

Fig. 5.109 *Venus and Cupid* by Diego Rodríguez de Silva y Velásquez—National Gallery, London.

Fig. 5.110 *The Bar at the Folies Bergères* by Édouard Manet—Courtauld Institute Galleries, London.

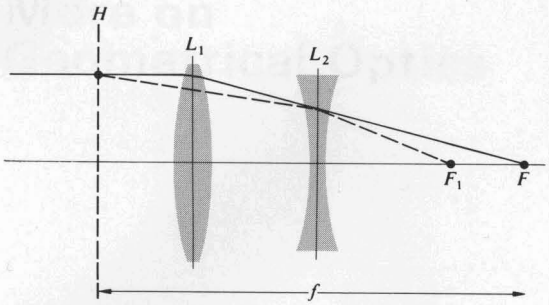


Fig. 5.111

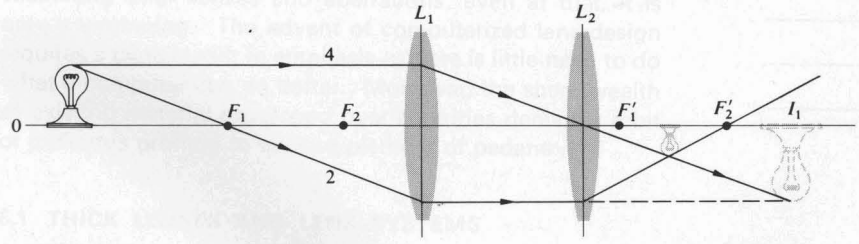


Fig. 5.112

5.41* Figure 5.114 shows an arrangement such that the beam is deviated through a constant angle σ, equal to twice the angle β between the plane mirrors, regardless of the angle of incidence. Prove that this is indeed the case.

5.42 An object 20 m from the objective ($f_o = 4$ m) of an astronomical telescope is imaged 30 cm from the eyepiece ($f_e = 60$ cm). Find the total linear magnification of the scope.

5.43* Figure 5.115 which purports to show an erecting lens system, is taken from an old, out-of-print optics text. What's wrong with it?

5.44* If a photo of a moving merry-go-round is perfectly exposed, but blurred, at $\frac{1}{30}$ s and $f/11$, what must the diaphragm setting be if the shutter speed is raised to $\frac{1}{120}$ s in order to "stop" the motion?

5.45 The field of view of a simple two-element astronomical telescope is restricted by the size of the eye lens. Make a ray sketch showing the vignetting which arises.

5.46 A *field lens*, as a rule, is a positive lens placed at (or near) the intermediate image plane in order to collect the rays which would otherwise miss the next lens in the system. In effect, it increases the field of view without changing the power of the system. Redraw the ray diagram of the previous problem to include a field lens. Show that as a consequence the eye relief is reduced somewhat.

5.47* Describe completely the resulting image of a bug which is sitting at the vertex of a thin positive lens. How does this relate directly to the manner in which a field lens works (see previous problem)?

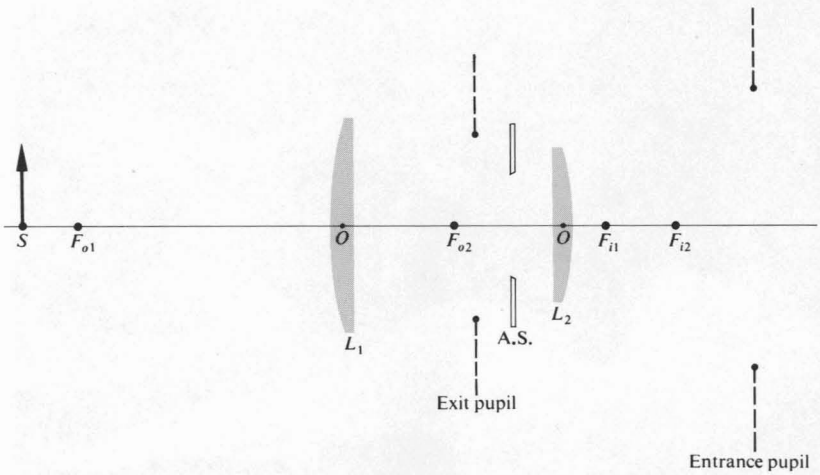


Fig. 5.113

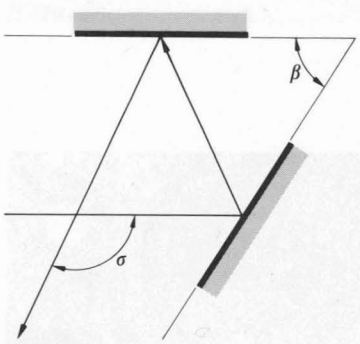


Fig. 5.114

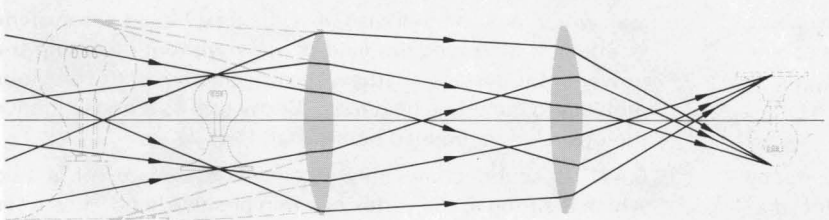


Fig. 5.115